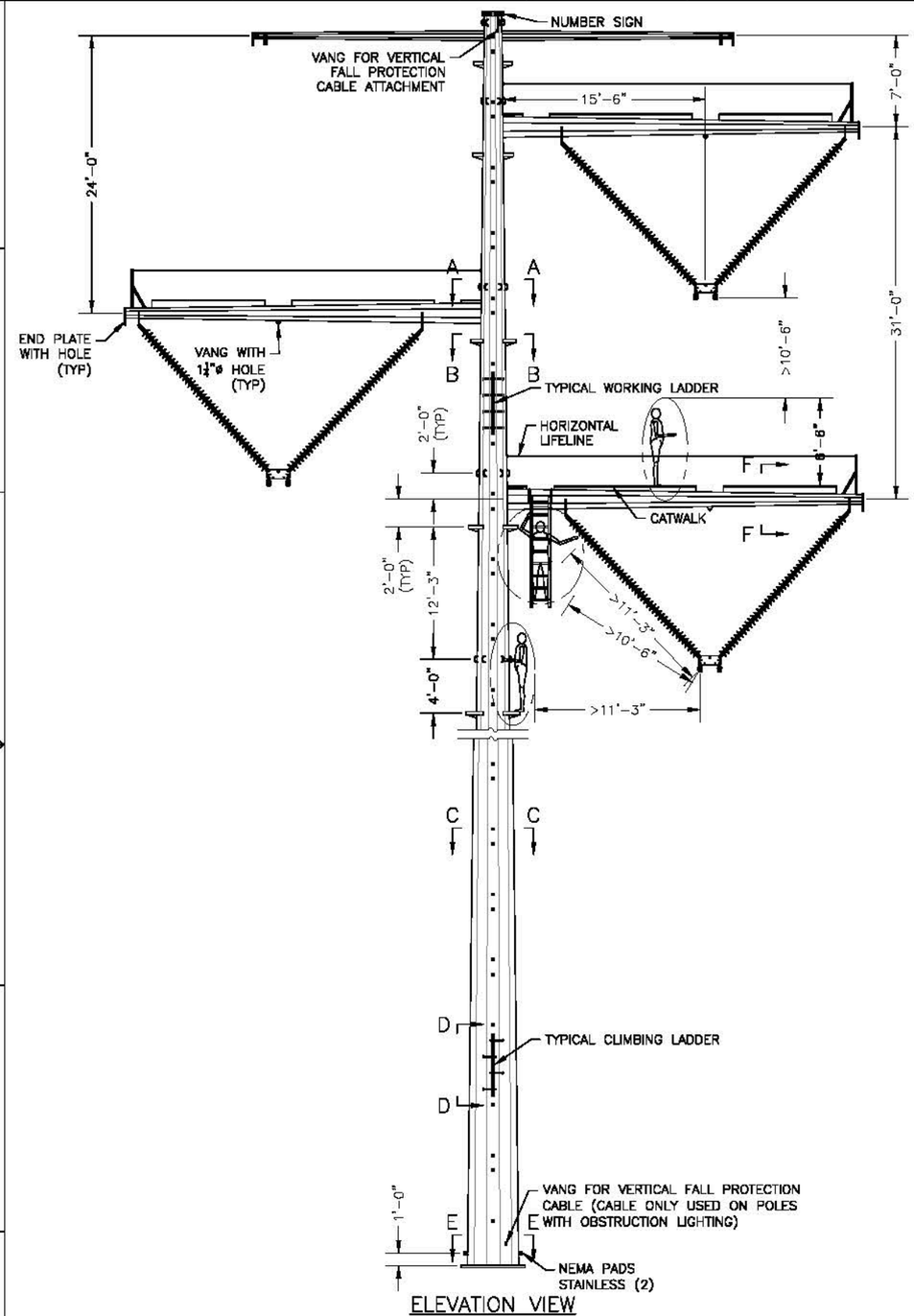
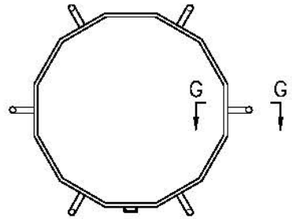
APPENDIX S

Engineering Drawings

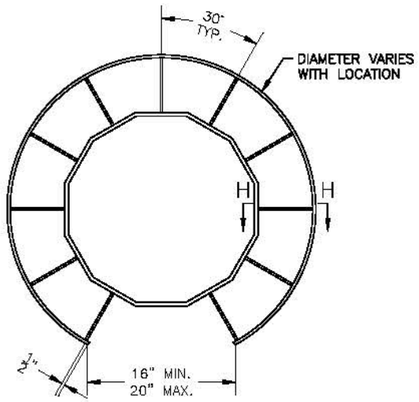
THIS DRAWING IS THE PROPERTY OF CRICKET VALLEY ENERGY AND CONTAINS INFORMATION WHICH IS CONFIDENTIAL AND PROPRIETARY. IT SHALL NOT BE DUPLICATED, USED, OR DISCLOSED WITHOUT WRITTEN AUTHORIZATION FROM CRICKET VALLEY ENERGY.



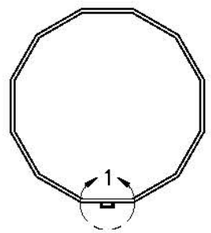
ELEVATION VIEW



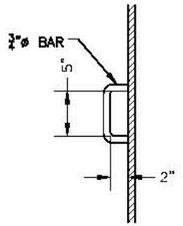
SECTION A–A
HAND HOLD ORIENTATION



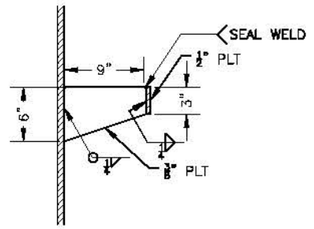
SECTION B–B
MEYERS RING PLATFORM



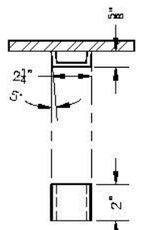
SECTION C–C
CLIMBING LADDER LUGS



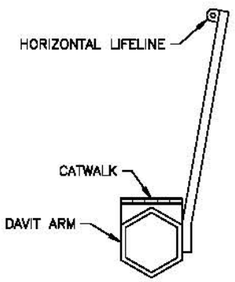
SECTION G–G



SECTION H–H



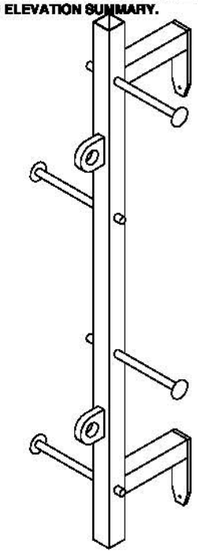
DETAIL 1
CLIMBING LADDER LUG DETAIL



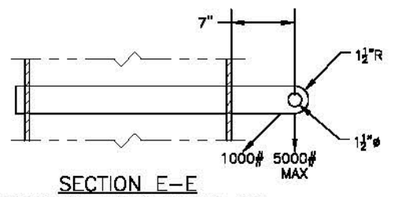
SECTION F–F
HORIZONTAL FALL PROTECTION BENT TUBE

LIVE LINE MAINTENANCE & FALL PROTECTION FEATURES:

1. MCGREGOR PORTABLE CLIMBING & WORKING LADDERS WITH VANGS FOR CONNECTING FALL PROTECTION LANYARD TO POLE.
2. RING PLATFORMS LOCATED 4 FT BELOW CONDUCTOR ATTACHMENT LOCATION AND 2 FT BELOW ℄ ARM.
3. HAND HOLDS 4 FT ABOVE RING PLATFORMS (EXCEPT AT TOP OF POLE).
4. VANG FOR VERTICAL FALL PROTECTION CABLE TO WITHSTAND MINIMUM 5000 LBS VERTICAL IMPACT LOAD (CABLE ONLY PERMANENTLY ATTACHED TO POLES WITH OBSTRUCTION LIGHTING).
5. MINIMUM 11'–3" CLEARANCE USING HOT–LINE TOOLS AND 10'–6" CLEARANCE WITHOUT HOT–LINE TOOLS BETWEEN ENERGIZED PARTS AND GROUNDED LINE WORKER.
6. CATWALK AND HORIZONTAL LIFELINE CABLE ON DAVIT ARMS
7. ANCHORAGE ATTACHMENTS SHALL SUPPORT A 5000# IMPACT LOAD
8. **STRUCTURE HEIGHTS SHALL BE DETERMINED SUCH THAT A MINIMUM 30 FT CONDUCTOR TO GROUND CLEARANCE IS ACHIEVED UNDER THE MAXIMUM OPERATING TEMPERATURE OF 356°F AND THE HEIGHT OR TOP ELEVATION OF STRUCTURE IS EQUAL OR LESS THAN THE VALUES PRESENTED IN TABLE 1- REDUCTION IN STRUCTURE HEIGHT AND ELEVATION SUMMARY.**



SECTION D–D
CLIMBING LADDER WITH FALL PROTECTION VANG



SECTION E–E
VERTICAL FALL PROTECTION THRU ANG

Cricket Valley Energy

DRWN. BY:	MPS
DSGN. BY:	PGC
CK'D. BY:	PGC
APP. BY: PGC	XX/XX/XXXX
DGA PROJ. NO.:	2013-195

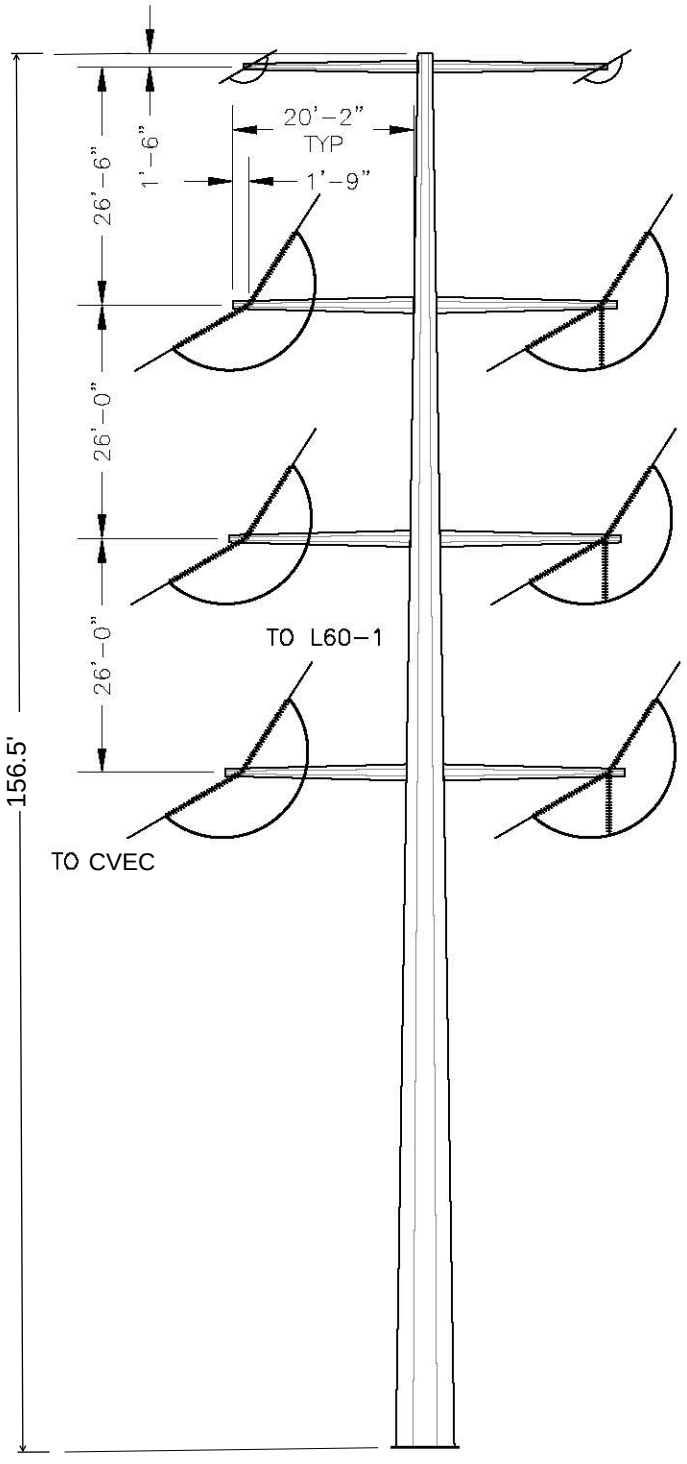
DiGioia Gray & Associates
DiGioia, Gray & Associates, LLC
570 Beatty Road
Monroeville, PA 15146
Phone: (412) 372-4500 Fax: (412) 372-1972

NO.	DATE	ACCT.		BY	REVIEWED	APPROVED

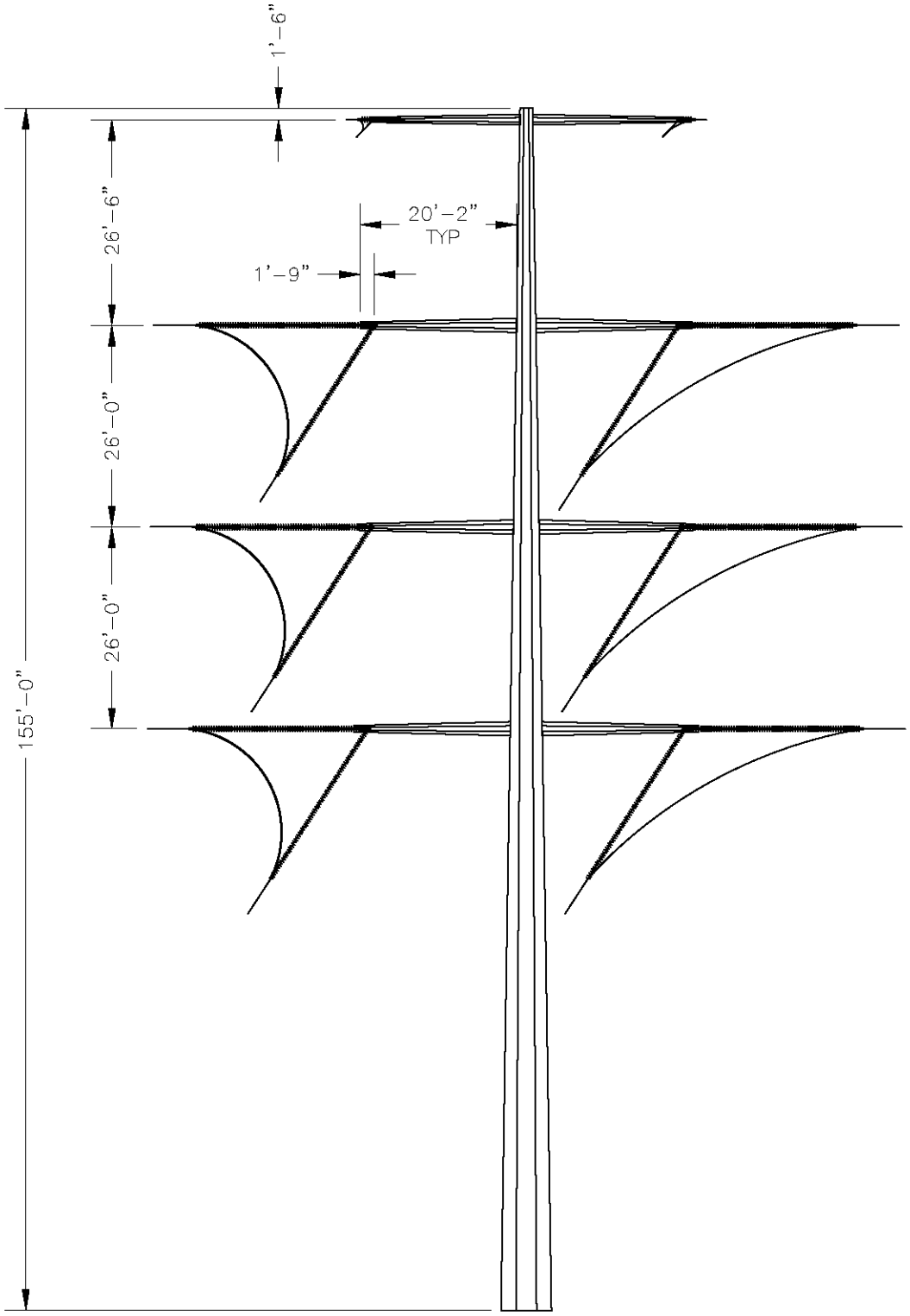
CRICKET VALLEY 345kV TRANSMISSION LINE
DUTCHESS COUNTY, NY
LIVE LINE CLEARANCE AND FALL PROTECTION
ELEVATION VIEW, SECTIONS & DETAILS
SUSPENSION POLE

APPROVED	DATE	Cricket Valley Energy, LLC Dover Plains, New York
CRICKET VALLEY ENERGY DRAWING NO.	SHEET NO.	REV.
2013–195–T	21	2

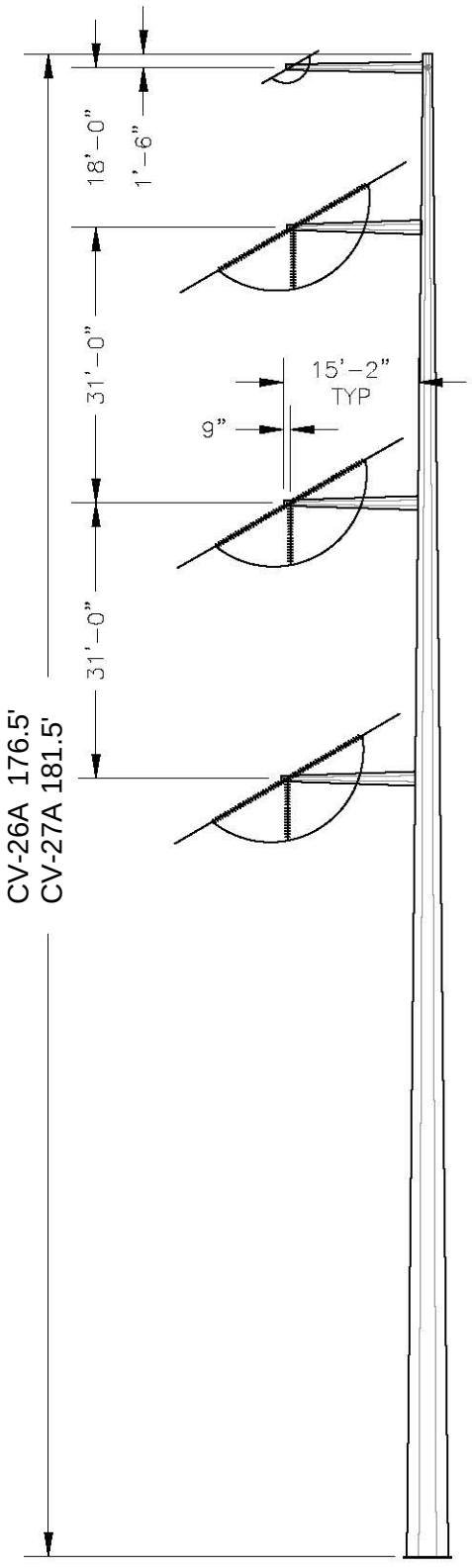
DSIZE
FRACTIONAL
DECIMAL



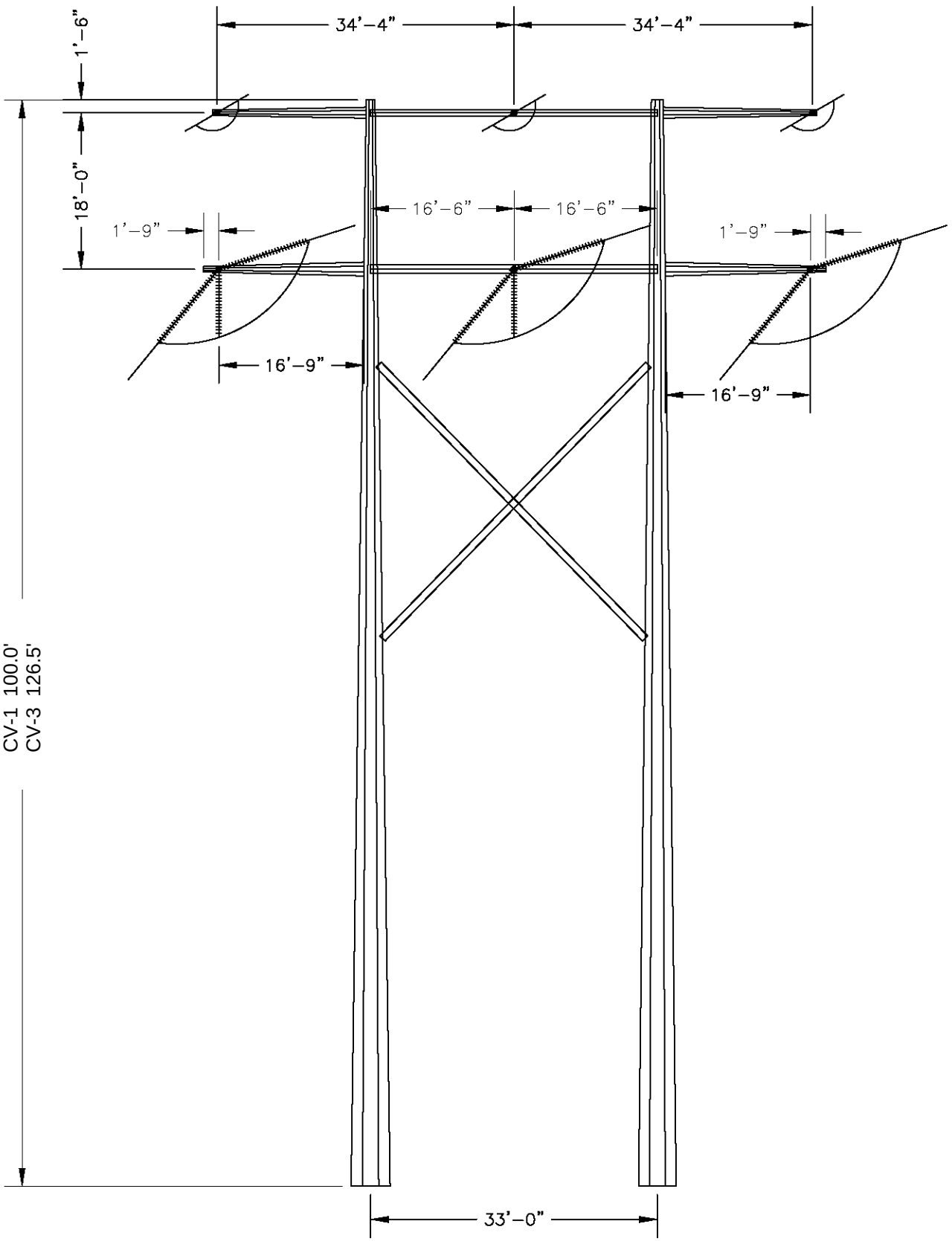
STR L 60-2



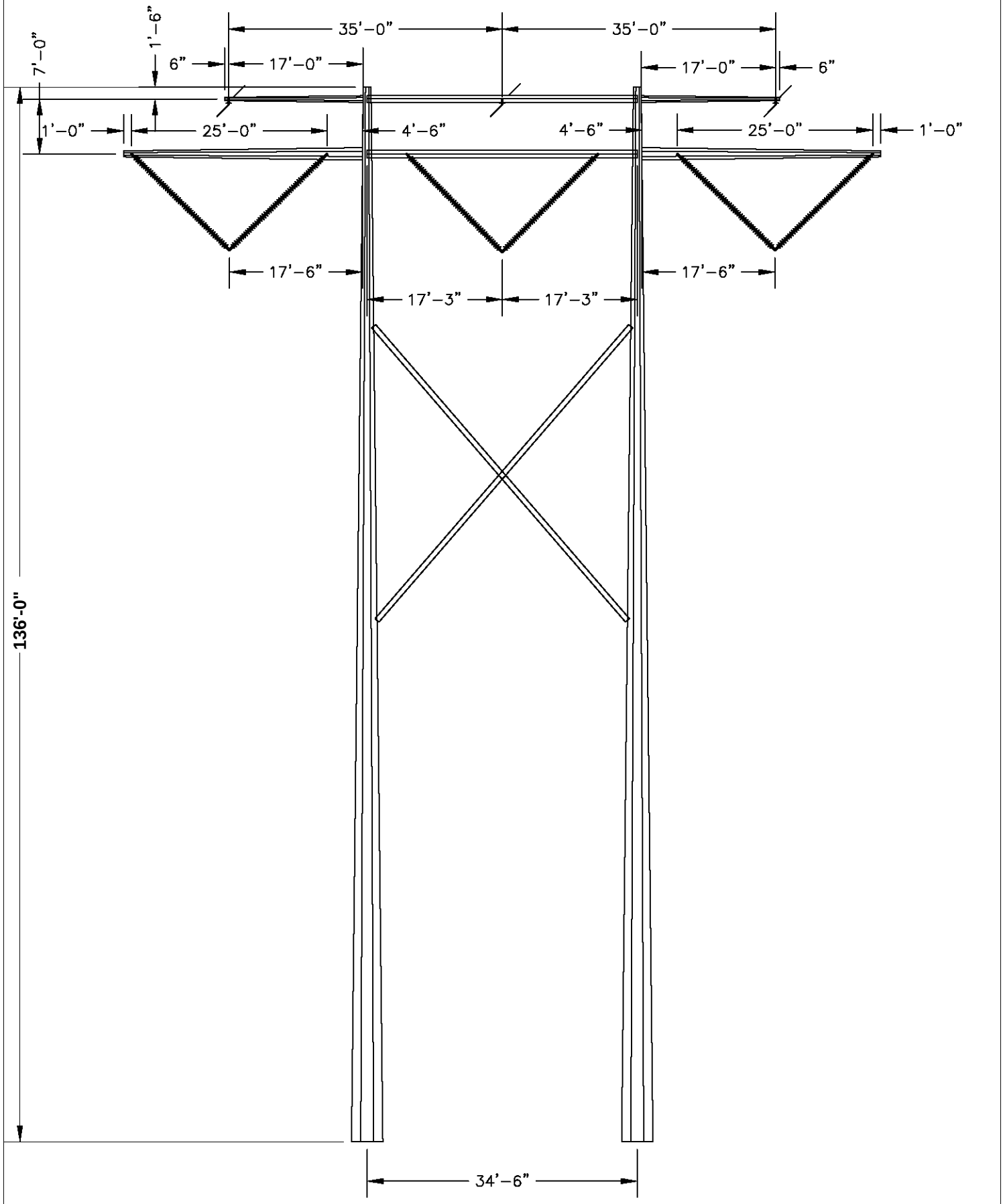
STR L-60-1 DEAD END T



STR CV-26 A & CV-27 A

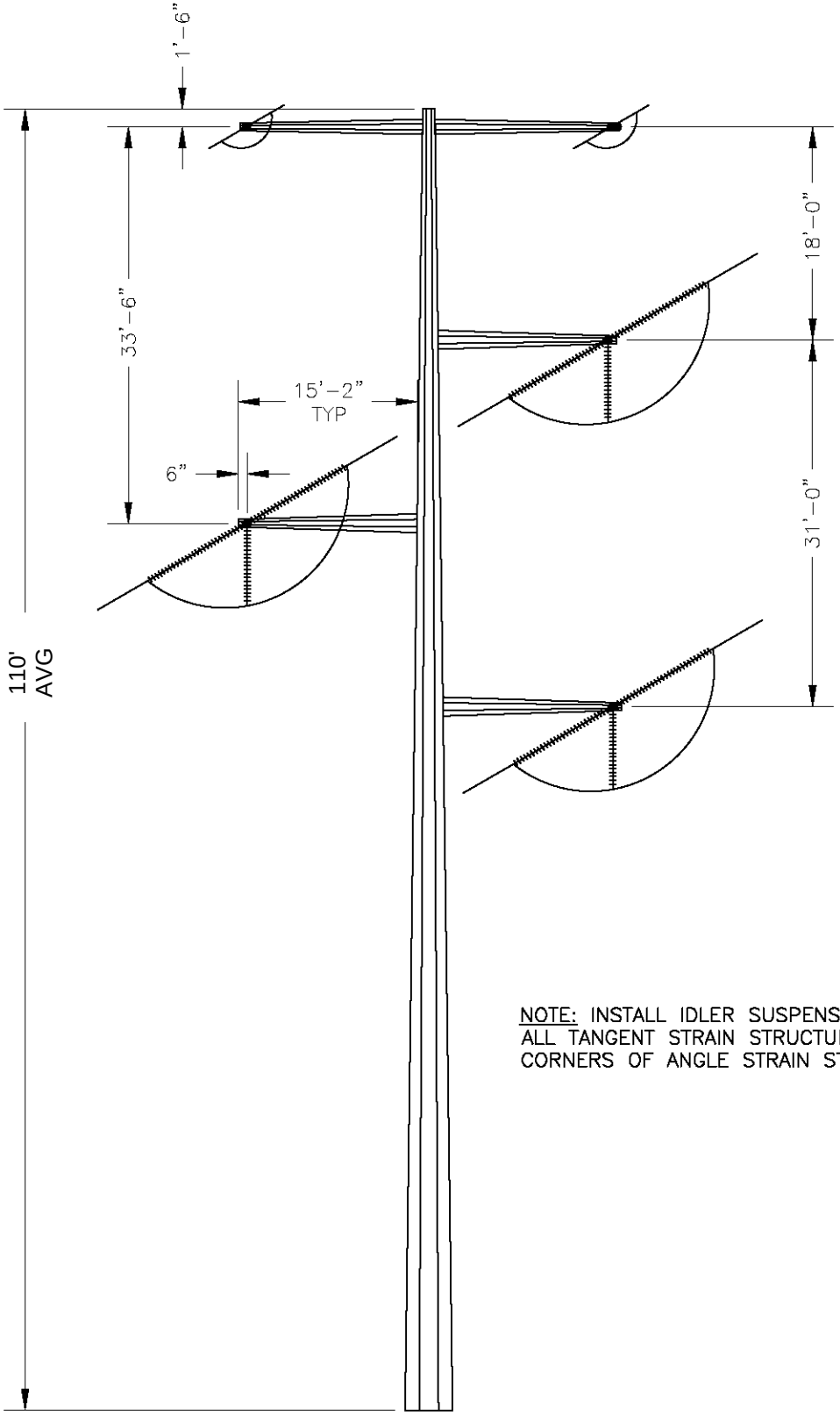


STRAIN H–FRAME
CV–1 & CV–3



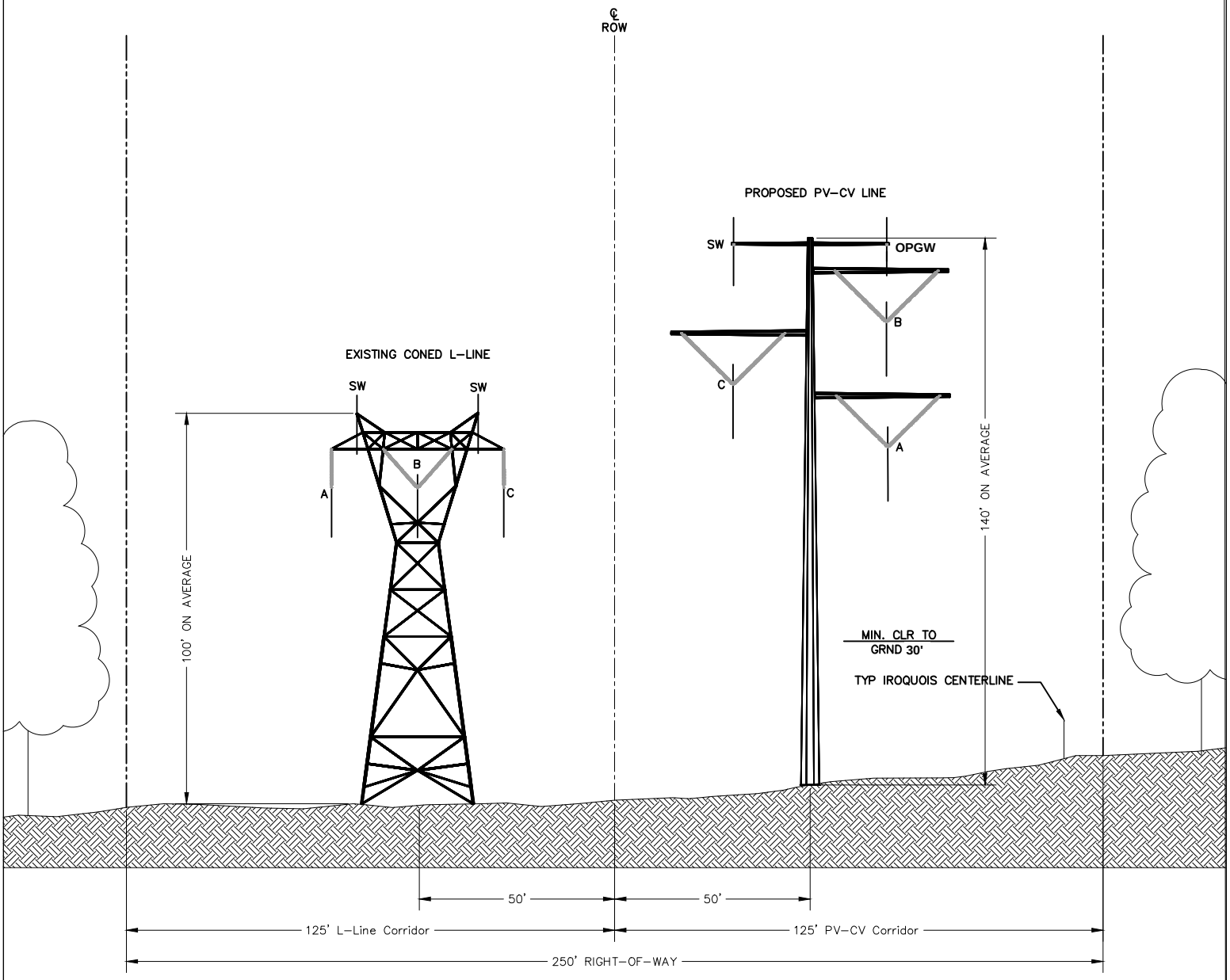
SUSPENSION H–FRAME

CV-2



NOTE: INSTALL IDLER SUSPENSION ASSEMBLY AT ALL TANGENT STRAIN STRUCTURES, AND OUTSIDE CORNERS OF ANGLE STRAIN STRUCTURES.

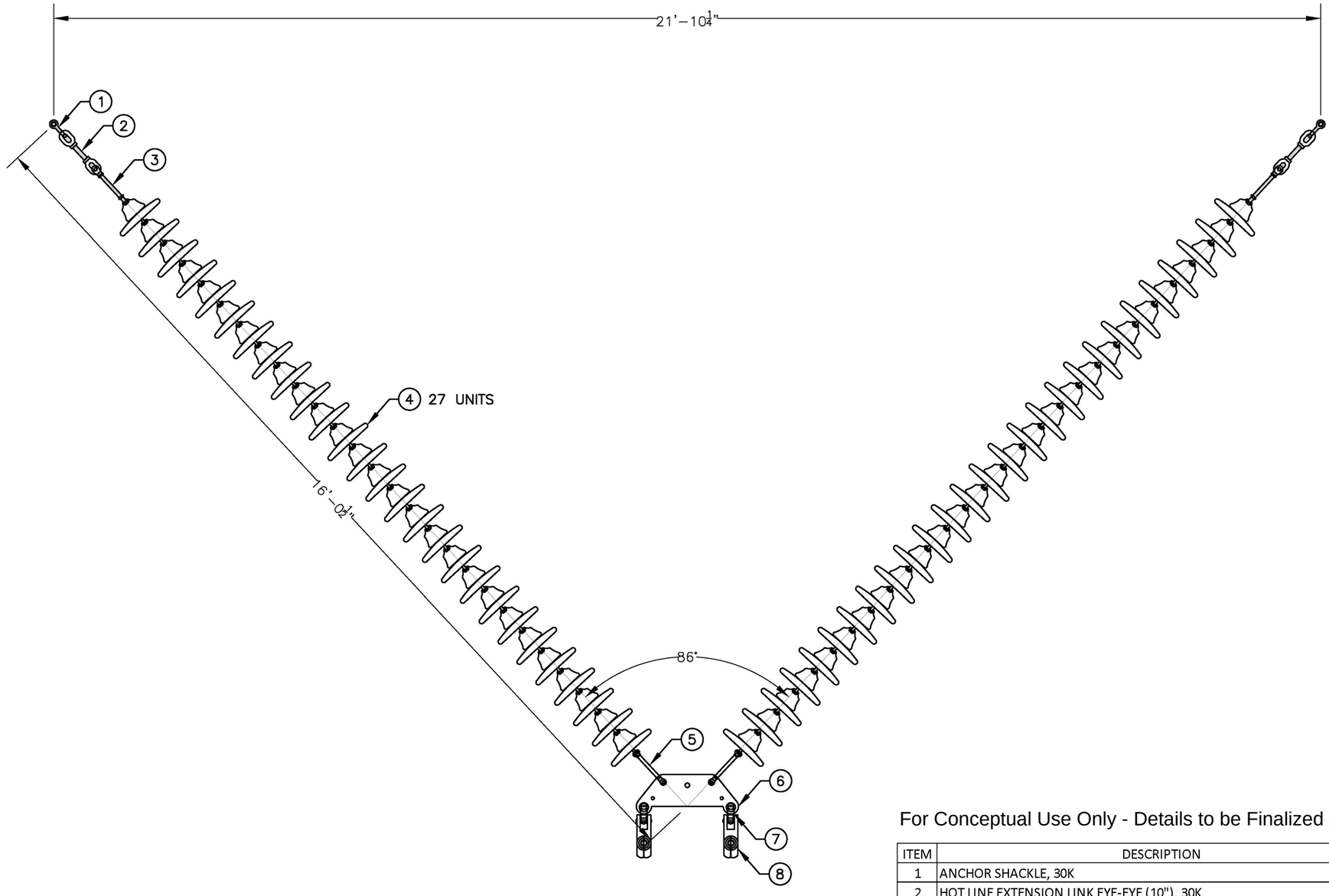
TYPICAL STRAIN POLE



TYPICAL CROSS SECTION
OF L–LINE & PV–CV LINE
FROM PLEASANT VALLEY TO CRICKET VALLEY

DiGioia Gray & Associates
Consulting Engineers and Scientists
DiGioia, Gray & Associates, LLC

CRICKET VALLEY 345kV
TRANSMISSION LINE
DUTCHESS COUNTY, NY
TYPICAL CROSS SECTION

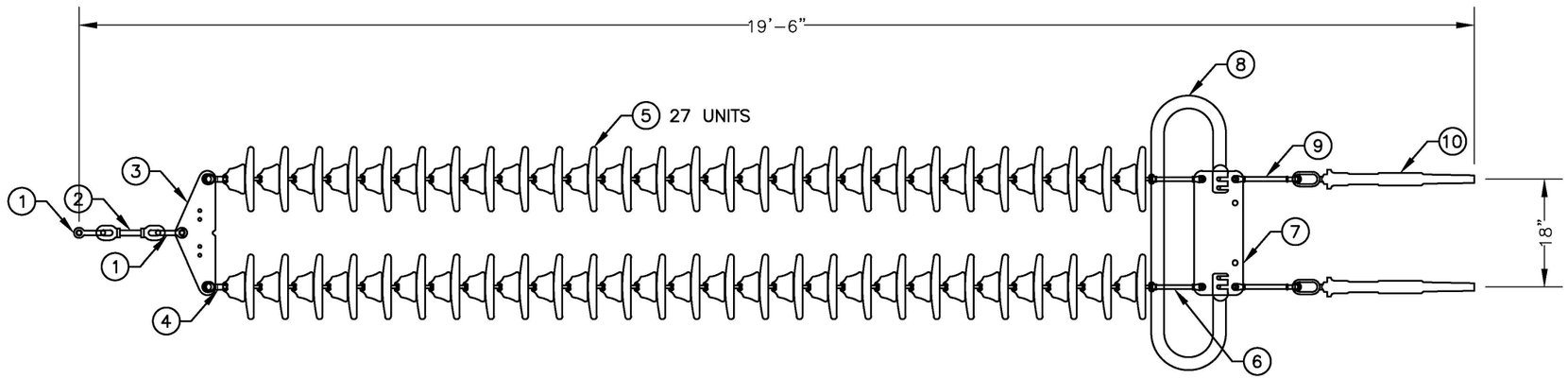


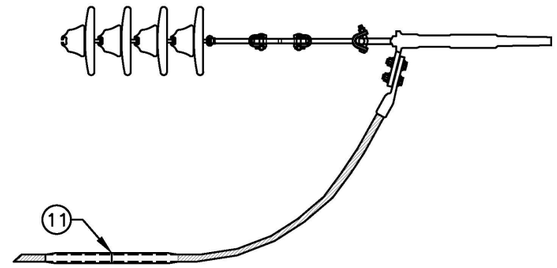
STANDARD V–STRING SUSPENSION
(0 TO 3 DEG)

For Conceptual Use Only - Details to be Finalized at a Later Time

ITEM	DESCRIPTION	MANUFACTURER	PART NUMBER	QUANTITY
1	ANCHOR SHACKLE, 30K	HUBBELL	AS25WBNK	2
2	HOT LINE EXTENSION LINK EYE-EYE (10"), 30K	LINDSEY	3484-10	2
3	HOT LINE BALL CLEVIS, 30K	HUBBELL	HBC30	2
4	PORCELAIN SUSPENSION INSULATORS, 30K ULTIMATE	LAPP	5960A-70	54
5	HOT LINE EXTENSION SOCKET CLEVIS, 30K	HUBBELL	HSC100	2
6	YOKE PLATE, 40K PER ATTACHMENT	HUBBELL	930064004	1
7	STANDARD CLEVIS EYE, 30K	AFL	SCE-55-1375	2
8	HIBUS, CONDUCTOR SUSPENSION CLAMP, 25K	AFL	HCSHT1116LL	2

30K ULTIMATE CAPACITY PER STRING
15K WORKING CAPACITY PER STRING
WEIGHT OF ASSEMBLY 862.69 LB



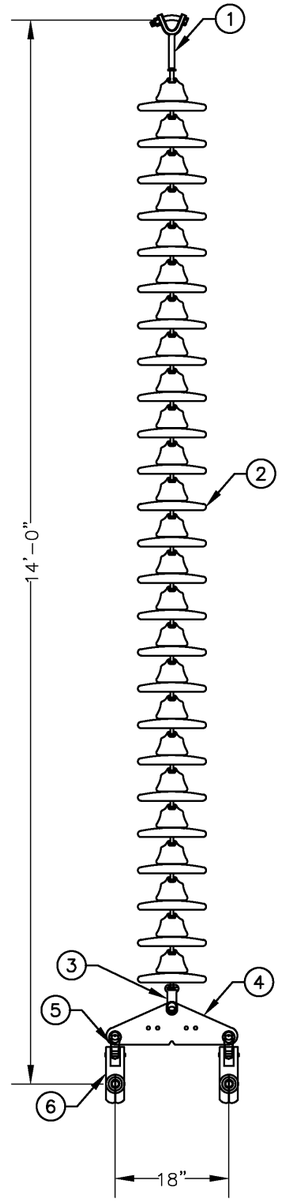


STANDARD STRAIN ASSEMBLY

For Conceptual Use Only - Details to be Finalized at a Later Time

ITEM	DESCRIPTION	MANUFACTURER	PART NUMBER	QUANTITY
1	ANCHOR SHACKLE, 60K	HUBBELL	AS25WBNK	2
2	HOT LINE EXTENSION LINK EYE-EYE (10"), 60K	LINDSEY	3484-10	1
3	YOKE PLATE, 60K PER ATTACHMENT	LINDSEY	370	1
4	HOT LINE BALL CLEVIS, 50K	HUBBELL	HBC50	2
5	PORCELAIN SUSPENSION INSULATORS, 40K ULTIMATE	LAPP	61236H-70	60
6	HOT LINE EXTENSION SOCKET CLEVIS, 50K	HUBBELL	HSC110501	2
7	YOKE PLATE, 60K PER ATTACHMENT	HUBBELL	YPR6019401	1
8	CORONA RING	LINDSEY	3855/45	1
9	HOT LINE Y-CLEVIS CLEVIS, 30K	HUBBELL	HYCC30	2
10	HI TEMP COMPRESSION DEAD END ASSEMBLY (MALLARD ACSS)	AFL	E33143HTEHV	2
11	HI TEMP JUMPER CONNECTOR (MALLARD ACSS)	AFL	5030.122HT	1

60K ULTIMATE CAPACITY PER ASSEMBLY
40K WORKING CAPACITY PER ASSEMBLY
WEIGHT OF ASSEMBLY 1104.975 LB

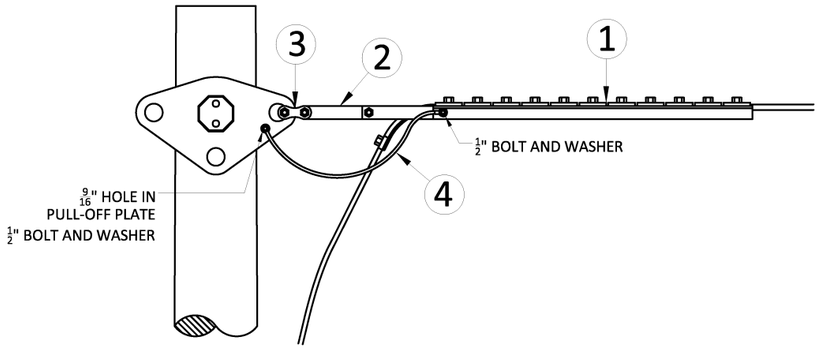


STANDARD IDLER INSULATOR ASSEMBLY

For Conceptual Use Only - Details to be Finalized at a Later Time

ITEM	DESCRIPTION	MANUFACTURER	PART NUMBER	QUANTITY
1	HOT LINE BALL Y-CLEVIS, 30K	LINDSEY	3478	1
2	PORCELAIN SUSPENSION INSULATORS, 30K ULTIMATE	LAPP	5960A-70	25
3	SOCKET CLEVIS, 30K	LINDSEY	3183	1
4	YOKE PLATE, 40K PER ATTACHMENT	LINDSEY	3570	1
5	STANDARD CLEVIS EYE, 30K	AFL	SCE-55-1375	2
6	HIBUS, CONDUCTOR SUSPENSION CLAMP, 25K	AFL	HCSHT1116	2

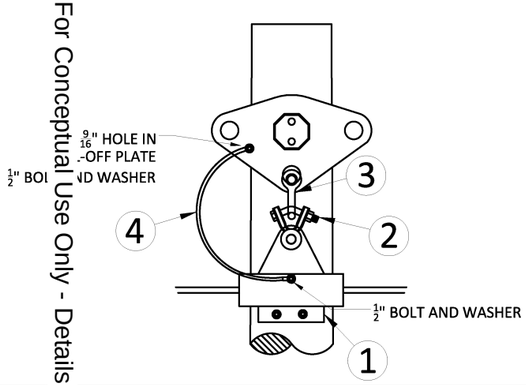
30K ULTIMATE CAPACITY PER STRING
15K WORKING CAPACITY PER STRING
WEIGHT OF ASSEMBLY 130.18



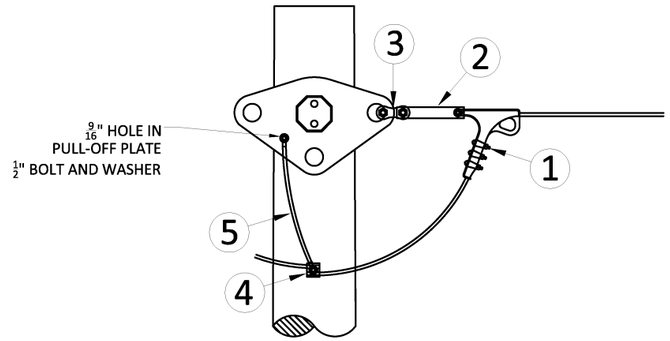
ITEM	DESCRIPTION	MANUFACTURER	NUMBER	QUANTITY
1	BOLTED DEAD END FOR OPGW, 29.6K	AFL	90/699G11	1
2	EXTENSION LINK (10")	AFL	ELP10	1
3	Y-CLEVIS CLEVIS	AFL	YCC	1
4	GROUNDING LEAD	AFL	0H/50H-36	1

OHGW (SHIELD WIRE) DEAD END ASSEMBLY

NOTE: For Conceptual Use Only - Details to be Finalized at a Later Time



ITEM	DESCRIPTION	MANUFACTUI	PART NUMBER	QUANTITY
1	IBUS, OPGW SUSPENSION CLAMP, 20K FOR OPGW	AFL	HOS682/696	1
1	IBUS, OPGW SUSPENSION CLAMP, 20K FOR OHGW	AFL	HOS532/546	1
2	-CLEVIS EYE	HUBBELL	YCS0690	1
3	NCHOR SHACKLE	HUBBELL	AS25WBNK	1
4	ROUNDING LEAD	AFL	BWAL50H/50H-36	1



ITEM	DESCRIPTION	MANUFACTURER	PART NUMBER	QUANTITY
1	BOLTED QUADRANT STRAIN CLAMP, ALUMINUM	HUBBELL	SD86	1
2	EXTENSION LINK (10")	HUBBELL	ES30781910	1
3	Y-CLEVIS CLEVIS	HUBBELL	YCC30	1
4	BONDING CLAMP	AFL	OBCF2F21M	1
5	GROUNDING LEAD	AFL	BWAL50H/50H-36	1

D

INSULATOR ASSEMBLIES

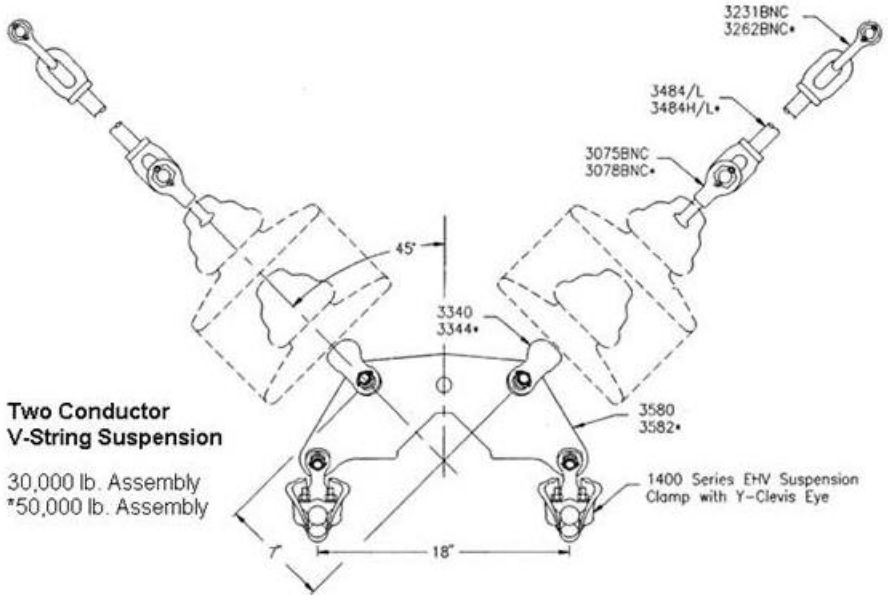
Objective

Provide replacement insulator assemblies for bundled conductors. List of Drawings – Existing Insulator Assemblies

- Existing Insulator Assemblies
    - EO-11350C – Insulator Details Tangent 0° to 3° Angle Tower
    - EO-11353C – Insulator Details 15° to 30° Angle Tower
    - EO-11355C – Insulator Details Terminal Tower

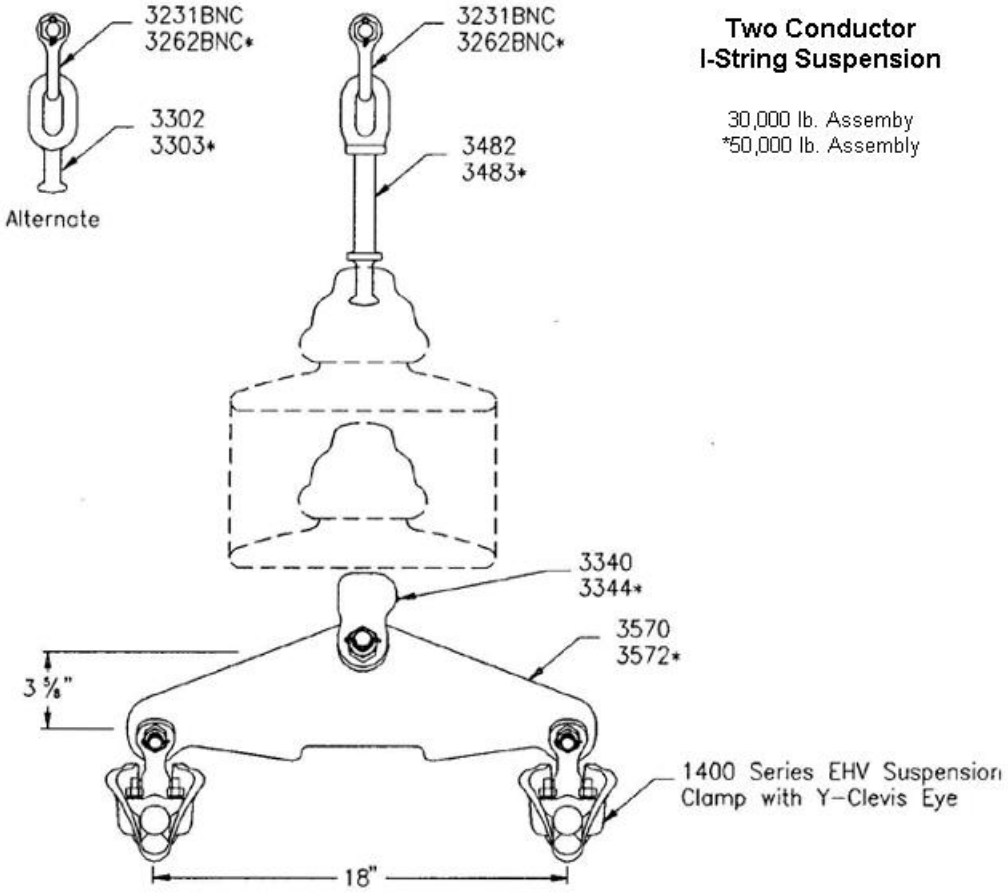
Proposed Insulator Assemblies (Reference Lindsay Manufacturing Co)

- Replacement V-String Suspension Insulator Assembly (36 – 25,000# suspension insulators) 25,000)



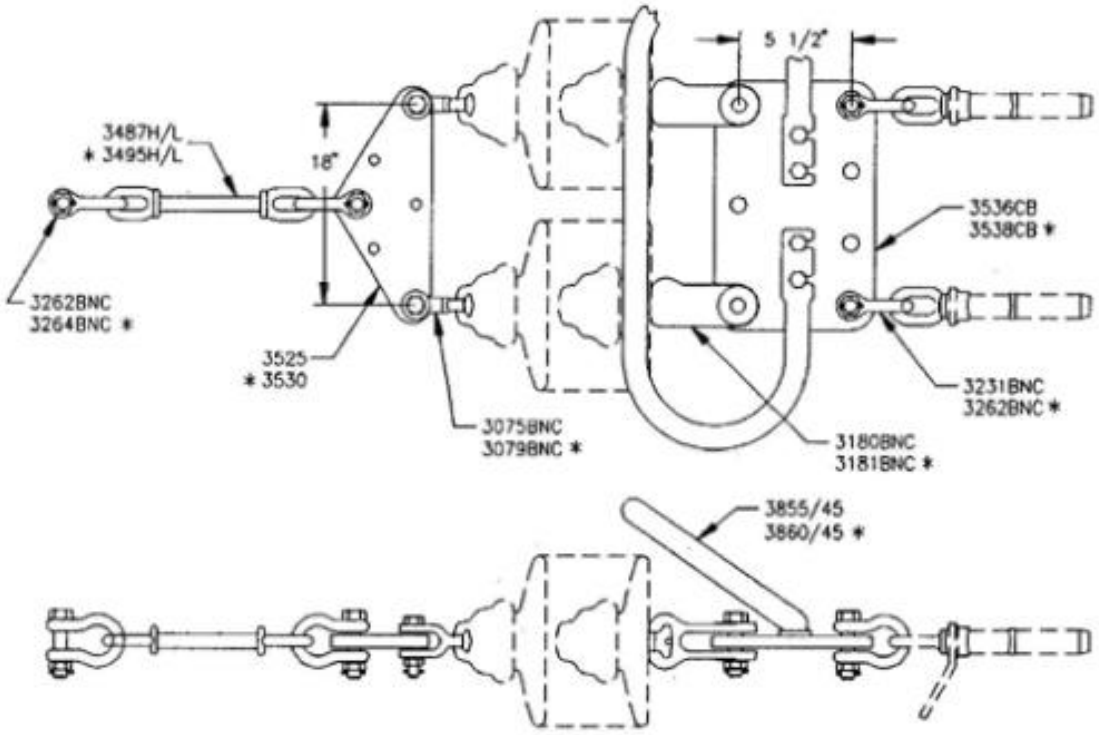
- Replacement I-String Suspension Insulator Assembly (18 – 25,000# suspension insulators)

EHV Suspension Clamps



- Replacement Dead-End Insulator Assembly (36 – 30,000# suspension insulators)

Two Conductor Tension Assembly
60,000 lb. Assembly
*100,000 lb Assembly
345 kV to 500kV Class



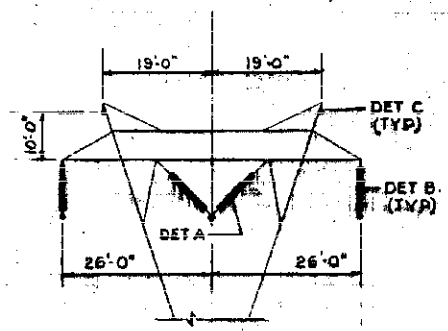
EO-11350-C

BILL OF MATERIAL

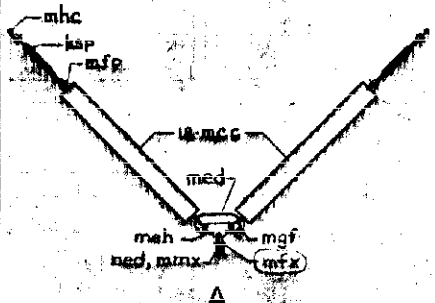
ITEM	DESCRIPTION	SPEC APPROVED EQUAL	QTY
mcc	SUSP INS, BALL & SOC 10" DISC, 5 3/4" SPC'G, ULT STR 25,000 LBS, GRAY GLAZE, ASA 52-5	LAPP 30525S1-TZ	
meh	YOKE PL, 15" SPCG, 5/8" THK, CAT NO BT-8510 HT	BT Dwg V-91948	1
ksp	TURNBUCKLE 7/8" X 18" LG, JAW-JAW WITH JAM NUTS, ULT STR 35,000 LBS, LUBEX TREATED	BT-3136	2
maz	SOC EYE, A = 2 1/8" THK, B = 1 1/2" ULT STR 35000 LBS GE CONNECTOR #93220-Z-GG	GE DWG. V-93220-Z	2
mfp	OVAL BALL EYE, ULT STR 25,000 LBS	BT-3014 HT	4
mfx	Y-CLV EYE, 90° B=2 1/2" HOLE, ULT STR 30000 LBS	BT 3031 HT SPECIAL	1
mgf	HOT LINE SOC CLV, ULT STR 35,000 LBS	BT-3087 HT	2
mhb	ANCHOR SHACKLE ULT STR 35,000 LBS	BT-3023 HT	2
mhc	ANCHOR SHACKLE ULT STR 45,000 LBS	BT-3025 HT	4
mmw	ARMOR GRIP SUSP FOR 9/16" 7 NO. 5 ALUMOWELD WITH CLV CONN, RODS 60" LG	PREF. LP *AGS 415-3076	2
mmx	ARMOR GRIP SUSP FOR 2.156 MCM ACSR 84/19 NO. CONN	PREF LP AGS-US-5531	3
ned	2,156 MCM, ACSR, 84 AL, 19 STL STRD, 1.762" OD, "BLUE BIRD" CONDUCTOR		
nfa	9/16" 7 NO. 5 ALUMOWELD, 0.546" OD, SHLD WIRE		
pua	PREFORMED ALUMOWELD SPLICE FOR 9/16" 7 NO. 5 ALUMOWELD	PREF AMS-795	AS REQ
pyb	AL COMPR SPLICE FOR 2,156 MCM, ACSR, 84/19 STRD COND	T&B 25042	
pza	STL COMPR SPLICE FOR 2,156 MCM ACSR 84/19 STRD COND	T&B 25043	
* meb	CORONA SHLD FOR SUSP ASSEMBLY WITH HDW GE DETAIL DWG. W-93821-R-1 GE ASSY. DWG. W-93821	GE DWG. V-93983	2
* mee	CORONA SHLD FOR ~~SUSPENSION~~ ASSEMBLY WITH HDW		1 SET

NOTES:
1-EXT TURNBUCKLE (ITEM KSP) TO FULL LENGTH BEFORE INSTALLING.
* 2-ON TOWERS 64, 65 & 75 CORONA SHIELDS CHANGED TO GE DWG. V-93253-Z SINGLE SUSP & GE DWG. V-94241 V-STRING SUSP.

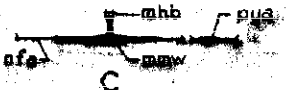
REFERENCES:
STL PWR'S T&M TRANS. ~~EO 11350-A~~



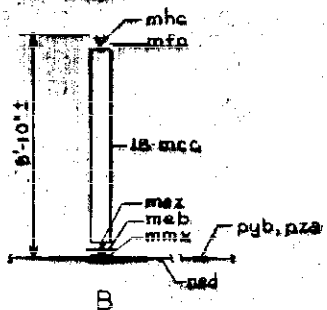
ELEVATION



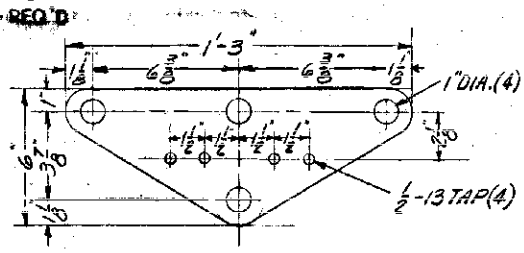
A
1-REQ'D



C
2-REQ'D



B
2-REQ'D



DETAIL OF ITEM meh
YOKE PLATE - BTC #8510HT MODIFIED AS ABOVE

APPROVED William C. Feeney
LICENSED PROFESSIONAL ENGINEER, NO. 40207
STATE OF NEW YORK

NO.	DESCRIPTION	CHKD	CORR	APPD	DATE
4	B/M REVISED		HK		
3	B/M REVISED		HK		
2	NOTE & B/M REVISED				
1	NOTE & ITEMS ADDED, B/M REV.				

STONE & WEBSTER ENGINEERING CORP.
BOSTON, MASS.

INSULATOR DET'S. TANGENT & 0° TO 3° ANGLE TOWER
345KV OVERHEAD TRANSMISSION LINE
PLEASANT VALLEY SUBSTATION TO CONNECTICUT STATE LINE

CONSOLIDATED EDISON COMPANY OF N.Y., INC.

DATE 7-18-63
LAST REV. 10-3-66
DWG. NO. EO-11350-C REV. 7

NO.	DATE	REVISIONS	APPROVED
7	10-3-66	ADD DETAIL meh, YOKE PLATE	
6	3-21-65	ADD GE DWG NOS TO CORONA SHLDS & YOKE PLATE	
5	11-65	AS CONSTRUCTED	

FOR APPROVAL

DIV.	SIGNATURE	DATE
GE	C Taylor	10/2/63

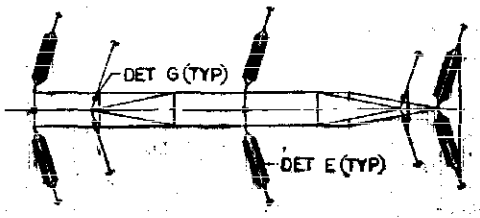
DRAWN BY | DATE | CHECKED BY | DATE
TRACED BY | DATE | SCALE
APPROVED
DESIGN ENGINEER | DATE 10/2/63
DISTR | ENG | DATE

J.O. 10963-GE-1A

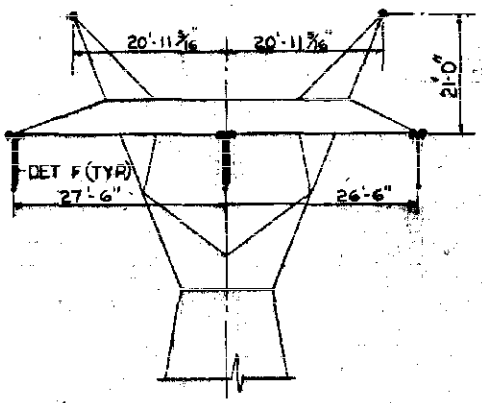
CHK'D BY: C.M.D. | CHK'D BY: | CHK'D BY: | CHK'D BY: | DRAWN BY: V.L.W.

EXTRA PRINTS
PLUMMER ① PLUMMER ② PLUMMER ③ KNOWLES ④ KNOWLES

EO-11353-C



PLAN

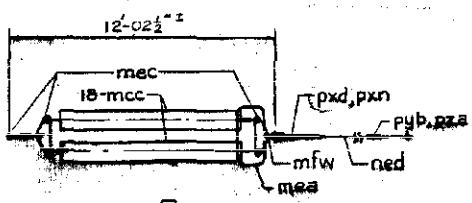


ELEVATION

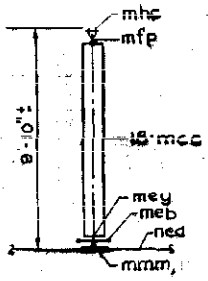
BILL OF MATERIAL

ITEM	DESCRIPTION	OR APPROVED EQUAL	QTY
knj	AL PREFORMED GUY GRIP DEAD E FOR 7 NO.5 ALUMOWELD (9/16")	PREF "ANDE-7X"	4
kns	THIMBLE CLV FOR 7 NO.5 ALUMOWELD (9/16")	MID-CONTN	4
mcc	SUSP INS, BALL & SOC, 10" DISC, 5 3/4" SPCG, ULT STR 25,000 LBS, GRAY GLAZE, ASA 52-4	LOCKE 3052-55	252
mec	STL DOUBLE STRAIN YOKE SET, BALL & SOC TYPE, ULT STR 50000 LBS, BT 303 GHT SHACKLE & BT-309 INT Y BALL CLV'S ON UPPER PL, LOWER PL TO HAVE BT-308 7HT SOC CLV'S REPLACE SHACKLE WITH CLV EYE, BT-300 4 HT	BT-6150 HP AS PER GE DWG V-93982	6
mhc	ANCHOR SHACKLE ULT STR 45,000 LBS	BT-302 5HT	6
mmm	AL SUSP CLAMP FOR 1.375"-1.82" COND RANGE	BT-8648 J	2
mea	CORONA SHLD FOR DEAD E ASSEMBLY WITH MOW GE DETAIL DWG. W-98548-R GE ASSY. DWG. V-93982		6
ned	2,156 MCM, ACSR, 84 AL, 19 STL STR 0.1766" OD "OLIVE BIRD" CONDUCTOR		AS REQ
pxd	COMPR DEAD E CONSISTING OF AL DEAD E BODY & ADJUSTABLE STL CLV END FOR 2,156 MCM, ACSR, 84/19 STRD COND, SINGLE TONGUE	T&B-28142	6
pxn	AL COMPR JUMPER T&B 94 FOR 2156 MCM ACSR 84/19 STRD COND WITH MOW	T&B-28643	6
mfp	OVAL BALL EYE, ULT STR 25,000 LBS	BT-301 4HT	2
mfw	CORONA SHLD FOR SUSP ASSY WITH MOW GE DETAIL DWG. W-93821-R GE ASSY. DWG. W-93821		2
nfa	9/16" 7 NO. 5 ALUMOWELD, O. 546" OD, SHLD WIRE	PREF LP "AWLS 7"5	AS REQ
pya	PREFORMED ALUMOWELD SPLICE FOR 9/16" 7 NO. 5 ALUMOWELD	T&B-28042	
pyb	AL COMPR SPLICE FOR 2156 MCM ACSR, 84/19 STRD COND	T&B-28043	
pza	STL do		
mey	SOC EYE, A=1 7/8" THK, B=1 1/2", ULT STR 35,000 LBS CAT. GE CONNECTOR # 93220-Z-GE	GE DWG. V-93220-Z	2

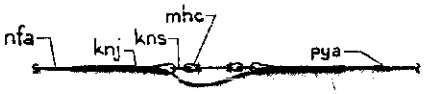
REFERENCES
STL DETS 15° TO 35° ANGLE TWR'S EO-11351-A



E
6-REQ'D



F
2-REQ'D



G
2-REQ'D

APPROVED William C. Finnell
LICENSED PROFESSIONAL ENGINEER NO. 40207
STATE OF NEW YORK

NO.	DESCRIPTION	CHKD	CORR	APPD	DATE
3	B/M REVISED				
2	B/M REVISED				
1	ITEMS ADDED; B/M REV DIM. CHANGED				
0	ORIGINAL ISSUE				

STONE & WEBSTER ENGINEERING CORP.
BOSTON, MASS.

INSULATOR DET'S 15° TO 35° ANGLE TOWER

345KV OVERHEAD TRANSMISSION LINE
PLEASANT VALLEY SUBSTATION TO CONNECTICUT STATE LINE

CONSOLIDATED EDISON COMPANY OF N.Y., INC.

DATE 7-10-68
LAST REV. 9-22-65
DWG. NO. EO-11353-C REV. 5

NO.	DATE	REVISIONS	APPROVED
5	9-22-65	ADD GE DWG NOS TO CORONA SHLDS & YOKE PLATES	
4	1-11-65	AS CONSTRUCTED	

FOR APPROVAL

DIV.	SIGNATURE	DATE
GEE		10/25/68

DRAWN BY | DATE | CHECKED BY | DATE
TRACED BY | DATE | SCALE
APPROVED
DESIGN ENGINEER | DATE
DISTR STR ENG | DATE

JO 10963-0E-10

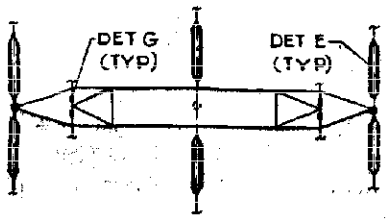
CHKD BY: C.M. COX DRAWN BY: C.W.C.

EXTRA PRINTS: ① PLUMMER ② PLUMMER ③ PLUMMER HANDLES

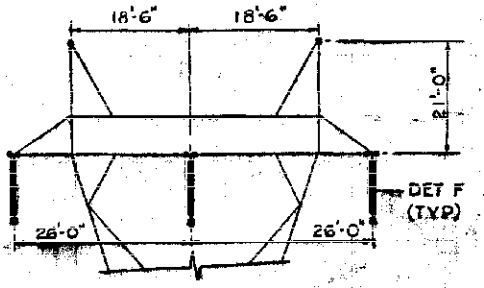
EO-11355-C

BILL OF MATERIAL

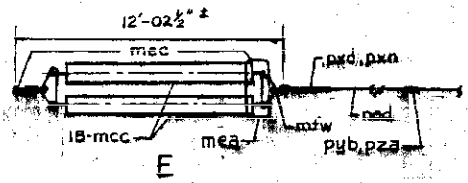
ITEM	DESCRIPTION	OR APPROVED EQUAL	QTY
knj	AL PREFORMED GUY GRIP DEAD E FOR 7 NO. 5 ALUMOWELD	PREF LP *AWDE-7x5	4
kns	THIMBLE CLV FOR 7 NO 5 ALUMOWELD	MIF-PA 272	4
mcc	SUSP INS, BALL & SOC. 10"DISC, 5¾ SPCG, ULT STR 25,000 LBS, GRAY GLAZE ASA 52-5	LAPP 30125S	220
mec	STL DOUBLE STRAIN YOKE SET, BALL & SOC TYPE, ULT STR 50,000 LBS, BT3036 HT SHACKLE & BT-3091 HT Y. BALL CLV'S ON UPPER PL, LOWER PL TO HAVE BT 3087 HT. SOC CLVS, REPLACE SHACKLE WITH CLV EYE, BT-3064 HT	BT-8750 HT REV AS PER GE DWG V-93382	6
mey	SOC EYE A-1 W SHK, ULT STR 35,000 LBS, CAT 6 = ⅝" GE CONNECTOR #93220-I-G5	GE DWG V-932202	3
mhc	ANCHOR SHACKLE, ULT STR 45,000 LBS	BE-3026 HT	7
mmm	AL SUSP CLAMP FOR 1.375"-1.82" COND RANGE	BT-9548-U	3
ned	2,156 MCM, ACSR, 84 AL, 19 STL STRD COND, 1.762" OD, "BLUEBIRD" CONDUCTOR		AS REQ
pza	STL COMPR SPLICE FOR 2,156 MCM, ACSR, 84/19 STRD COND	T&B 29143	6
pxd	COMPR DEAD E CONSISTING OF AL DEAD E BODY & ADJUSTABLE STL CLV END FOR 2,156 MCM ACSR, 84/19 STRD COND, SINGLE TONGUE	T&B 28142 & 29145	6
pxn	AL COMPR JUMPER TERM FOR 2,156 MCM, ACSR, 84/19 STRD COND WITH HDW	T&B 28143	6
mfp	CLEVIS EYE, 18T STR 25,000 LBS	BT-3014 HT	3
pya	PREFORMED ALUMOWELD SPLICE FOR 9/16" 7 NO 5 ALUMOWELD	PREF LP *AWLS-7x5	AS REQ
pyb	AL COMPR SPLICE FOR 2,156 MCM, ACSR, 84/19 STRD COND	T&B 28042	
nfa	9/16" 7 NO. 5 ALUMOWELD, 0.546" OD, SHIELD WIRE		
mea	CORONA SHLD FOR DEAD E ASSEMBLY WITH HDW GE DETAIL DWG. W-93643-R GE ASSY. DWG. V-93382		6
mcb	CORONA SHLD FOR SUSP ASSEMBLY WITH HDW GE DETAIL DWG. W-93361-R-1 GE ASSY. DWG. W-93351		3



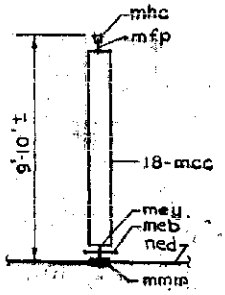
PLAN



ELEVATION

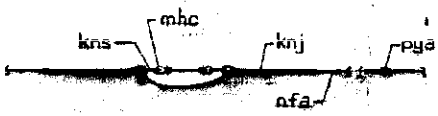


E
6-REQ'D



F
3-REQ'D

REFERENCES:
STL DET'S DEAD E TWR EO-11307-A



G
2-REQ'D

APPROVED William G. Linnard
LICENSED PROFESSIONAL ENGINEER
STATE OF NEW YORK
No. 40207

NO.	DESCRIPTION	ISSUE	CHKD	APPD	DATE
3	B/M REVISED				
2	B/M REVISED				
1	ITEMS ADDED; B/M REV				

STONE & WEBSTER ENGINEERING CORP.
BOSTON, MASS.

INSULATOR DETAILS TERMINAL TOWER

345KV OVERHEAD TRANSMISSION LINE
PLEASANT VALLEY SUBSTATION TO CONNECTICUT STATE LINE

CONSOLIDATED EDISON COMPANY OF N.Y., INC.

DATE 7-19-63
LAST REV. 9-23-65

DWG. NO. EO-11355-C REV. 5

NO.	DATE	REVISIONS	APPROVED
5	9-23-65	ADD GE DWG NOS TO CORONA SHLDS & YOKE PLATES	
4	1-11-65	AS CONSTRUCTED	

FOR APPROVAL

DIV.	SIGNATURE	DATE
CAE		10/14/63

DRAWN BY DATE CHECKED BY DATE
TRACED BY DATE SCALE
APPROVED
DESIGN ENGINEER DATE
DISTR STR ENG

JO 10963-QE-IF

CHK'D BY: C.M. Cox DRAWN BY: V.L.W.

EXTRA PRINTS: 1 PLUMMER 2 PLUMMER 3 PLUMMER KNOWLES

MICROPILE FOUNDATIONS IN TRANSMISSION

Micropiles are defined as small-diameter drilled and grouted piles, a subset of cast-in-place replacement piles. With conventional cast-in-place replacement piles, in which most of the load is resisted by concrete as opposed to steel, small cross-sectional area is synonymous with low structural capacity. This is not the case with micropiles, however. Innovative drilling and grouting methods permit high grout/ ground bond values to be generated along the micropile's periphery. To exploit this benefit, high-capacity steel elements can be used as the principal load-bearing element, with the surrounding grout serving only to transfer, by friction, the applied load between the soil and the steel. End bearing is not relied on, and in any event is relatively insignificant given the pile geometries involved. Early micropile diameters were around 100 mm, but with the development of more powerful drilling equipment, diameters of up to 300 mm are now considered practical. Micropiles are capable of sustaining surprisingly high loads (compressive loads of more than 5000 kN have been recorded) or, conversely, can resists lower loads with minimal movement. Most micropiles are 100-250mm in diameter, 5-20m long and 300- 1000 kN in compressive or tensile service load.

Classifications

Micropiles have been sub classified according to diameter, construction process, and the nature of the reinforcement. The Federal Highway Administration (FHWA} developed a classification system for micropile designs on the basis of two criteria: (1) behavior, which dictates the basis of the overall design concept; and (2) the method of grouting, which mainly determines grout/ ground capacity. The classification for behavior is based primarily on the use of the micropile in the geo-structural sense.

Case 1 refers to micropiles which are designed to transfer axial and/or lateral structural loads a competent bearing strata. These micropiles are generally used as structural support to directly resist applied loads. This design concept relies mainly on substituting conventional large-diameter pile types with closely spaced, small-diameter, high-strength piles to resist applied loads and settlement.

Case 2 micropiles rely primarily on using a three-dimensional network of reticulated micropiles to a composite, reinforced soil system. This design concept utilizes piles that are not designed individually to directly to support the applied load but rather to circumscribe and internally reinforce the in situ soil, forming a composite gravity structure to support the applied loading with minimal movement.

There are four classification for micropiles based on the type and pressure of grouting

Type A

Gravity-Grouted Micropiles: The grout is placed in the pile under gravity head.

Type B

Low-Pressure-Grouted Micropiles: Pressures are typically in the range of 0.3- 1 MPa, and the neat cement grout is injected into the drilled hole as the temporary steel drill casing or auger is withdrawn.

Type C

High-Pressure-Grouted Micropiles: The neat cement grout is placed in the hole like a Type A installation but before hardening of this primary grout, similar grout is injected once, via a preplaced sleeved grout pipe, at a pressure of at least 1 MPa.

Type D

Post-grouted Micropiles: The neat cement grout is placed in the hole as for Type A. When the primary grout has hardened, similar grout is injected via a preplaced sleeved post grout pipe. The use of a packer inside allows specific horizons to be treated several times if necessary, at pressures of 2-8 MPa.

CRUX subsurface, inc.

Helicopter Portable Installation Equipment

Monopole Foundation in Rock

Monopole Foundation on Steep Slope

Case 1 micropiles have been used extensively within the Power Transmission industry to support both tubular steel poles and lattice tower structures.

Contractors commonly install Type A piles in consolidated formations (rock) and type B piles in unconsolidated formation (soils). The proposed micropile foundations supporting transmission towers for this project will consist of a varying number of micropiles typically arranged in a circular array. The micropiles are battered away from the center of the array and will derive their capacity through interaction with the native soils/bedrock. A steel casing will be installed in the upper section of the pile to provide lateral capacity to the pile. Below the cased section, an uncased section composed of neat cement grout and high-strength reinforcing bar will derive the piles axial capacity through a bond between the grout and surrounding geotechnical material. A Foundation Schedule will be constructed to provide foundation capacities in a range of geotechnical materials and foundation reveals. Geotechnical conditions will be characterized at each foundation site and will be sued in conjunction with the Foundation Schedule to determine the optimal foundation installation. If the Foundation Schedule does not incorporate geotechnical conditions identified at a site, the foundation will be placed on hold until Owner approves revision(s) to the Foundation Schedule.

Steel Cap Micropile Foundations for Lattice Towers
US patent number 13.087294

CRUX subsurface, inc.

EROSION AND SEDIMENTATION CONTROL
ACCESS & GAS LINE PROTECTION
DRAWINGS

EROSION AND SEDIMENT CONTROL NOTES:

APPLICATION OF EROSION AND SEDIMENT POLLUTION CONTROL DEVICES SHALL BE INITIATED BY THE END OF EACH WORKING DAY AND COMPLETED WITHIN SEVEN DAYS IN ACCORDANCE WITH GP-0-15-002 OR CURRENT GP FOR CONSTRUCTION. THE CONTROL DEVICES SHALL BE INSPECTED AND MAINTAINED AS INDICATED IN THE DETAIL NOTES ON THESE DRAWINGS.

1. THE DISCHARGE OF STORMWATER RUNOFF FROM DISTURBED GROUND IN THIS PROJECT IS REGULATED UNDER THE NY SPDES PERMIT PROGRAM, GENERAL PERMIT GP-0-15-002 FOR STORMWATER DISCHARGES FROM CONSTRUCTION ACTIVITY. REFER THE SPDES PERMIT AND THE EMCP NARRATIVE FOR ADDITIONAL STORMWATER REQUIREMENTS.
2. IT IS A VIOLATION OF STATE LAW TO MANAGE STORMWATER AND EROSION AND SEDIMENT CONTROL IN A MANNER INCONSISTENT WITH THE STORMWATER POLLUTION PREVENTION PLAN, INCLUDING THESE EM&CP DRAWINGS.
3. THE CONTRACTOR SHALL MODIFY, IN CONSULTATION WITH THE ENVIRONMENTAL MONITOR, THE EROSION AND SEDIMENT CONTROL PRACTICES AS NECESSARY THROUGHOUT THE PROJECT TO ENSURE COMPLIANCE WITH THE REQUIREMENTS AND GOALS OF THE SWPPP AND THE EROSION CONTROL BLANKETS SHALL BE INSTALLED WHERE VEGETATION IS BEING ESTABLISHED IN SWALES, DITCHES AND ON SLOPES STEEPER THAN 3:1 (HORIZONTAL: VERTICAL).
4. ALL EROSION AND SEDIMENT CONTROL PRACTICES SHALL BE CHECKED FOR STABILITY AND OPERATION BY A QUALIFIED INSPECTOR ONCE EVERY SEVEN DAYS, ANY NEEDED OR REPLACEMENT REPAIRS SHALL BE MADE IMMEDIATELY TO MAINTAIN ALL PRACTICES AS DESIGNED. IF FIVE OR MORE ACRES ARE DISTURBED AT ANY TIME, THE QUALIFIED INSPECTOR SHALL PERFORM TWO INSPECTIONS (EACH SEPARATED BY TWO DAYS).
5. ALL SEEDED AREAS SHALL BE FERTILIZED, SEEDED, AND MULCHED ACCORDING TO SPECIFICATIONS IN THE VEGETATIVE PLAN TO ACHIEVE A VIGOROUS PERENNIAL VEGETATIVE COVER WITH A MINIMUM DENSITY OF 80%.
6. ACTUAL LOCATIONS OF EROSION AND SEDIMENT CONTROL FEATURES SHOWN MAY VARY BASED ON SITE CONDITIONS ENCOUNTERED AT THE TIME OF CONSTRUCTION AND AT THE DISCRETION OF THE ENVIRONMENTAL MONITOR.
7. ADDITIONAL EROSION AND SEDIMENT CONTROL FEATURES SHOWN MAY BE REQUIRED AS DETERMINED BY THE ENVIRONMENTAL MONITOR AT THE TIME OF CONSTRUCTION
8. INSTALLATION OF TEMPORARY EROSION AND SEDIMENT CONTROLS INCLUDING, BUT NOT LIMITED TO, MAINTENANCE AND PLACEMENT OF ADDITIONAL CONTROLS SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR. ALL TEMPORARY EROSION AND SEDIMENT CONTROLS SHALL BE INSTALLED AND MAINTAINED IN ACCORDANCE WITH THE PROJECT EM&CP AND THE LATEST EDITION OF THE NEW YORK STATE STANDARDS AND SPECIFICATIONS FOR EROSION AND SEDIMENT CONTROL.
9. TO THE EXTENT PRACTICAL, THE CONTRACTOR SHALL MINIMIZE THE AREA OF SOIL DISTURBANCE AT ANY GIVEN TIME.
10. THE CONTRACTOR SHALL IMPLEMENT AND MAINTAIN APPROPRIATE DUST CONTROL MEASURES AT ALL TIMES TO MINIMIZE THE GENERATION AND POTENTIAL OFF-SITE MIGRATION OF FUGITIVE DUST.
11. WHERE PRACTICABLE, TRACKED GRADING EQUIPMENT SHALL TRACK UP AND DOWN SLOPES TO MINIMIZE THE POTENTIAL FOR EROSION (I.E., TRACK GROUSERS ORIENTED PARALLEL TO SLOPE CONTOURS).

SPECIFIC STREAM RESTRICTIONS:

1. THE LOCATIONS OF KNOWN STREAMS IN THE VICINITY OF APPROVED WORK AREAS ARE DEPICTED ON THE CONSTRUCTION DRAWINGS.
2. DISTURBANCE TO THE BED AND BANKS OF STREAMS SHALL BE KEPT TO THE MINIMUM NECESSARY TO COMPLETE THE PROJECT.
3. ANY TEMPORARY GRAVEL ROAD AND ASSOCIATED CONSTRUCTION MATERIAL SHALL BE REMOVED FOLLOWING CONSTRUCTION AND IN ALL CASES THE PREDISTURBANCE FLOW REGIME SHALL BE RESTORED AND/OR MAINTAINED. THE CONTRACTOR SHALL EXERCISE ALL NECESSARY AND REASONABLE PRECAUTIONS TO MINIMIZE STREAM SEDIMENTATION AND SOIL EROSION IN WORK AREAS. FURTHER THE CONTRACTOR SHALL TAKE PROMPT AND EFFECTIVE ACTION TO CONTROL EXCESSIVE SEDIMENTATION TRANSPORT AND EROSION IN THE EVENT IT DOES OCCUR.
4. TEMPORARY EROSION CONTROL, PERMANENT STABILIZATION AND OTHER STREAM PROTECTION MEASURES SHALL BE IMPLEMENTED AS DEPICTED IN THE EM&CP AND AS DIRECTED BY THE ENVIRONMENTAL MONITOR.
5. DURING PERIODS OF WORK ACTIVITY, THE NATURAL FLOW OF WATER SHALL BE MAINTAINED AT ALL TIMES TO AVOID IMPACTS TO DOWNSTREAM HABITAT.
6. SOILS AND/OR EXCAVATED MATERIALS SHALL BE STORED SUFFICIENTLY DISTANT FROM STREAMS TO PRECLUDE RE-ENTRY OF SILT LADEN WATER INTO THE STREAM OR CAUSE THE BANK TO COLLAPSE.

DUST CONTROL

APPLICATION

1. FOR CONTROL OF DUST FROM LAND-DISTURBING ACTIVITIES OR ON CINSTRUCTION ROADS AS THE RESULT OF VEHICLE TRAFFIC.

METHODS

1. SPRINKLING – SPRAY SURFACE WITH WATER UNTIL SURFACE IS WET. APPLY AS NEEDED TO PROVIDE SHIRT TERM LIMITED DUST CONTROL.
2. TEMPORARY SEEDING – ON GRADED OR SMOOTHED AREAS INCLUDING SLOPES.
3. EROSION CONTROL FABRIC ON GRADED SLOPES.

SPECIFIC WETLAND RESTRICTIONS:

1. THE LOCATIONS OF KNOWN REGULATED WETLANDS IN THE APPROVED WORK AREAS ARE DEPICTED ON THE CONSTRUCTION DRAWINGS.
2. THE BOUNDARIES OF STATE REGULATED WETLANDS AS DEPICTED ON THE CONSTRUCTION DRAWINGS SHALL BE MARKED IN THE FIELD BY EITHER LATH MARKERS, SURVEYORS RIBBON, PIN FLAGS, OR SUITABLE EQUIVALENT PRIOR TO CONSTRUCTION.
3. ANY TEMPORARY GRAVEL ROAD AND ASSOCIATED CONSTRUCTION MATERIAL SHALL BE REMOVED FOLLOWING CONSTRUCTION AND IN ALL CASES THE PRE-DISTURBANCE FLOW REGIME SHALL BE RESTORED AND/OR MAINTAINED.
4. THE CONTRACTOR SHALL EXERCISE ALL NECESSARY AND REASONABLE PRECAUTIONS TO PREVENT STREAM SEDIMENTATION AND SOIL EROSION IN WORK AREAS. FURTHER THE CONTRACTOR SHALL TAKE PROMPT AND EFFECTIVE ACTION TO CONTROL EXCESSIVE SEDIMENTATION TRANSPORT AND EROSION IN THE EVENT IT DOES OCCUR. NOTE: ANY STREAM SEDIMENTATION IS A VIOLATION OF THE SPDES GENERAL PERMIT GP-0-15-002.
5. PROJECT RELATED TRAFFIC SHALL BE LIMITED TO THE ABSOLUTE MINIMUM NECESSARY FOR CONSTRUCTION AND SHALL BE CONFINED TO THE TEMPORARY WORK AREA AND ACCESS ROADS (EITHER PROPOSED OR EXISTING)
6. ALL WETLANDS SHALL BE RETURNED TO THE PRE-CONSTRUCTION CONTOURS FOLLOWING CONSTRUCTION ACTIVITIES.
7. IN ALL CASES, THE PRE-DISTURBANCE FLOW REGIME SHALL BE MAINTAINED FOR WETLANDS AND WATERBODIES.
8. NO SPOIL MATERIAL SHALL BE PLACED IN A WETLAND AND ALL EXCESS MATERIAL SHALL BE DISPOSED OF AT AN UPLAND SITE AS APPROVED BY THE ENVIRONMENTAL MONITOR (NO BULL-DOZING BACK-BLADING OR OTHERWISE SPREADING OF EXCESS SPOIL OVER THE WETLANDS SURFACE).
9. NO STUMP GRINDING IS ALLOWED WITHIN ANY REGULATED WETLAND.
10. WETLAND CROSSING EROSION CONTROL AND OTHER WETLAND PROTECTION MEASURES SHALL BE IMPLEMENTED AS DEPICTED ON THE CONSTRUCTION DRAWINGS AND AS DIRECTED BY THE ENVIRONMENTAL MONITOR.
11. THE CONTRACTOR SHALL INSTALL AND MAINTAIN FENCING AND SEDIMENT BARRIERS WHEREVER EXCAVATION OR FILLING ACTIVITIES OCCUR ADJACENT TO OR WITHIN WETLAND AREAS AT THE DIRECTION OF THE ENVIRONMENTAL MONITOR.

VEGETATIVE RESTORATION PLAN:

1. THE CONTRACTOR SHALL INITIATE STABILIZATION MEASURES IN PORTIONS OF THE SITE WHERE CONSTRUCTION ACTIVITIES HAVE TEMPORARILY OR PERMANENTLY CEASED WITHIN 7 DAYS.
2. WATERING MAY BE ESSENTIAL TO ESTABLISH A NEW SEEDING WHEN A DROUGHT CONDITION OCCURS SHORTLY AFTER NEW SEEDING EMERGES.
3. ALL SEEDED AREAS SHALL BE FERTILIZED, RESEEDED AS NECESSARY, AND MULCHED ACCORDING TO SPECIFICATIONS IN THE VEGETATIVE PLAN TO ACHIEVE A VIGOROUS PERENNIAL VEGETATIVE COVER WITH A MINIMUM DENSITY OF 80%.

RIGHT-OF-WAY PERMANENT SEED MIXS	
NATIVE RIGHT-OF-WAY WOODS – SEED MIX WITH ANNUAL RYEGRASS – ERNMX-132-1	
USE ALONG ROADSIDE AREAS OR OTHER RIGHTS-OF-WAY	
TYPE	RATE (LBS/ACRE)
ERNMX -132-1 SEED MIX	30
FERTILIZER (5-10-10)	850
LIME	PERFORM SOIL TEST TO DETERMINE LIME RATES
MULCH, HAY, OR STRAW	2 TONS/ACRE

CON EDISON GENERAL MIX – CONE00401	
USE IN WETLAND OR WET SHADED AREAS	
TYPE	RATE (LBS/ACRE)
CONE00401 SEED MIX	15
FERTILIZER (5-10-10)	600
LIME	PERFORM SOIL TEST TO DETERMINE LIME RATES
MULCH	2 TONS/ACRE

WETLAND MEADOW MIX – ERNMX-122	
USE IN WET MEADOWN AND RIPARIAN SITES	
TYPE	RATE (LBS/ACRE)
ERNMX – 122 SEED MIX	15
FERTILIZER (5-10-10)	600
LIME	PERFORM SOIL TEST TO DETERMINE LIME RATES
MULCH	2 TONS/ACRE

TEMPORARY SEED MIX	
TYPE	RATE (LBS/ACRE)
ANNUAL OR PERENNIAL RYEGRASS (FOR SPRING/ SUMMER/ EARLY FALL PLANTING)	30
"AROOSTOOK" WINTER RYE (FOR LATE FALL/ EARLY WINTER PLANTING)	100
MULCH, HAY, OR STRAW	2 TONS/ACRE

OPTIMAL SEEDING TIME	
SPRING	SEED UNTIL MAY 15TH
FALL	SEED AFTER AUGUST 15TH

NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
1	1-23-17	REMOVED ROAD SECTION DETAIL	MJG	PGC	PGC	PGC

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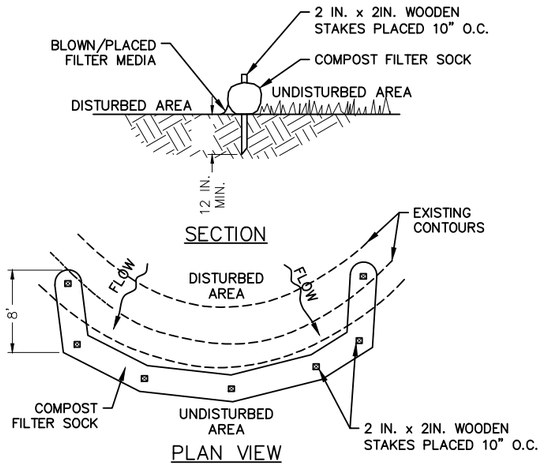
CRICKET VALLEY ENERGY TRANSMISSION

EROSION AND SEDIMENT CONTROLS

PROJ. NO.: 2015-185
DRG. NO.: 2015-185-D-01
SCALE:
DRN. BY: MPS
CHK'D. BY: PGC
APP. BY: PGC
DATE: 5/21/2016

STATE OF NEW YORK * PAUL G. CRAIG * 086926 * 1/23/17 * LICENSED PROFESSIONAL ENGINEER

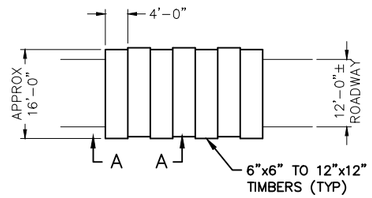


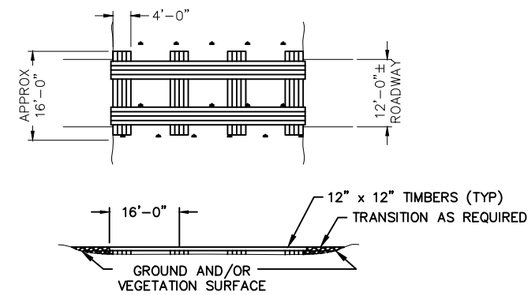
DIA. (IN.)	SLOPE%						
	2	5	10	20	25	33	50
8	225	200	100	5	2	–	–
12	250	225	125	65	50	40	25
18	275	250	150	70	55	45	30
24	350	275	200	130	100	60	35
32	450	325	275	150	120	75	50

MAXIMUM SLOPE FOR LENGTH FOR COMPOST FILTER SOCKS

NOTES:
1. THE COMPOST INFILL SHALL MEET NYS CRITERIA IN TABLE 5.2 OF NYS EROSION AND SEDIMENT CONTROL STANDARDS. IT SHALL BE WELL DECOMPOSED, WEED FREE ORGANIC MATTER
2. COMPOST SOCK FABRIC MATERIAL SHALL MEET NYD CRITERIA IN TABLE 5.1 OF NYS EROSION AND SEDIMENT CONTROL STANDARDS FOR PHOTODEGRADABLE 5 MIL HDPE WITH A 38" MESH OPENINGS.
3. COMPOST FILTER SOCKS MUST HAVE CONTINUOUS GROUND CONTACT.

COMPOST FILTER SOCK





NOTES:
1. TO BE INSTALLED IF NECESSARY TO PREVENT RUTTING, TO ACCESS STRUCTURES.
2. THIS DETAIL SHOWS TYPICAL DIMENSIONS. SOME CONTRACTORS SWAMP MATS ARE DIMENSIONALLY DIFFERENT FROM WHAT IS SHOWN HERE.
3. DEPENDENT ON SITE CONDITIONS, MULTIPLE LAYERS OF SWAMP MATS MAY BE INSTALLED..

PRE–FABRICATED MATS ARE PADS THAT CAN SUPPORT THE WEIGHT OF EQUIPMENT AND PREVENT BOTH RUTTING AND DISTURBANCE OF SOILS. THESE MATS CAN BE USED FOR CROSSING WETLANDS OR WATERCOURSES, OR FOR PROVIDING A STABLE WORK PLATFORM TO CONDUCT REPAIRS AND MAINTENANCE ACTIVITIES. MATS ARE CONSTRUCTED OF SEVERAL DIFFERENT TYPES OF MATERIALS INCLUDING WOOD AND PLASTIC:

GUIDELINES:
- PLACEMENT OF MATS SHOULD BE IN A LOCATION THAT WOULD MINIMIZE THE LENGTH NEEDED FOR CROSSING.
- MATS MAY BE USED TO CROSS SMALL STREAMS PROVIDED THE MAT DOES NOT DISTURB THE BED OR BANKING.
- WHERE POSSIBLE, MATS SHOULD BE PLACED SO THAT THEY ARE NOT OVERTOPPED WITH WATER AND EQUIPMENT CAN WORK IN THE DRY.
- MATS SHALL BE REMOVED FROM WETLAND AREAS ONCE THE CROSSING OR MAINTENANCE ACTIVITY IS COMPLETED.

LIMITATION:
- MATS SHALL NOT BE PLACED SO THAT THEY RESTRICT THE NATURAL FLOW OF WATER.
- MATS THAT HAVE BEEN PLACED IN AREAS OF INVASIVE SPECIES SHALL NOT BE USED AGAIN UNTIL THEY HAVE BEEN CLEANED AND INSPECTED.

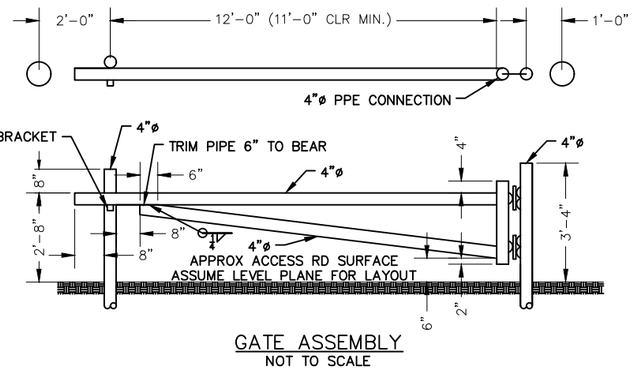
MATERIALS:
- 12 X 12 TIMBER
- EQUIVALENT CAPACITY COMMERCIAL MATS

INSTALLATION:
- INSTALL AND REMOVE IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS, IF APPLICABLE.
- EVERY EFFORT SHOULD BE MADE TO MINIMIZE IMPACT TO WETLAND AREAS DURING INSTALLATION, USE AND REMOVAL.

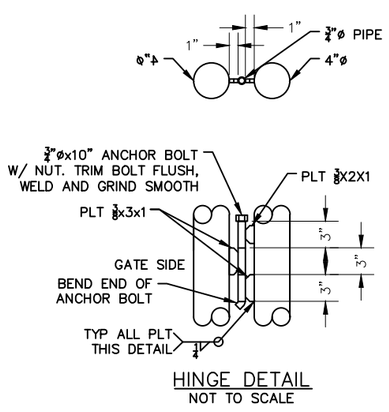
INSPECTIONS:
- LOOK FOR ANY DEFECTS OR STRUCTURAL PROBLEMS
- INSPECT FOR ANY PLANT MATERIAL LEFT ON MATS, REMOVE IMMEDIATELY.

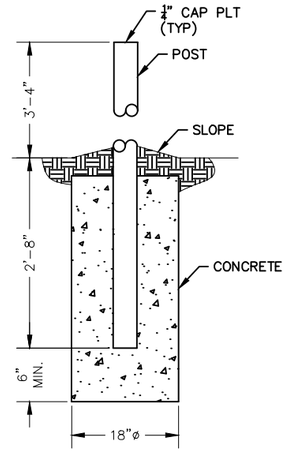
REMOVAL:
- MATS SHOULD BE CLEANED BEFORE BROUGHT TO ANOTHER WETLAND LOCATION TO REMOVE SOIL AND ANY INVASIVE PLANT SPECIES, SEED STOCK OR PLANT MATERIALS.

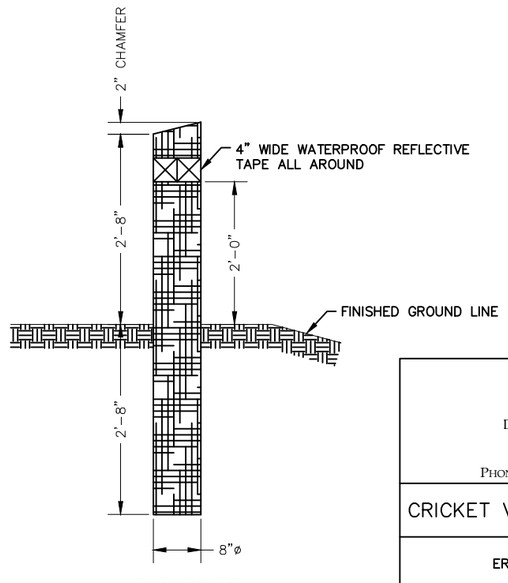
WETLAND MATTING ⑧



NOTE:
STEEL AND GATE MATERIAL TO BE GALVANIZED







ACCESS ROAD GATES ⑩

NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
1	1-23-17	ADDED NOTE 3 TO COMPOST FILTER SOCK	MJG	PGC	PGC	PGC

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CRICKET VALLEY ENERGY TRANSMISSION

EROSION AND SEDIMENT CONTROLS

PROJ. NO.	2015–185
DWG. NO.:	2015–185–D–02
SCALE:	
DRN. BY: MPS	CHK'D. BY: PGC
APP. BY: PGC	DATE: 5/21/2016

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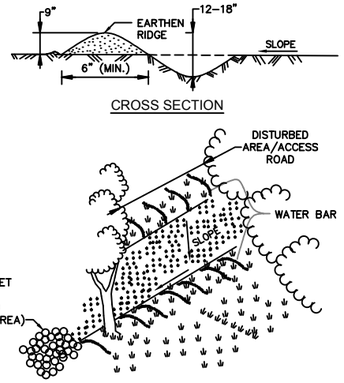
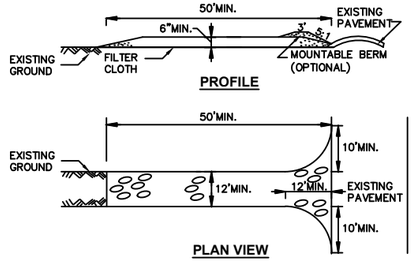


NOTES:

1. ALTERNATE DIRECTION OF WATERBAR DISCHARGE TO THE EXTENT PRACTICABLE.
2. DISK OR STRIP THE SOD FROM THE BASE FOR THE CONSTRUCTED RIDGE BEFORE PLACING FILL.
3. TRACK THE RIDGE TO COMPACT IT TO THE DESIGN CROSS SECTION.
4. THE OUTLET SHALL BE LOCATED ON AN UNDISTURBED AREA. FIELD SPACING WILL BE ADJUSTED TO USE THE MOST STABLE OUTLET AREAS. OUTLET PROTECTION WILL BE PROVIDED WHEN NATURAL AREAS ARE NOT ADEQUATE.
5. VEHICLE CROSSING MAY BE STABILIZED WITH GRAVEL. EXPOSED AREAS SHALL BE IMMEDIATELY SEEDED AND MULCHED.
6. PERIODICALLY INSPECT WATER BARS FOR EROSION DAMAGE AND SEDIMENT. CHECK OUTLET AREAS AND MAKE REPAIRS AS NEEDED TO RESTORE OPERATION.
7. ADJUST BERM DIMENSIONS TO FACILITATE VEHICLE TRAFFIC.
8. TEMPORARY WATERBARS WILL BE REMOVED FROM AGRICULTURAL AND RESIDENTIAL AREAS. REMAINING TEMPORARY WATERBARS, AT THE DISCRETION OF THE ENVIRONMENTAL MONITOR, SHALL BE VEGETATED TO CONVERT THEM TO PERMANENT WATERBARS.
9. WATERBARS/DRIVEABLE BERMS MAY NOT BE REQUIRED IF THERE IS AN EXISTING GRAVEL ROAD AND IT REMAINS STABLE WITH CONSTRUCTION TRAFFIC.

WATERBARS/DRIVEABLE BERM (1)

NOT TO SCALE

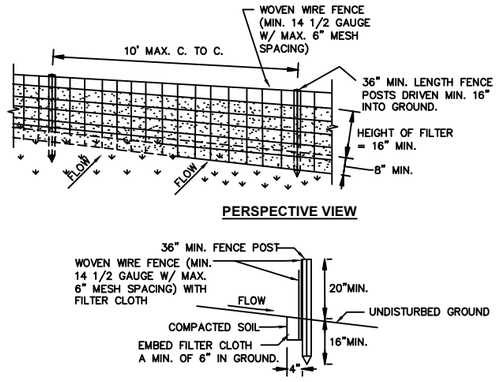


NOTES:

1. STONE SIZE – USE 2" STONE, OR RECLAIMED OR RECYCLED CONCRETE EQUIVALENT.
2. LENGTH – NOT LESS THAN 50 FEET (EXCEPT ON A SINGLE RESIDENCE LOT WHERE A 30 FOOT MINIMUM LENGTH WOULD APPLY).
3. THICKNESS – NOT LESS THAN SIX (6) INCHES.
4. WIDTH – TWELVE (12) FOOT MINIMUM, BUT NOT LESS THAN THE FULL WIDTH AT POINTS WHERE INGRESS OR EGRESS OCCURS. TWENTY-FOUR (24) FOOT IF SINGLE ENTRANCE TO SITE.
5. FILTER CLOTH – WILL BE PLACED OVER THE ENTIRE AREA PRIOR TO PLACING OF STONE.
6. SURFACE WATER – ALL SURFACE WATER FLOWING OR DIVERTED TOWARD CONSTRUCTION ENTRANCES SHALL BE PIPED ACROSS THE ENTRANCE. IF PIPING IS IMPRACTICAL, A MOUNTABLE BERM WITH 5:1 SLOPES WILL BE PERMITTED.
7. MAINTENANCE – THE ENTRANCE SHALL BE MAINTAINED IN A CONDITION WHICH WILL PREVENT TRACKING OR FLOWING OF SEDIMENT ONTO PUBLIC RIGHTS-OF-WAY, ALL SEDIMENT SPILLED, DROPPED, WASHED OR TRACTED ONTO PUBLIC RIGHTS-OF-WAY MUST BE REMOVED IMMEDIATELY.
8. WHEN WASHING IS REQUIRED, IT SHALL BE DONE ON A AREA STABILIZED WITH STONE AND WHICH DRAINS INTO AN APPROVED SEDIMENT TRAPPING DEVICE.
9. PERIODIC INSPECTION AND NEEDED MAINTENANCE SHALL BE PROVIDED AFTER EACH RAIN.

STABILIZED CONSTRUCTION ENTRANCE (2)

NOT TO SCALE

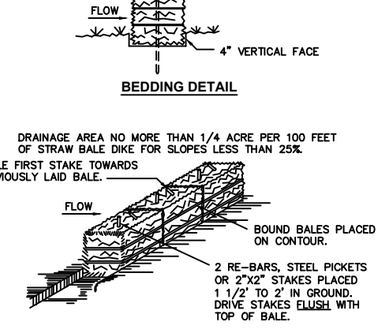


NOTES:

1. WOVEN WIRE FENCE TO BE FASTENED SECURELY TO FENCE POSTS WITH WIRE TIES OR STAPLES. POSTS SHALL BE STEEL EITHER "T" OR "U" TYPE OR HARDWOOD.
2. FILTER CLOTH TO BE TO BE FASTENED SECURELY TO WOVEN WIRE FENCE WITH TIES SPACED EVERY 24" AT TOP AND MID SECTION. FENCE SHALL BE WOVEN WIRE, 12 1/2 GAUGE, 6" MAXIMUM MESH OPENING.
3. WHEN TWO SECTIONS OF FILTER CLOTH ADJOIN EACH OTHER THEY SHALL BE OVERLAPPED BY SIX INCHES AND FOLDED. FILTER CLOTH SHALL BE EITHER FILTER X, MIRAFI 100X, STABILINKA T140N, OR APPROVED EQUIVALENT.
4. PREFABRICATED UNITS SHALL BE GEOFAB, ENVIROFENCE, OR APPROVED EQUIVALENT.
5. MAINTENANCE SHALL BE PERFORMED AS NEEDED AND MATERIAL REMOVED WHEN "BULGES" DEVELOP IN THE SILT FENCE.

SILT FENCE (3)

NOT TO SCALE

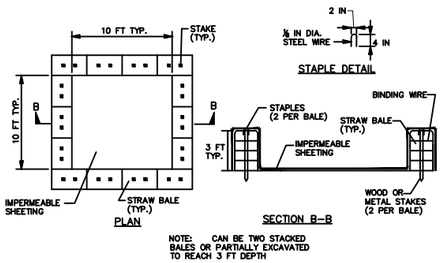


NOTES:

1. BALES SHALL BE PLACED AT THE TOE OF A SLOPE OR ON THE CONTOUR AND IN A ROW WITH ENDS TIGHTLY ABUTTING THE ADJACENT BALES.
2. EACH BALE SHALL BE EMBEDDED IN THE SOIL A MINIMUM OF (4) INCHES, AND PLACED SO THE BINDINGS ARE HORIZONTAL.
3. BALES SHALL BE SECURELY ANCHORED IN PLACE BY EITHER TWO STAKES OR REBARS DRIVEN THROUGH THE BALE. THE FIRST STAKE IN EACH BALE SHALL BE DRIVEN TOWARD THE PREVIOUSLY LAID BALE AT AN ANGLE TO FORCE THE BALES TOGETHER. STAKES SHALL BE DRIVEN FLUSH WITH THE BALE.
4. INSPECTION SHALL BE FREQUENT AND REPAIR REPLACEMENT SHALL BE MADE PROMPTLY AS NEEDED.
5. BALES SHALL BE REMOVED WHEN THEY HAVE SERVED THEIR USEFULLNESS SO AS NOT TO BLOCK OR IMPEDE STORM FLOW OR DRAINAGE.

STRAW BALE DIKE (4)

NOT TO SCALE

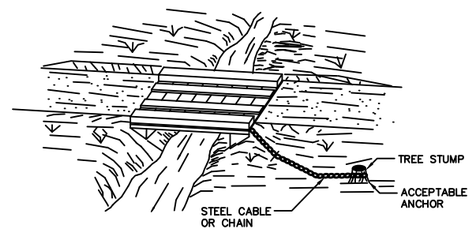


NOTES:

1. WASHOUT AREAS SHOULD NOT BE WITHIN 100' OF WETLANDS, HABITATS OF PROTECTED SPECIES, OR STORM DRAINS. LOCATE WASHOUT STRUCTURE AWAY FROM CONSTRUCTION TRAFFIC.
2. IF WASHOUT AREA IS NOT WITHIN VIEW FROM THE POUR LOCATION, SIGNAGE WILL BE NEEDED TO DIRECT THE TRUCK DRIVERS.
3. SIZE WASHOUT STRUCTURE FOR VOLUME NECESSARY TO CONTAIN WASH WATER AND SOLIDS AND MAINTAIN AT LEAST 4" OF FREEBOARD. TYPICAL DIMENSIONS ARE 10' X 10' X 3' DEEP.
4. PREPARE SOIL BASE FREE OF ROCKS OR OTHER DEBRIS THAT MAY CAUSE TEARS OR HOLES IN THE LINER. FOR LINER, USE 10 MIL OR THICKER UV RESISTANT, IMPERMEABLE SHEETING, FREE OF HOLES AND TEARS OR OTHER DEFECTS THAT COMPROMISE IMPERMEABILITY OF THE MATERIAL.
5. CONCRETE WASHOUT FACILITIES SHOULD BE INSPECTED DAILY AND AFTER HEAVY RAINS TO CHECK FOR LEAKS, IDENTIFY IF ANY PLASTIC LININGS AND SIDEWALLS HAVE BEEN DAMAGED BY CONSTRUCTION ACTIVITIES, AND DETERMINE WHETHER THEY HAVE BEEN FILLED TO OVER 75% CAPACITY.
6. WHEN THE WASHOUT CONTAINER IS FILLED TO OVER 75% OF ITS CAPACITY, THE WASHWATER SHOULD BE VACUUMED OFF OR ALLOWED TO EVAPORATE TO AVOID OVERFLOWS. WHEN THE REMAINING CEMENTITIOUS SOLIDS HAVE HARDENED, THE SOLIDS SHALL BE REMOVED FROM THE PROJECT SITE AREA AND RECYCLED..
7. DAMAGES TO THE CONTAINER SHOULD BE REPAIRED PROMPTLY. BEFORE HEAVY RAINS, THE WASHOUT CONTAINER'S LIQUID LEVEL SHOULD BE LOWERED OR THE CONTAINER SHOULD BE COVERED TO AVOID AN OVERFLOW DURING THE RAIN STORM.
8. UPON COMPLETION OF CONSTRUCTION IN THE CONCRETE WASHOUT AREA, THE RETAINING STRUCTURES SHALL BE DISMANTLED AND REMOVED FROM THE PROJECT SITE. ALL GRADE AND VEGETATION COVER MUST BE RESTORED TO PRE-CONSTRUCTION CONDITIONS.

CONCRETE WASHOUT AREA (5)

NOT TO SCALE



TEMPORARY ACCESS BRIDGE (6)

NOT TO SCALE

TEMPORARY ACCESS BRIDGE:

A TEMPORARY ACCESS BRIDGE IS A STRUCTURE MADE OF WOOD, METAL OR OTHER MATERIALS, WHICH PROVIDES ACCESS ACROSS A STREAM OR WATERWAY.

TEMPORARY ACCESS BRIDGE NOTES:

1. BRIDGE PLACEMENT – A TEMPORARY BRIDGE STRUCTURE SHALL BE CONSTRUCTED AT OR ABOVE BANK ELEVATION TO PREVENT THE ENTRAPMENT OF FLOATING MATERIALS AND DEBRIS.
2. ABUTMENTS – ABUTMENTS SHALL BE PLACED PARALLEL TO AND ON STABLE BANKS.
3. BRIDGE SPAN – BRIDGES SHALL BE CONSTRUCTED TO SPAN THE ENTIRE CHANNEL.
4. STRINGERS – STRINGERS SHALL EITHER BE LOGS, SAW TIMBER, PRE-STRESSED CONCRETE BEAMS, METAL BEAMS OR OTHER APPROVED MATERIALS.
5. DECK MATERIAL – DECKING SHALL BE OF SUFFICIENT STRENGTH TO SUPPORT THE ANTICIPATED LOAD. ALL DECKING MEMBERS SHALL BE PLACED PERPENDICULAR TO THE STRINGERS, BUTTED TIGHTLY AND SECURELY FASTENED TO THE STRINGERS. DECKING MATERIALS MUST BE BUTTED TIGHTLY TO PREVENT ANY SOIL MATERIAL TRACKED ONTO THE BRIDGE FROM FALLING INTO THE WATERWAY BELOW.
6. BRIDGE ANCHORS – BRIDGES SHALL BE SECURELY ANCHORED AT ONLY ONE END USING STEEL CABLE OR CHAIN. ANCHORING AT ONLY ONE END WILL PREVENT CHANNEL OBSTRUCTION IN THE EVENT THAT FLOODWATERS FLOAT THE BRIDGE.
7. STABILIZATION – ALL AREAS DISTURBED DURING INSTALLATION AND REMOVAL SHALL BE STABILIZED WITHIN 14 CALENDAR DAYS OF THE DISTURBANCE.
8. INSPECTION – PERIODIC INSPECTION SHALL BE PERFORMED BY THE USER TO ENSURE THAT THE BRIDGE, STREAMBED AND STREAM BANKS ARE MAINTAINED AND NOT DAMAGED.
9. MAINTENANCE – MAINTENANCE SHALL BE PERFORMED, AS NEEDED TO ENSURE THAT THE STRUCTURE COMPLIES WITH THE STANDARD AND SPECIFICATIONS.
10. REMOVAL – WHEN THE TEMPORARY BRIDGE IS NO LONGER NEEDED, ALL STRUCTURES INCLUDING ABUTMENTS AND OTHER BRIDGING MATERIALS SHALL BE REMOVED WITHIN 14 CALENDAR DAYS.
11. FINAL CLEAN-UP – FINAL CLEAN-UP SHALL CONSIST OF REMOVAL OF THE TEMPORARY BRIDGE FROM THE WATERWAY, PROTECTION OF BANKS FROM EROSION AND REMOVAL OF ALL CONSTRUCTION MATERIALS.

1 ADD NOTES 12/16/16

REVISIONS

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CRICKET VALLEY ENERGY TRANSMISSION

EROSION AND SEDIMENT CONTROLS

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APP. BY: PGC	DATE: 5/21/2016

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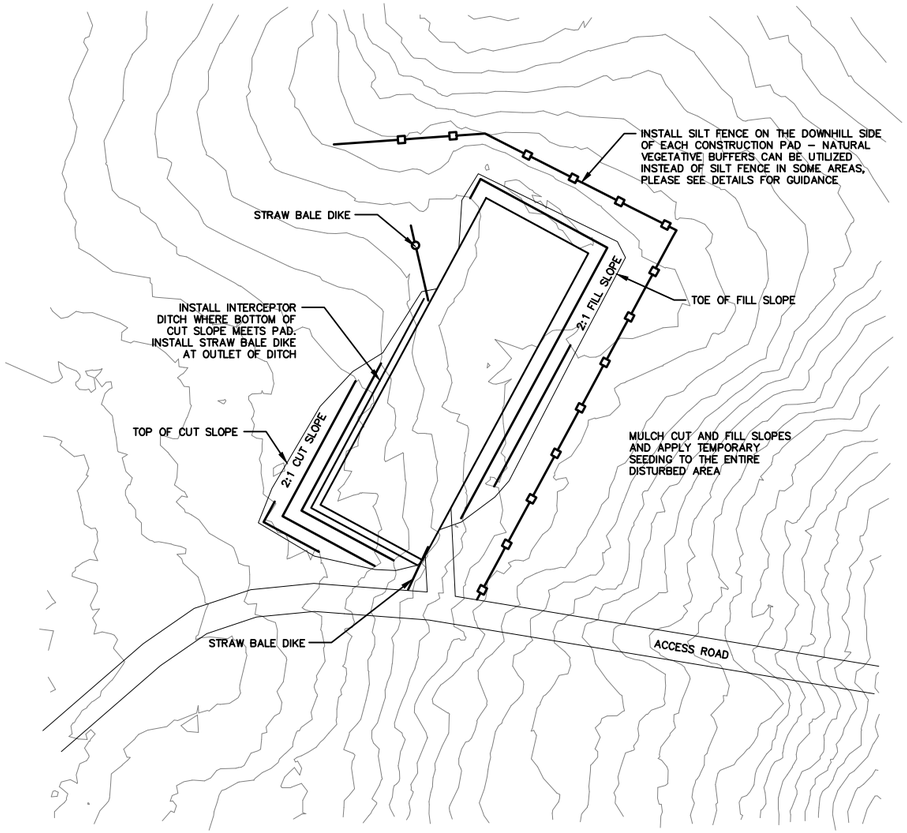
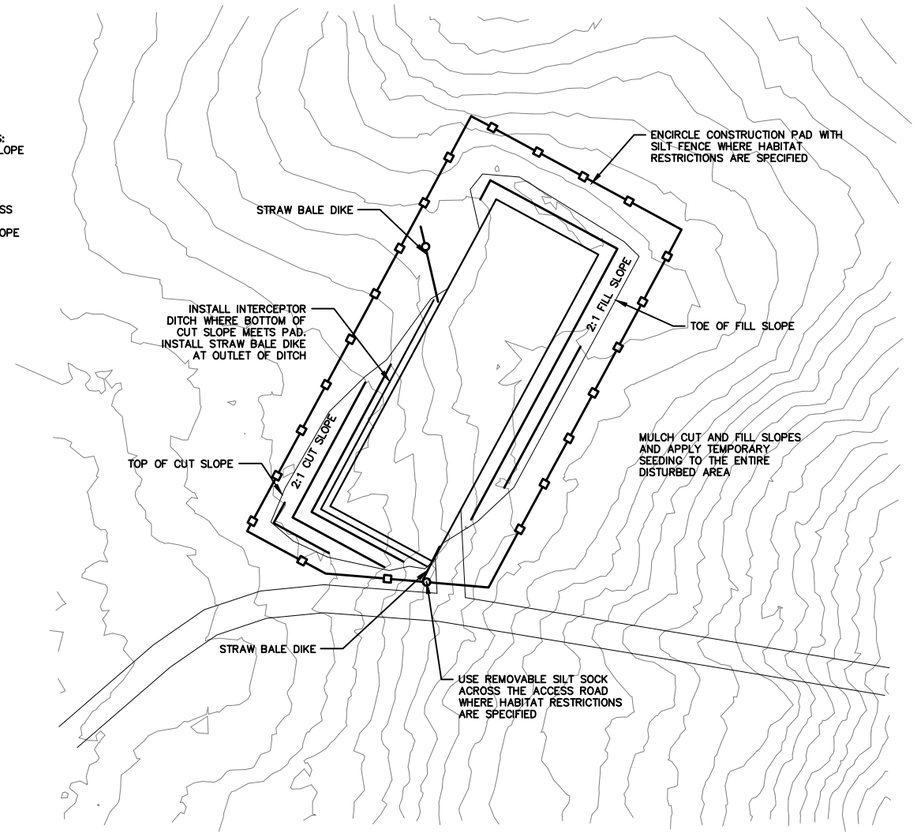


PAD SITE NOT IN HABITAT RESTRICTED AREAS

GRADING OF PAD AREAS
CONSTRUCTION PAD SITES:
PROVIDE 0.5% TO 2.5% SLOPE FOR DRAINAGE.

STRINGING PAD SITES:
PROVIDE 0% SLOPE ACROSS THE WIDTH
PROVIDE 0.5% TO 15% SLOPE ACROSS THE LENGTH



PAD SITE IN HABITAT RESTRICTED AREAS

E&S AND GRADING DETAILS FOR TYPICAL STRINGING SITES AND CONSTRUCTION PADS
NOT TO SCALE

DETAILS:

1. SILT FENCE VERSUS NATURAL VEGETATION BUFFERS – USE SILT FENCE IF NEAR A WETLAND OR STREAM OR A STEEP SLOPE OR IF NO SUITABLE NATURAL VEGETATIVE BUFFER EXISTS.
2. CONSTRUCTION PADS AND STRINGING SITE PADS ARE TO BE GRADED WITH BALANCED CUT AND FILL. CUT AND FILL SLOPES ARE TO BE A MAXIMUM OF 1V:2H.
3. INSTALL AN ACCESS ROAD TO EACH PAD AT A LOCATION WHERE MINIMUM GRADING IS REQUIRED.
4. MULCH CUT AND FILL SLOPES AND APPLY TEMPORARY SEEDING TO THE ENTIRE DISTURBED AREA.

REVISED NOTE 3 TO BE CONSISTENT WITH OTHER DRAWINGS
01 PGC 1/22/17
REVISIONS

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CRICKET VALLEY ENERGY TRANSMISSION

EROSION AND SEDIMENT CONTROLS

PROJ. NO.	2015–185
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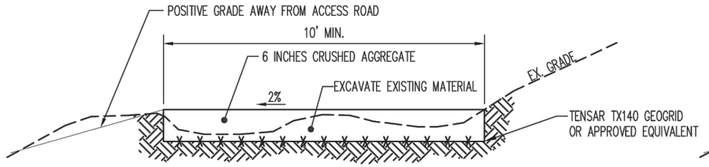
DRN. BY: LEB	CHK'D. BY: PGC
APP. BY: PGC	DATE: 5/21/2016

STATE OF NEW YORK PAUL G CASS 5/21/16 086902 LICENSED PROFESSIONAL ENGINEER

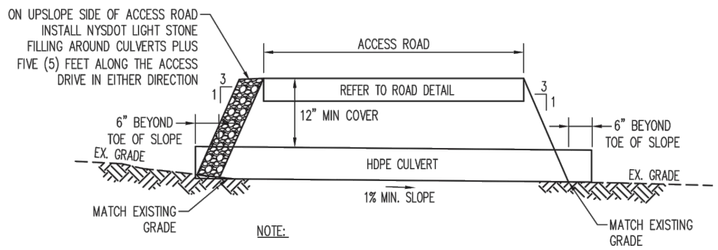


NOTES:

1. CONTRACTOR TO REMOVE EXISTING MATERIAL AND PROVIDE A LEVEL SURFACE FOR ROADWAY.
2. COMPACT SUB-GRADE AS NOTED. REMOVE SOFT AREAS AND REPLACE WITH FILL.
3. PLACE TENSAR AS PER MANUFACTURER SPECIFICATIONS.
4. UTILIZE 3/4" STONE AS CRUSHED AGGREGATE UNLESS LISTED AS A STEEP AREA.
5. UTILIZE 1-1/2" STONE WITHIN STEEP AREAS.

4 STANDARD/STEEP STONE ACCESS ROAD REBUILD DETAIL

N.T.S.

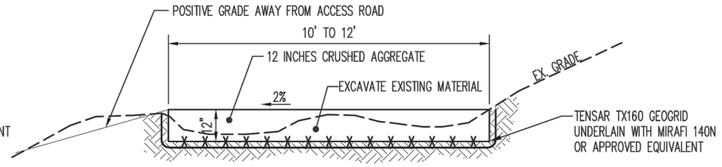


NOTE:

1. IF MINIMUM COVER CANNOT BE ATTAINED DUE TO SITE CONSTRAINTS, BUILD UP ACCESS DRIVE.

8 TYPICAL CULVERT DETAIL

N.T.S.

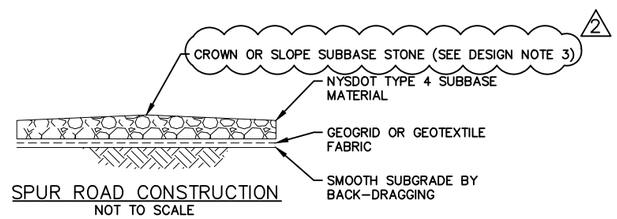


NOTES:

1. CONTRACTOR TO REMOVE EXISTING MATERIAL AND PROVIDE A LEVEL SURFACE FOR ROADWAY.
2. COMPACT SUB-GRADE AS NOTED. REMOVE SOFT AREAS AND REPLACE WITH FILL.
3. PLACE TENSAR AS PER MANUFACTURER SPECIFICATIONS.
4. UTILIZE 1-1/2" STONE AS CRUSHED AGGREGATE.

5 SOFT SOIL ACCESS ROAD REBUILD DETAIL

N.T.S.



SPUR ROAD CONSTRUCTION
NOT TO SCALE

CONSTRUCTION ROAD STABILIZATION

APPLICATION
1. FOR PREVENTION OF RUTTING AND EROSION AND TO INCREASE TRACTION FOR CONSTRUCTION EQUIPMENT AND VEHICLES.

MATERIALS
1. GEOGRID – TENSAR TX160 OR APPROVED EQUAL.
2. GEOTEXTILE FABRIC – MIRAFI 140N NON-WOVEN GEOTEXTILE FABRIC OR APPROVED EQUAL.
3. SUBBASE – NYSDOT TYPE 4 SUBBASE MATERIAL.

DESIGN
1. MAXIMUM 15 FT. ROAD WIDTH.
2. PROVIDE WATERBARS OR TRAFFIC MOUNTABLE BERMS WHERE SHOWN ON DRAWINGS.
3. WHERE GRADING FOR SPUR ROADS EITHER CROWN ROAD TO DRAIN IN BOTH DIRECTIONS OR SLOPE ROAD TO ALLOW SHEET FLOW TO DOWNSLOPE SIDE.

TABLE 733-04A SUBBASE GRADATION	
SIEVE SIZE DESIGNATION	PERCENTAGE PASSING BY WEIGHT
4 IN.	–
3 IN.	–
2 IN.	100
¼ IN.	30-65
NO. 400	5/1/1940
NO. 200	1/10/2016

NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
1	1-23-17	ADDED STABILIZED ROAD SECTION DETAIL	MJG	PGC	PGC	PGC
2	4-3-17	NOTES ON SPUR ROAD CONSTRUCTION	MJG	PGC	PGC	PGC

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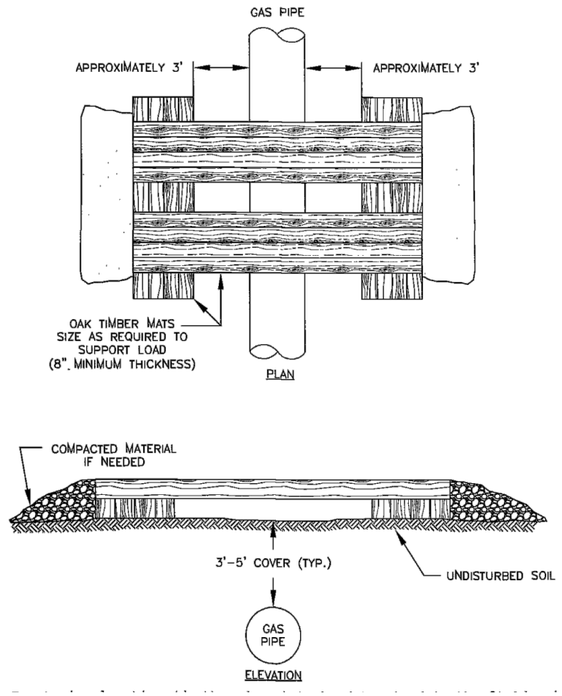
Phone: (412) 372-4500 Fax: (412) 372-1972

CRICKET VALLEY ENERGY TRANSMISSION

SPUR ROAD CONSTRUCTION &
EXISTING ACCESS ROAD STABILIZATION

PROJ. NO.: 2015-185
DWG. NO.: 2015-185-D-05
SCALE:
DRN. BY: LEB
CHK'D. BY: PGC
APP. BY: PGC
DATE: 5/21/2016

STATE OF NEW YORK · PAUL G. CAIN · 088902 · 1/23/17 · LICENSED PROFESSIONAL ENGINEER



TRENCHING AND BACKFILL AROUND GAS PIPELINES

CHG&E GAS CONSTRUCTION STANDARD G.01.03.002.0

NOTES

1. EXACT PIPE LOCATION (DEPTH AND RUN) TO BE DETERMINED IN THE FIELD PRIOR TO BRIDGE CONSTRUCTION. THE METHOD AND LOCATION OF DETERMINATION SHALL BE APPROVED BY CENTRAL HUDSON – GAS & ELECTRIC MECHANICAL ENGINEERING, PRIOR TO FIELD CONSTRUCTION.
2. ASSURE MATS ARE WIDE ENOUGH TO FIT VEHICLE TIRE WIDTH.
3. MATS ARE TO BE SECURED TO PREVENT ANY MOVEMENT.
4. FOR THIRD PARTY ACTIVITIES, THE TIMBER SIZE SHALL BE SPECIFIED TO SUPPORT THE MAXIMUM APPLIED LOAD. ENGINEERED DESIGN (PE STAMPED) TO BE REVIEWED BY CENTRAL HUDSON – GAS & ELECTRIC MECHANICAL ENGINEERING, PRIOR TO FIELD CONSTRUCTION.

CHG&E REQUIREMENTS FOR CONSTRUCTION NEAR THE MPI GAS TRANSMISSION LINE

CENTRAL HUDSON REQUIRES A MINIMUM OF TWELVE INCHES OF CLEARANCE BETWEEN THE GAS MAIN AND ANY NEW UTILITY AS SHOWN ON GAS CONSTRUCTION STANDARD G.01.03.004.0. TYPICALLY, THE GAS MAINS HAVE AT LEAST THREE FEET OF COVER. HOWEVER, IT IS UNCERTAIN AT THIS TIME THE EXACT DEPTH OF PIPE DUE TO EROSION OR OTHER CONSTRUCTION ACTIVITIES SINCE THIS LINE'S ORIGINAL INSTALLATION.

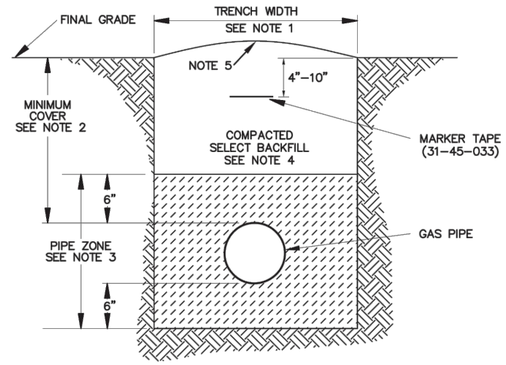
BEFORE ANY EXCAVATION IS PERFORMED, THE SITE CONTRACTOR MUST FOLLOW SEVERAL REQUIREMENTS:

- DIG SAFELY NY SHALL BE CALLED FOR A MARK–OUT REQUEST. CENTRAL HUDSON MUST BE CONTACTED AT LEAST 48 HOURS IN ADVANCE OF WHEN ANY EXCAVATION WORK WILL BE WITHIN 25 FEET OF THE GAS TRANSMISSION LINE.
- DIG SAFELY NY MAY ALSO BE CALLED IN ADVANCE FOR A "DESIGN" MARK–OUT REQUEST FOR VERIFICATION OF PIPE LOCATION.
- WHEN EXCAVATION WORK WILL BE WITHIN 10–FEET OF THE GAS TRANSMISSION LINE, A CENTRAL HUDSON REPRESENTATIVE MUST BE PRESENT ON SITE.
- WHENEVER ANY GAS MAIN IS EXPOSED, THE CENTRAL HUDSON REPRESENTATIVE SHALL HAVE ACCESS TO FULLY INSPECT THE MAIN AND PERFORM ANY REQUIRED PREVENTIVE MAINTENANCE. THE SITE CONTRACTOR SHALL HAND CLEAN ALL TRENCH SPOIL OFF THE SURFACE OF THE PIPELINE.
- A HAND DUG TEST HOLE MUST BE COMPLETED WITHIN THE AREA OF CROSSING UTILITIES TO VERIFY THE DEPTH AND DIAMETER OF THE PIPE AND RECORD FOR SURVEY PURPOSES TO BETTER FACILITATE THE DEVELOPMENT'S DESIGN. THE EXCAVATION DEPTH SHALL BE AT LEAST 12–INCHES BELOW THE PIPE INVERT ELEVATION. THE EXCAVATION WIDTH SHALL BE AT LEAST 12–INCHES ON EITHER SIDE OF THE PIPELINE.
- CENTRAL HUDSON CREWS (OR THEIR DESIGNEES) SHALL BE ALLOWED TWO FULL DAYS UNRESTRICTED BY INCLEMENT WEATHER TO ACCOMPLISH THEIR INSPECTION, EVALUATION, AND REPAIR WORK TO THE EXPOSED PIPE. THE SITE CONTRACTOR SHALL MAINTAIN THE EXCAVATION AS NECESSARY DURING THIS PERIOD, INCLUDING MAINTAINING SAFETY FENCING AROUND IT.
- THE MARKED LOCATION OF THE GAS MAIN SHALL BE MAINTAINED THROUGHOUT CONSTRUCTION.
- ANY EXPOSED SECTIONS OF GAS MAIN MUST BE SUFFICIENTLY SUPPORTED AS PER GAS CONSTRUCTION STANDARD G.01.07.003.0 (SEE ATTACHED).
- BACKFILL REQUIREMENTS FOR THE EXPOSED GAS MAIN SHALL FOLLOW GAS CONSTRUCTION STANDARD G.01.03.002.0 (SEE ATTACHED).
- ANY CROSSINGS OF THE PIPELINE SHOULD BE PERPENDICULAR IN NATURE UTILIZING AIR BRIDGING AS PER GAS CONSTRUCTION STANDARD G.01.03.004.20) (SEE ATTACHED).
- TRAVELING IN PARALLEL TO THE CENTER OF THE PIPE SHOULD BE DONE AT LEAST 5' AWAY FROM THE CENTERLINE OF THE PIPE. FOR THE DURATION OF THE PROJECT THE LICENSEE IS RESPONSIBLE FOR THE INSTALLATION OF CONSTRUCTION FENCING (OR SIMILAR TYPE) AT LEAST 5' AWAY FROM THE CENTER OF THE PIPE TO ACT AS A VISIBLE AND PHYSICAL BARRIER FROM THE PIPE IN KEEPING WITH TRAFFIC STAYING AT LEAST 5' AWAY FROM THE PIPE.
- CENTRAL HUDSON'S PIPE SHALL NOT BE DAMAGED. THIS INCLUDES THE PIPE'S COATING. ANY DAMAGE DONE TO THE STEEL OR COATING SHALL BE REPAIRED IMMEDIATELY BY CENTRAL HUDSON. THE SITE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL COSTS INCURRED TO COMPLETE THE REPAIR.
- ALL COSTS TO PERFORM THE WORK ABOVE SHALL BE INCURRED BY THE SITE CONTRACTOR EXCEPT FOR ANY PREVENTATIVE MAINTENANCE OF THE GAS MAIN BY CENTRAL HUDSON.

CENTRAL HUDSON REQUIRES NOTIFICATION OF AT LEAST TWO WEEKS SHOULD ANY BLASTING BE CONDUCTED BY THE SITE CONTRACTOR.

THROUGHOUT THE COURSE OF CONSTRUCTION NEAR THE GAS MAIN, CENTRAL HUDSON PERSONNEL SHALL BE PERMITTED WITHIN THE CONSTRUCTION AREA TO INSPECT ALL ACTIVITY INVOLVING OUR GAS MAIN. ALL WORK MUST BE PERFORMED TO THE SATISFACTION OF THE CENTRAL HUDSON REPRESENTATIVE. PLEASE RESPOND WITH ANY ADDITIONAL REQUIREMENTS FOR OUR PERSONNEL TO BE ON SITE SUCH AS CHECKING IN WITH A SITE SUPERINTENDENT.

IT MUST BE UNDERSTOOD BY THE LICENSEE AND ALL OTHER INVOLVED PARTIES THAT CENTRAL HUDSON RETAINS ITS RIGHTS TO OPERATE, MAINTAIN, AND REPLACE THE GAS LINE AT ITS OWN DISCRETION. IT IS CRITICAL THAT LICENSEE MAINTAIN CONTACT WITH CENTRAL HUDSON TO PROVIDE A SAFE WORKING ENVIRONMENT FOR YOUR EMPLOYEES, CONSTRUCTION CREWS, AND THE NEIGHBORING COMMUNITY. THE PRINCIPLE CONTACT FOR CENTRAL HUDSON IS DAVID SCHNECK, HIS DIRECT PHONE NUMBER IS (845)486–5747 AND CELL PHONE IS (845)242–1939.



AIR BRIDGE OVER GAS PIPELINE

CHG&E GAS CONSTRUCTION STANDARD G.01.03.004.20

NOTES:

1. THE MINIMUM TRENCH WIDTH SHALL BE AS FOLLOWS:

PIPE SIZE	TRENCH WIDTH
¾" – 2"	12"
3" – 16"	PIPE O.D. + 12"

2. MINIMUM DEPTH OF COVER FROM FINAL GRADE TO TOP OF PIPE SHALL BE AS FOLLOWS:

	IN EARTH	IN ROCK
TRANSMISSION MAINS AND SERVICES	36"	24"

3. SANDPADDING IS REQUIRED IN THE PIPE ZONE. IT SHALL BE THOROUGHLY COMPACTED IN 12' LIFTS.
4. COMPACTED SELECT BACKFILL MAY BE ON–SITE MATERIAL PROVIDED IT CONTAINS NO ROCKS OR STONES OVER 6" IN DIAMETER, ROOTS, STUMPS OR CONSTRUCTION DEBRIS. IT SHALL BE THOROUGHLY COMPACTED IN MINIMUM 12" LIFTS.
5. THE BACKFILLED TRENCH SHALL BE CROWNED SLIGHTLY TO ALLOW FOR FUTURE SETTLEMENT.
6. REFER TO GAS CONSTRUCTION STANDARD G.02.01.037.0 TO DETERMINE THE NEED FOR TRENCH SHORING AND SLOPING.
7. THE MINIMUM CLEARANCE BETWEEN GAS PIPELINE AND UNDERGROUND OBSTRUCTION IS 12"
8. IN STEEP TERRAIN, EROSION CONTROL MAY BE REQUIRED.

NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP

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CRICKET VALLEY ENERGY TRANSMISSION

GAS LINE PROTECTION FOR CHG&E FACILITIES – SHEET 1

PROJ. NO.	2015–185
DWG. NO.:	2015–185–D–06
SCALE:	
DRN. BY: MJG	CHK'D. BY: PGC
APP. BY: PGC	DATE: 1/23/2017

STATE OF NEW YORK * PAUL G. CASS * 086902 * LICENSED PROFESSIONAL ENGINEER * 1/23/17

TEMPORARY SUPPORT AND PROPER RE-EMBEDMENT OF EXPOSED OR DISTURBED UNDERGROUND GAS FACILITIES FOR CONSTRUCTION OF CRICKET VALLEY 345KV LINES CHG&E CONSTRUCTION STANDARD G.01.07.003.0

THIS STANDARD APPLIES IN ALL SITUATIONS WHERE VERTICAL AND/OR LATERAL SUPPORT OF EXISTING UNDERGROUND GAS MAINS OR SERVICES OF ANY TYPE OR SIZE HAS BEEN COMPROMISED. THIS OCCURS WHENEVER SUCH FACILITIES ARE EXPOSED BY EXCAVATION FOR A LINEAR DISTANCE IN EXCESS OF TEN (10) FEET (THREE FEET FOR CAST IRON) OR FOR ALL EXPOSURES THAT INCLUDE A FITTING, A CONNECTION, OR AN APPURTENANCE OF ANY KIND. THIS STANDARD IS ALSO APPLICABLE IN CASES WHERE EMBEDMENT MATERIAL HAS BEEN DISTURBED, EVEN THOUGH EXPOSURE HAS NOT OCCURRED.

ACCEPTABLE MEANS OF TEMPORARY SUPPORT ARE PROVIDED AS WELL AS SOUND, PERMANENT EMBEDMENT PROCEDURES ONCE THE INITIATING CAUSE OF THE EXPOSURE OR DISTURBANCE HAS BEEN CORRECTED.

I. GENERAL

(A) AS STATED IN NEW YORK STATE INDUSTRIAL CODE RULE #53 (12 NYCRR53), ANYONE PERFORMING EXCAVATION OR DEMOLITION (HEREINAFTER REFERRED TO AS "EXCAVATOR") IS REQUIRED TO USE EXTREME CAUTION AND PROVIDE:
- NOTIFICATION OF THE PROPOSED EXCAVATION WORK TO CENTRAL HUDSON AT LEAST 48 HOURS, BUT NO MORE THAN 10 DAYS, PRIOR TO CONSTRUCTION, AND
- ALL NECESSARY SHORING, SHEATHING, AND/OR SUPPORT TO ASSURE THAT NO DAMAGE OCCURS IN THE EVENT OF EXPOSURE.

(B) MAINS AND SERVICES ENCOUNTERED IN THE CENTRAL HUDSON GAS SYSTEM WILL BE OF NOMINAL SIZE AND CONSIST OF ANYONE OR COMBINATION OF THE FOLLOWING PIPE MATERIALS.
STEEL
WROUGHT IRON
CAST IRON
PLASTIC (HDPE)
THE MAINS AND SERVICES MAY INCLUDE VALVES, FITTINGS, DRIPS, STATIONS, CATHODIC PROTECTION OFFS, AND/OR ANY OTHER SUCH APPURTENANCES.

(C) THE LOCATION OF THE UNDERGROUND GAS FACILITIES WITHIN THE PROPOSED CONSTRUCTION AREA WILL BE PROVIDED TO THE EXCAVATOR BY CENTRAL HUDSON PRIOR TO START OF WORK. IF KNOWN, THE SIZE AND MATERIAL OF THE FACILITIES WILL ALSO BE PROVIDED.

II. TEMPORARY SUPPORT

(A) ANY GAS MAIN OR SERVICE UNDERMINED BY EXCAVATION (OR EXPOSED TO THE EXTENT THAT LATERAL AND/OR VERTICAL SUPPORT IS JEOPARDIZED) AS DESCRIBED IN THE FIRST PARAGRAPH ON SHEET 1 MUST BE PROVIDED WITH IMMEDIATE, TEMPORARY SUPPORT. SUCH SUPPORT IS MEANT FOR SHORT DURATION ONLY. PERMANENT RE-EMBEDMENT (AND/OR MANDATORY REPLACEMENT FOLLOWED BY REEMBEDMENT WHERE APPLICABLE IN CAST IRON SITUATIONS) SHOULD OCCUR AS SOON AS POSSIBLE AFTER EXPOSURE.

THE ENGINEERING SKETCHES PROVIDED AT THE END OF THIS STANDARD SHOW THE SPECIFIC TECHNICAL DETAILS OF TEMPORARY SUPPORT SYSTEMS FOR STEEL PIPE, CAST AND WROUGHT IRON PIPE, AND PLASTIC PIPE. ALTHOUGH ONLY FLAT-SURFACED, WOODEN STRUCTURAL SUPPORT MEMBERS ARE DEPICTED, OTHER MATERIALS MAY ALSO BE ACCEPTABLE. STEEL I-BEAMS, BOX BEAMS, CHANNEL, AND PIPE MAY BE UTILIZED PROVIDED THE STEEL IS IN SOUND CONDITION (NO VISIBLE DEFORMATION, WARPS, BENDS, OR SIGNIFICANT CORROSION), THAT IT HAS A MINIMUM DIMENSION OF 6" IN THE DIRECTION OF SUPPORT, THAT THE THICKNESS OF THE LOAD BEARING PORTION IS NO LESS THAN 1/4", AND THAT AT LEAST TWO SQUARE FEET OF SURFACE AREA DISTRIBUTE THE LOAD TO THE GROUND ON EITHER SIDE OF THE EXCAVATION. IN THE CASE OF PIPE, I-BEAMS, AND CHANNEL, THIS REQUIRES FLAT LUMBER (1-1/2" MINIMUM THICKNESS) OR PLATE (1/2" MINIMUM THICKNESS) BETWEEN THE SUPPORT MEMBER AND THE GROUND. FOR PIPE, CHOCKS OR STAKES ARE ALSO NECESSARY TO PREVENT ROLLING.

IF WOODEN SUPPORTS ARE TO BE USED, STRUCTURAL GRADE, FLAT-SURFACED LUMBER AS SHOWN IN THE SKETCHES (AND SIZED IN THE TABLES THEREON) IS PREFERRED. THE LUMBER MUST BE IN AS-NEW CONDITION, WITH NO VISIBLE SIGNS OF DISTRESS (SPLINTERING, CRACKS, SPLITS, ROT, HOLES, APPRECIABLE KNOTS, EXCESSIVE DRYNESS, ETC.) AND VISIBLY STRAIGHT AND TRUE. POLE STOCK MAY ALSO BE USED, AS LONG AS IT MEETS THE SAME CRITERIA FOR CONDITION, SIGNS OF DISTRESS, AND DIMENSIONAL CONSISTENCY. POLE STOCK MUST BE AT LEAST 10" IN DIAMETER FOR SUPPORT OF PIPING 8" OR LESS, AND AT LEAST 12" IN DIAMETER FOR 10" AND GREATER PIPING. ALSO, AS WITH USING STEEL PIPE FOR SUPPORT, POLE STOCK MUST BE LAID ON AT LEAST TWO SQUARE FEET OF FLAT PLATE (1/2" MINIMUM THICKNESS) OR LUMBER (MINIMUM OF 1-1/2" THICKNESS) ON EACH SIDE OF THE EXCAVATION AND BE CHOCKED OR STAKED TO PREVENT ROLLING. WHEN USING WOOD, ANY TYPE (OAK, PINE, ETC .) IS ACCEPTABLE, AS LONG AS THE FORM IT TAKES WAS OBVIOUSLY MEANT FOR STRUCTURAL DUTY (E.G. STRAPPING TWELVE LAYERS OF 1/2" PLYWOOD OR 8 LAYERS OF 1 X 6 BOARD TOGETHER TO MAKE A 6" X 6" SUPPORT IS NOT APPROPRIATE).

THE USE OF ANY OTHER MATERIAL FOR TEMPORARY SUPPORT OF EXPOSED UNDERGROUND GAS FACILITIES IS NOT ALLOWED WITHOUT ENGINEERING CONSULTATION AND APPROVAL.

(B) STRUCTURAL MEMBERS USED FOR THE TEMPORARY UNDERGROUND GAS FACILITIES SHALL NOT BE USED TO SUPPORT ANY OTHER STRUCTURE OR PIPE.

(C) MAXIMUM EXCAVATION SPANS FOR SUPPORT MEMBERS AND MAXIMUM SPACING OF SLINGS (3/8" WIRE ROPE OR 3/8" GUY WIRE) ARE SHOWN IN THE ENGINEERING SKETCHES. ENGINEERING SHALL BE CONSULTED IF THESE LIMITS CAN NOT BE MET.

(D) THE GROUND UPON WHICH SUPPORT MEMBER(S) WILL REST SHALL BE LEVEL AND FIRM TO PROVIDE A UNIFORM BEARING SURFACE. IF THE SURFACE IS UNEVEN, NOT FIRM, OR NON-UNIFORM, IT SHALL BE LEVELED AND COMPACTED UNTIL ACCEPTABLE.

(E) WHERE A FITTING, COMPRESSION COUPLING, GAS SERVICE, REPAIR CLAMP, VALVE, DRIP OR ANY OTHER APPURTENANCE IS INCLUDED ON THE EXPOSED GAS FACILITY, AN ADDITIONAL CROSS MEMBER AND S11NG SHALL BE INSTALLED TO SUPPORT THE PIPING AT THIS LOCATION.

(F) UNLESS AN APPURTENANCE IS PRESENT, ANY CAST IRON OR WROUGHT IRON PIPE EXPOSED FOR LESS THAN THREE (3) FEET, OR STEEL OR PLASTIC PIPE EXPOSED FOR LESS THAN TEN (10) FEET DOES NOT REQUIRE TEMPORARY SUPPORT. A CROSS MEMBER AND SLING SHALL BE PROVIDED, THOUGH, WHEREVER SUCH EXPOSED APPURTENANCE(S) EXIST.

(G) THE ENGINEERING SKETCHES INDICATE THAT SLINGS SHALL BE 3/8" WIRE ROPE OR GUY WIRE. THE USE OF SUBSTITUTE MATERIAL IS NOT ALLOWED WITHOUT ENGINEERING CONSULTATION AND APPROVAL.

(H) ONCE A GAS MAIN AND/OR SERVICE HAS BEEN UNDERMINED/EXPOSED, NO FURTHER CONSTRUCTION IS ALLOWED IN THE AREA UNTIL:
- APPROPRIATE TEMPORARY SUPPORT HAS BEEN INSTALLED AS SPECIFIED HEREIN.
- SUFFICIENT TENSION HAS BEEN APPLIED TO THE SLING(S) TO ASSURE THAT NO DEFLECTION FROM THE ORIGINAL PIPE LOCATION AND ROUTE IS POSSIBLE (EXCESSIVE TENSION HAS BEEN APPLIED IF THE PIPE IS OBSERVED TO BEGIN TO RISE).

(I) SLINGS SHALL BE RE-TENSIONED REGULARLY AS NECESSARY TO MAINTAIN CONTINUOUS SUPPORT. THE ORIGINAL POSITION OF THE GAS MAIN AND/OR SERVICE MUST REMAIN AS FOUND.

III. RE-EMBEDMENT OF UNDERGROUND GAS FACILITIES

(A) THE EXCAVATOR SHALL PROVIDE, IF SPECIFIED BY CENTRAL HUDSON, 24 HOUR MINIMUM NOTIFICATION PRIOR TO BACKFILLING EXPOSED/SUSPENDED/DISTURBED UNDERGROUND GAS FACILITIES SO THAT CENTRAL HUDSON HAS THE OPPORTUNITY FOR INSPECTION. IF DAMAGE TO FACILITIES OR ASSOCIATED COATINGS OR APPURTENANCES IS OBSERVED, REPAIRS SHALL BE ACCOMPLISHED BEFORE BACKFILLING/RE-EMBEDMENT IS COMPLETED.

(B) WHEN USING SELECT GRANULAR MATERIAL TO EMBED AND/OR BACKFILL GAS FACILITIES, PROCEED AS FOLLOWS

(1) BACKFILL MATERIAL IN THE PIPE ZONE (6" BELOW, AROUND, AND ABOVE UNDERGROUND GAS FACILITIES) SHALL BE SELECT GRANULAR MATERIAL, SUCH AS SAND OR STONE DUST, FREE OF STONES, ROCK, OR OTHER SUCH OBJECTS WHICH COULD DAMAGE THE PIPE, COATING, OR APPURTENANCES.

(2) BACKFILL MATERIAL SHALL BE PLACED ADJACENT TO THE GAS FACILITIES AND WORKED UNDER THEM IN SUCH A WAY AS NOT TO DAMAGE OR DISTURB THEM. UNDER NO CIRCUMSTANCES SHALL BACKFILL MATERIAL BE DROPPED DIRECTLY UPON GAS FACILITIES.

(3) ALL BACKFILL MATERIAL UP TO AND INCLUDING THE PIPE ZONE SHALL BE PLACED IN 6" LIFTS. EACH LIFT SHALL BE THOROUGHLY COMPACTED TO ACCEPTABLE INDUSTRY STANDARDS. PLACEMENT AND COMPACTION OF BACKFILL ABOVE THE PIPE ZONE CAN BE ACCOMPLISHED IN 12" LIFTS.

(4) WATER SETTLEMENT IS THE PREFERRED MEANS OF COMPACTION UP TO AND INCLUDING THE PIPE ZONE. THIS METHOD ELIMINATES AIR POCKETS AND VOIDS AND PROVIDES SOLID EMBEDMENT AND SUPPORT TO THE GAS FACILITIES. WHERE FREEZING WEATHER OR LACK OF WATER PREVENTS THIS METHOD, MECHANICAL COMPACTION WITH MECHANIZED TAMPING MACHINES IS AN ACCEPTABLE ALTERNATE METHOD.

(C) FLOWABLE FILL, A STRUCTURAL MATERIAL MUCH LIKE CONCRETE IN APPEARANCE AND PHYSICAL CHARACTERISTICS WHOSE MAIN CONSTITUENT IS COAL-FIRED FLYASH, MAY ALSO BE USED TO EMBED AND/OR BACKFILL GAS FACILITIES. ITS APPLICATION MAY BE COST EFFECTIVE FOR LARGER PROJECTS, OR WHEN TAMPING MUST BE AVOIDED, AND/OR WHEN RE-EMBEDMENT IS COMPLICATED BY PHYSICAL OR SCHEDULE CONSTRAINTS. IT MAY BE PLACED IN THE PIPE ZONE AND THE AREA ABOVE THE PIPE ZONE. PROCEED AS FOLLOWS:

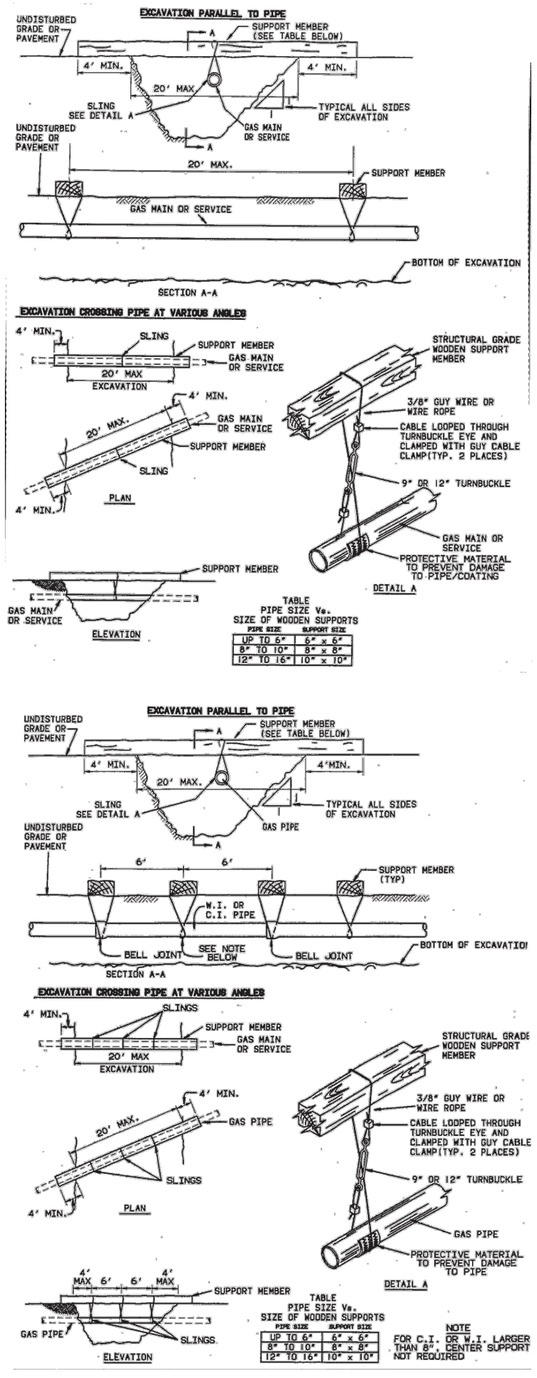
(1) FLOWABLE FILL MATERIAL SHALL BE PLACED ADJACENT TO THE GAS FACILITIES AND ALLOWED TO FLOW UNDER THEM SO AS NOT TO DAMAGE OR DISTURB THEM. UNDER NO CIRCUMSTANCES SHALL IT BE POURED DIRECTLY UPON GAS FACILITIES.

(2) FLOWABLE FILL SHALL BE A MIXTURE OF CLASS F COAL-FIRED FLYASH (SUCH AS IS PRODUCED AT THE DANSKAMMER GENERATING STATION), CEMENT, FINE AGGREGATE, AND WATER WITH AN UNCONSTRAINED COMPRESSIVE STRENGTH OF NO MORE THAN 175 PSI. IT SHALL BE PLACED BY PERSONNEL FAMILIAR WITH ITS APPLICATION, USING APPROPRIATE EQUIPMENT. AT SITES WHERE FLOWABLE FILL CAN BE DIRECTLY OFF-LOADED FROM A READY-MIX TYPE DELIVERY APPROXIMATELY 100 LBS. OF CEMENT, 250 LBS. OF FLYASH, 2,850 LBS. OF SAND, AND 500 LBS. OF WATER. THIS MATERIAL CURES IN ABOUT 24 HOURS (CONSULT GAS & MECHANICAL ENGINEERING IF ACCELERATED CURE RATES ARE NECESSARY). IN THE EVENT THE SITE IS INACCESSIBLE TO A DELIVERY TRUCK AND PUMPING IS REQUIRED, EACH CUBIC YARD OF MIXTURE SHALL CONSIST OF APPROXIMATELY 100 LBS. OF CEMENT, 2,000 LBS. OF FLYASH, AND 725 LBS. OF WATER (NO SAND SHALL BE USED).

(3) NEITHER PLACEMENT IN LIFTS NOR TAMPING IS REQUIRED SINCE FLOWABLE FILL DOES NOT SETTLE OR SHRINK.

(D) ONCE SATISFACTORY BACKFILL PLACEMENT AND COMPACTION HAS BEEN ACCOMPLISHED UP TO AND INCLUDING THE CENTERLINE OF THE GAS FACILITIES, THE TEMPORARY SUPPORT SYSTEM CAN BE REMOVED. SUCH ' REMOVAL SHALL BE PERFORMED SO AS NOT TO DAMAGE THE PIPE, COATING, OR APPURTENANCES. WHEN FLOWABLE FILL IS UTILIZED, TEMPORARY SUPPORTS MUST REMAIN UNTIL THE MATERIAL HAS CURED.

(E) IF SPECIFIED BY CENTRAL HUDSON, THE EXCAVATOR SHALL ALLOW CENTRAL HUDSON SUFFICIENT TIME TO PERFORM A LEAK SURVEY ON THE GAS MAIN(B) AND/OR SERVICE(S) PRIOR TO ANY BACKFILL MATERIAL BEING PLACED ABOVE THE GAS PIPE.



NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP

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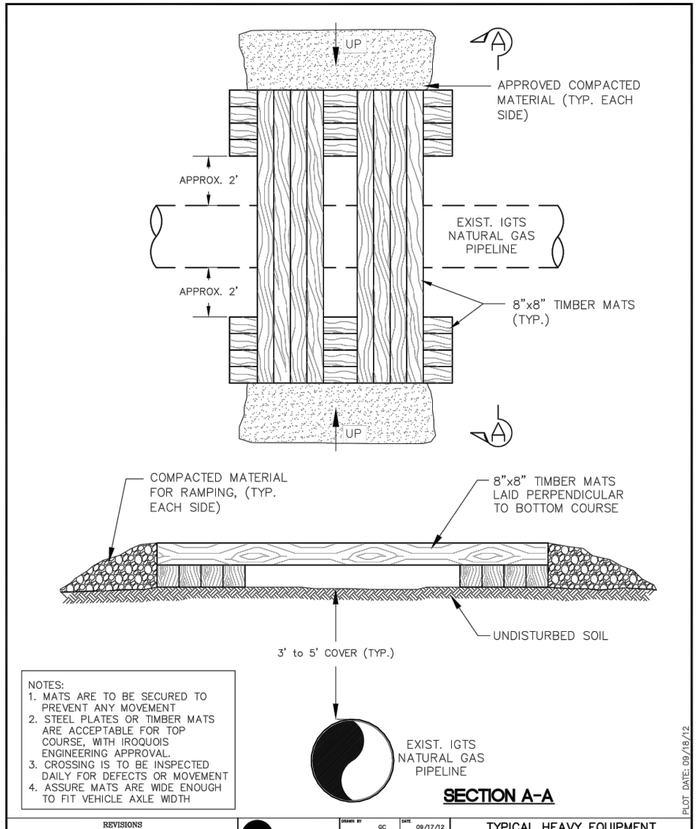
CRICKET VALLEY ENERGY TRANSMISSION

GAS LINE PROTECTION FOR CHG&E FACILITIES – SHEET 2

PROJ. NO.	2015-185
DWG. NO:	2015-185-D-07
SCALE:	
DRN. BY:	MJG
CHK'D. BY:	PGC
APP. BY:	PGC
DATE:	1/23/2017

STATE OF NEW YORK · PAUL G. CASS · LICENSED PROFESSIONAL ENGINEER · 086902 · 1/23/17



NOTES:
1. MATS ARE TO BE SECURED TO PREVENT ANY MOVEMENT
2. STEEL PLATES OR TIMBER MATS ARE ACCEPTABLE FOR TOP COURSE, WITH IROQUOIS ENGINEERING APPROVAL.
3. CROSSING IS TO BE INSPECTED DAILY FOR DEFECTS OR MOVEMENT
4. ASSURE MATS ARE WIDE ENOUGH TO FIT VEHICLE AXLE WIDTH

SECTION A-A

PLOT DATE: 09/18/12

REVISIONS						
REV.	DATE	DESCRIPTION	BY	CHK'D	ENG	APP

Iroquois GAS TRANSMISSION SYSTEM
IROQUOIS PIPELINE OPERATING COMPANY, OPERATOR
APPROVED BY: Michael Kirsh
MANAGER OF ENGINEERING

DRAWN BY	CC	DATE 09/17/12
CHECKED BY	JWD	DATE 09/18/12
DRAFT. SUP.		DATE
ENGR.		DATE
APP'D		VALVE SECTION MLV # – MLV #
FILE NO.	S-ENG-A-CON-001	SCALE NONE

TYPICAL HEAVY EQUIPMENT PIPELINE CROSSING

DWG. NO.: S-ENG-A-CON-001 REVISION 0

Iroquois GAS TRANSMISSION SYSTEM
IROQUOIS PIPELINE OPERATING COMPANY, OPERATOR

AGREEMENT ON CROSSING IROQUOIS PIPELINE

1. THE ACCESS ROAD CROSSINGS MAY BE DONE WITHOUT ANY CUTS TO THE EXISTING GRADE.
2. ANY PROPOSED ADDITIONS OF GRADE, OTHER THAN TEMPORARY BRIDGES DESCRIBED AS FOLLOWS, MUST BE APPROVED IN ADVANCE IN WRITING BY IROQUOIS.
3. THE CROSSINGS MUST BE TEMPORARILY BRIDGED AS SHOWN ON THE DRAWING SUBMITTED BY IROQUOIS, TITLED S-ENG-A-CON-001 'TYPICAL HEAVY EQUIPMENT PIPELINE CROSSING'. UPON COMPLETION OF THE CONSTRUCTION PHASE THE TEMPORARY BRIDGING WILL BE REMOVED BY YOU AND THE EASEMENT SHALL BE RETURNED TO ACCEPTABLE CONDITION AS PER IROQUOIS INSPECTORS. ALL DISTURBED AREAS WITHIN THE EASEMENT SHALL BE SEEDED AND MULCHED TO THE ACCEPTANCE OF IROQUOIS.
4. THE CROSSINGS MUST BE MAINTAINED BY YOU AT OUR REQUEST FOR THE DURATION OF THEIR INTENDED USE.
5. IF THE CONDITION OR USE OF REQUESTOR'S PROPOSED WORK UNREASONABLY INTERFERES WITH THE OPERATION, MAINTENANCE OR UPGRADES OF IROQUOIS PIPELINE, IROQUOIS MAY REQUIRE REQUESTOR, AT REQUESTORS' EXPENSE PROMPTLY TO MOVE, ALTER, OR REPAIR THE TEMPORARY CROSSINGS AND WORK LOCATIONS WITHIN IROQUOIS EXISTING RIGHT OF WAY."
6. THE WORK LOCATIONS SHALL BE, AS DESCRIBED AND INDICATED ON THE DRAWINGS, TEN FEET (10') FROM THE CENTERLINE OF THE PIPELINE. THE LOCATIONS SHALL BE TEMPORARILY FENCED AND IDENTIFIED WITH IROQUOIS PERMANENT MARKERS FOR THE DURATION OF THE WORK.
7. THE TEMPORARY CROSSING BOUNDARIES MUST BE DEFINED WITH TEMPORARY FENCING TO KEEP EQUIPMENT OFF THE EASEMENT DURING CONSTRUCTION ACTIVITIES. THE CROSSINGS WILL BE IDENTIFIED WITH IROQUOIS PERMANENT MARKERS.
8. NO STAGING OF EQUIPMENT OR MATERIALS WHATSOEVER WILL BE ALLOWED ON IROQUOIS EASEMENT.
9. IROQUOIS REQUIRES 48 HOURS' NOTICE PRIOR TO ANY ACTIVITIES WITHIN THE EASEMENT OTHER THAN THE AGREED UPON WORK LOCATIONS AND CROSSINGS THAT ARE IN THE DEFINED AREAS APPROVED THROUGH THE AGREEMENT.
10. CRICKET VALLEY ENERGY TRANSMISSION, ITS SUCCESSORS AND ASSIGNS, HOLDS IROQUOIS HARMLESS FROM ALL CLAIMS, ACTIONS AND CAUSES OF ACTION RESULTING FROM ANY WORK CONDUCTED BY CRICKET VALLEY ENERGY TRANSMISSION ON IROQUOIS' GAS PIPELINE RIGHT OF WAY.
11. ANY OTHER ENCROACHMENTS NOT SHOWN ON THE DRAWINGS OR NOT PREVIOUSLY SUBMITTED TO IROQUOIS WILL REQUIRE PRIOR APPROVALS FROM IROQUOIS.
12. DIGSAFE, 811 CALLS MUST BE MADE PRIOR TO ANY EXCAVATION ACTIVITIES.
13. A PRE JOB MEETING ON SITE WITH THE REQUESTOR, ITS CONTRACTORS AND IROQUOIS IS REQUIRED PRIOR TO COMMENCEMENT OF ANY CONSTRUCTION ACTIVITY. ANY AGREEMENTS, MODIFICATIONS, ADDITIONS OR CONFLICTS TO THIS AGREEMENT, MADE DURING THE PRE-JOB SHALL BE DOCUMENTED BY IROQUOIS AND SUBMITTED TO THE REQUESTOR AND ITS CONTRACTOR FOR AGREEMENT AND SIGNATURE. A COPY OF WHICH SHALL BE FILLED WITH THE AGREEMENT AFTER EXECUTED BY ALL PARTIES.
14. REQUESTOR SHALL PERFORM STUDIES ON THE EFFECT OF PROPOSED FACILITIES ON IROQUOIS PIPELINE INCLUDING INDUCED AC.
15. CVET SHALL REIMBURSE IROQUOIS COSTS TO ENGAGE ENGINEERING CONSULTANTS TO REVIEW STUDIES IDENTIFIED ABOVE AND FOR ANY MODIFICATIONS CVET REQUESTS IROQUOIS PERFORM OR THAT IROQUOIS DETERMINES MUST BE PERFORMED ON IROQUOIS FACILITIES. CVET SHALL ALSO REIMBURSE IROQUOIS FOR MODIFICATIONS TO THEIR FACILITIES AS REQUIRED BY CVET CONSTRUCTION OR FACILITIES INSTALLED.
16. THIS AGREEMENT MUST BE ASSIGNED BY CVET TO ANY SUCCESSORS OR ANY PARTIES ASSUMING OWNERSHIP OR CONTROL OF THE CONSTRUCTED FACILITIES.

NO ACTIVITY WHATSOEVER MAY BE PERFORMED ON IROQUOIS' EASEMENT WITHOUT AN IROQUOIS REPRESENTATIVE ON SITE. AN APPOINTMENT WITH A REPRESENTATIVE CAN BE MADE BY CONTACTING THE NEAREST IROQUOIS OFFICE 48 HOURS IN ADVANCE.

IROQUOIS FURTHER ADVISES YOU THAT THERE MAY BE OTHER PERMITS OR APPROVALS REQUIRED BY OTHER JURISDICTIONS BEFORE YOU PROCEED WITH YOUR PROPOSED WORK.

IROQUOIS HAS NOT RESEARCHED NOR MADE ANY INQUIRES AS TO THESE POSSIBLE REQUIREMENTS AND FURTHER SHALL NOT BE LIABLE IN ANY MANNER SHOULD YOU FAIL TO ACQUIRE ANY ADDITIONAL REQUIRED AUTHORIZATION.

NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP

DiGioia Gray & Associates
DiGioia, Gray & Associates, LLC
570 Beatty Road
Monroeville, PA 15146
Phone: (412) 372-4500 Fax: (412) 372-1972

CRICKET VALLEY ENERGY TRANSMISSION

GAS LINE PROTECTION FOR IROQUOIS GAS FACILITIES

PROJ. NO.	2015-185
DWG. NO.:	2015-185-D-08
SCALE:	
DRN. BY: MJG	CHK'D. BY: PGC
APP. BY: PGC	DATE: 1/23/2017

STATE OF NEW YORK * PAUL G. CASS * 086902 * 1/23/17 * LICENSED PROFESSIONAL ENGINEER

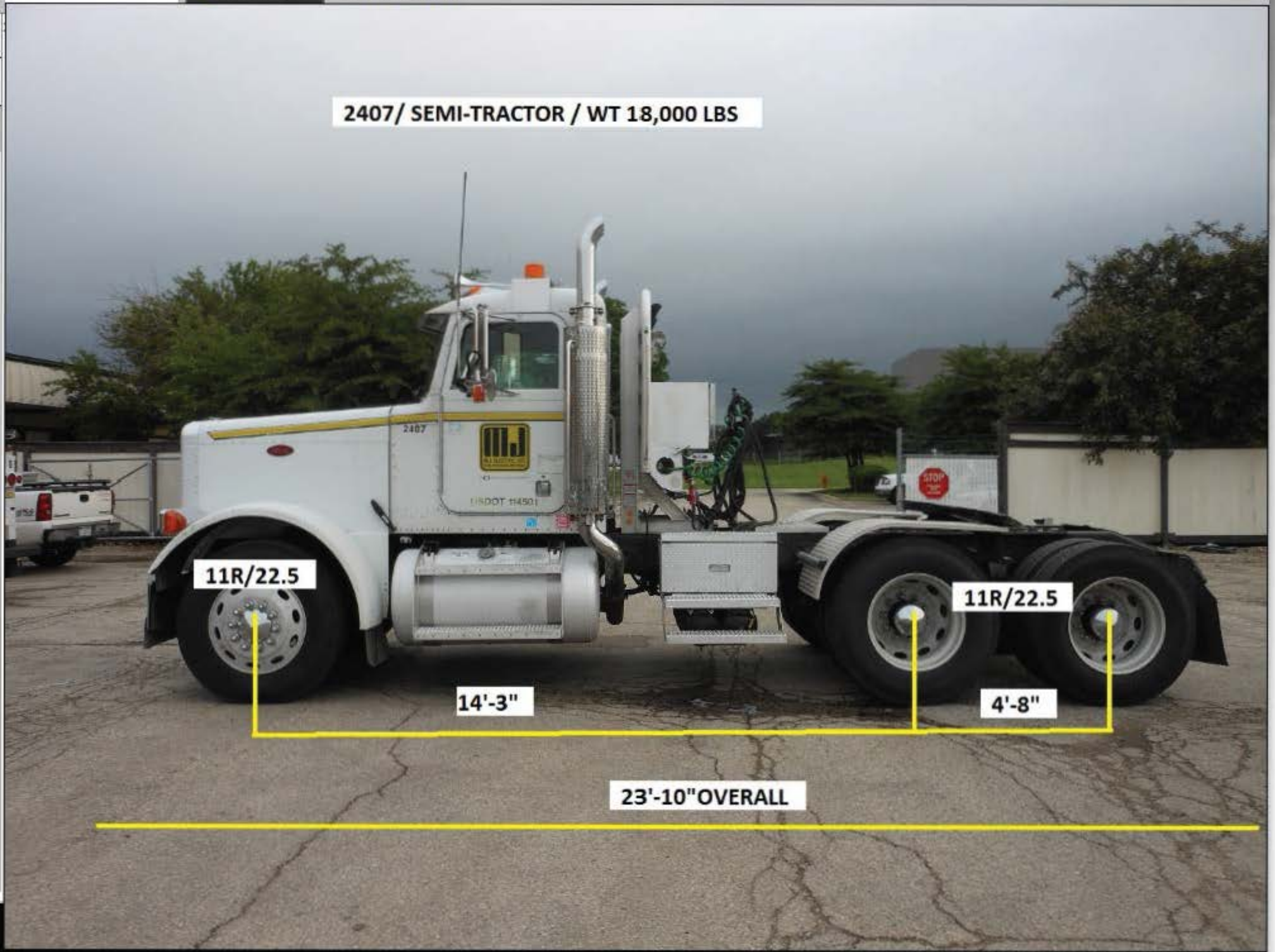
EQUIPMENT AND VEHICLE TURNING RADIUS

REFERENCES AND CALCULATIONS

4174/EXTENDABLE POLE TRLR /WT 17,000LBS/HT 64"/W 95"/ L 55' TO 83'
255/75R/22.5
4'-3"
55' TO 83' OVERALL

3392/AH 100/ALTEC 100' BUCKET ON A GT3000 TRACK BOMBER W/ROTATING BASKET/WT 60,000LBS/HT 12' W 10'-2"/L-40'/18'-9"TRACK LENGTH
TRACK LENGTH 18'-9"

MJ15059/AC38127S/38T ALTEC CRANE W/127' OF MAIN BOOM ON A PINOTH TRACK BOMBER/ WT 76,000LBS L 33'/ W 11'-6"/HT 12'-1"


2407/ SEMI-TRACTOR / WT 18,000 LBS
11R/22.5
11R/22.5
14'-3"
4'-8"
23'-10"OVERALL

November 30, 2016
DiGioia Gray & Associates

Vehicle Turning Radius and Access Road Widening in Corners

Access Roads – General condition and site specific conditions were evaluated

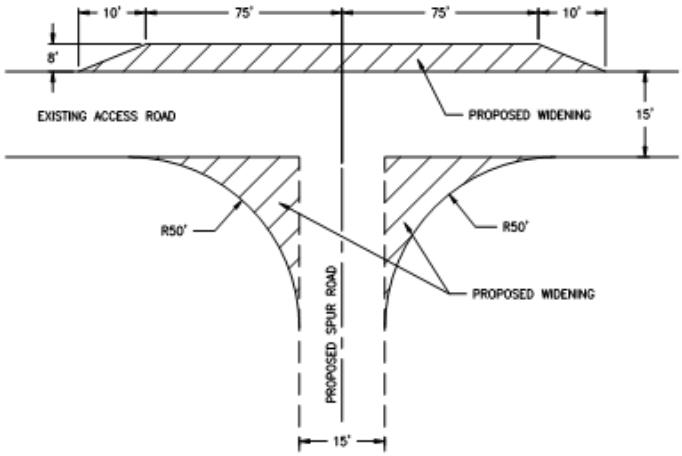
1. General cases of a proposed 15' wide spur road intersecting an existing 15' wide access road to determine the inside curve and widening at the turn for the access road.
2. Turning at specific sites: CV-01, CV-02, CV-20, CV-25, CV-32 & CV-37

Investigated 3 Vehicle Types

1. ASHTO SU-40 – Single Unit truck 40 ft long and 8.5 ft wide representing a large line truck
2. A Wb-67 Double steering tractor trailer representing a 73 ft long tractor trailer, the trailer is 43.5 ft long for carrying up to 55 ft long pole sections (a 10' overhang). The trailer has rear end steering. For this analysis, we assumed that the rear wheel steering is linked 1:1 with the front-end steering. (The photo of the trailer did not reveal a seating location for a second driver.)
3. Tracked bucket trucks and tracked cranes. Because they are tracked and one track can turn independently of the other it is assumed that they can easily turn tight corners with minimal radius widening. No modeling was conducted.

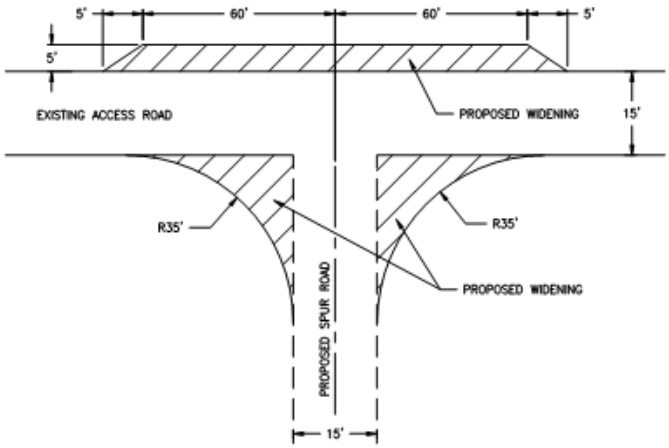
Results

General configuration for Wb-67 double steering tractor trailer or any other construction equipment – Widen road in the vicinity of the intersection by 8 feet and use a 50ft inside curve radius for 90° corners or intersections.



TYPICAL ACCESS ROAD TO SPUR ROAD CORNER DETAIL 1 (N.T.S.)
(INCLUDES TRACTOR WITH STEERABLE POLE TRAILER)

General configuration for trucks or construction vehicles up to 40 feet long - Widen road in the vicinity of the intersection by 5 feet and use a 35ft inside curve radius for 90° corners or intersections.



TYPICAL ACCESS ROAD TO SPUR ROAD CORNER DETAIL 2 (N.T.S.)
(INCLUDES ALL VEHICLES UNDER 40', EXCLUDES TRACTOR WITH STEERABLE POLE TRAILER)

Site specific curves and vehicle cornering analysis see Attachments B & C

Attachments

Attachment A - References for construction vehicles proposed for this job

Attachment B – SU-40 Bucket Truck Vehicle Parameters and Turning Analyses

Attachment C - Wb67 Double Steering Tractor Trailer Parameters and Turning Analyses

Attachment A

References for construction vehicles proposed for this job

Altec CRANES

Altec AC38-127S-HL
GT4500
Hydraulic Telescopic Crane

- Prinoth GO-TRACK™ 4500 Carrier
- 38 Ton (34.5 t)
- 127 ft (38.7 m), 5-Section Boom
- 137 ft (41.8 m) Sheave Height
- 192 ft (58.5 m) Sheave Height with 55 ft (16.8 m), 2-Piece Jib
- Altec LMAP (Load Moment & Area Protection) System
- Outrigger Boom Interlock System
- Winch Drum Rotation Indicator
- Optional 800 lb (363 kg) Capacity Personnel Platform
- Optional Radio Remote Controls
- Optional Platform Test Lift Package

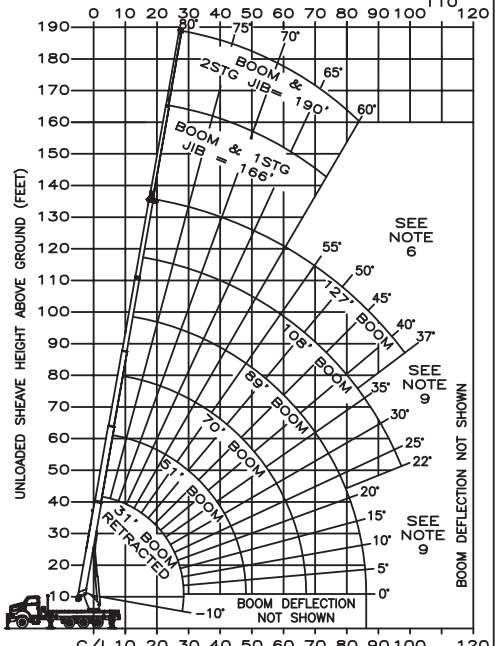
Altec CRANES

For Service: 1-877-GO-ALTEC
For Sales: 1-866-966-2969

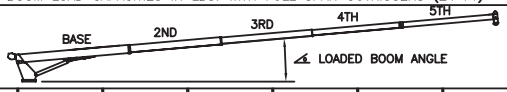
MODEL:
AC38–127S–HL–O/D–EJ

Altec Industries Inc.

RANGE DIAGRAM – FULL SPAN OUTRIGGERS



BOOM LOAD CAPACITIES IN LBS. WITH FULL SPAN OUTRIGGERS (24 FT)

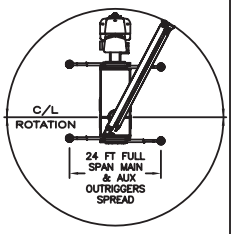


LOAD RADIUS (FT)	31 FT BOOM ∠	31 FT BOOM LBS	51 FT BOOM ∠	51 FT BOOM LBS	70 FT BOOM ∠	70 FT BOOM LBS	89 FT BOOM ∠	89 FT BOOM LBS	108 FT BOOM ∠	108 FT BOOM LBS	127 FT BOOM ∠	127 FT BOOM LBS
6	73	76000										
8	69	62600										
10	65	54000	76	35000								
12	60	47500	74	33000								
15	56	39500	70	31500	76	30000						
20	45	28000	64	26450	72	23000	77	17000				
25	28	21000	57	21000	68	20500	73	16500	77	14000		
30			50	17250	63	15650	70	14400	75	12000	77	8400
35			42	13800	58	13700	66	11850	72	10500	75	7900
40			33	11400	53	11000	62	9800	69	9150	73	7600
45			20	9200	47	9950	59	8100	66	7650	71	7350
50					41	8100	55	7200	63	6550	69	6500
55					34	6700	51	6700	60	5500	66	5600
60					26	5500	47	5700	56	5000	63	4900
65					14	4400	43	4750	53	4650	60	4200
70							37	3750	49	3850	58	3600
75							30	3050	46	3250	55	3100
80							23	2450	41	2600	52	2650
85							9	1900	37	2050	49	2200
90									32	1600	46	1800
95									26	1150	42	1300
100									22	850	37	950
105											34	800
110											30	650
	0	15500	0	6450	0	2900	0	1450	0	700	SEE NOTE 9	⇦0° BOOM CAPACITIES
												STOWED JIB LOAD DEDUCTIONS
		500		350		300		250		200	150	⇦FIXED 31FT JIB DEDUCTIONS
		1000		650		450		350		300	250	⇦TELESCOPIC JIB DEDUCTIONS

NOTE: RATINGS ABOVE THE HEAVY LINE ARE BASED ON STRUCTURAL COMPETENCE AND NOT ON MACHINE STABILITY

AREA OF OPERATION

360° CAPACITY WORKING AREA WITH FULL SPAN AND FRONT BUMPER OUTRIGGERS



DEDUCTIONS FROM RATED LOADS FOR HANDLING DEVICES

OVERHAUL BALL: 230 LBS

1–SHEAVE LOADBLOCK: 360 LBS

2–SHEAVE LOADBLOCK: 500 LBS

3–SHEAVE LOADBLOCK: 600 LBS

JIB LOAD CAPACITIES (LBS) FOR ALL BOOM LENGTHS. JIB CAPACITIES ARE FOR FULL SPAN OUTRIGGERS (24 FT)

		LOADED BOOM ANGLE	60°✳	65°	70°	75°	80°
		FIXED 31 FT JIB	1900	2600	3100	3400	3900
TELESCOPING JIB		RETRACTED 31 FT JIB	1100	2300	3100	3400	3900
		EXTENDED 55 FT JIB	700	1500	2000	2200	2500

✳DO NOT OPERATE JIBS BELOW THIS ANGLE UNLESS BOOM IS FULLY RETRACTED. SEE NOTE 6.

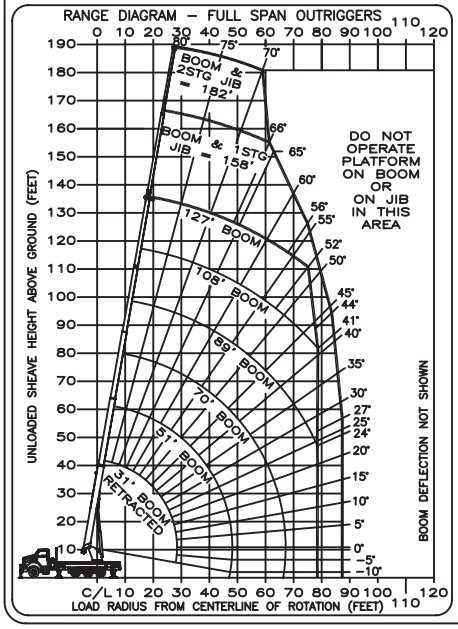
970171978 A (SHEET 1)

Altec Industries Inc.

PLATFORM OPERATION–BOOM & JIB
CRANE MODEL AC35/38–127S–EJ

NON–INSULATED PLATFORM ASSEMBLY

RANGE DIAGRAM – FULL SPAN OUTRIGGERS



ALTEC P/N 970019868

ONLY FOR USE ON ALTEC BOOM TRUCKS.

PLATFORM ASSEMBLY WEIGHT = 510 LBS.

PLATFORM LOAD CAPACITY = 800 LBS.

LOAD LINE MUST BE STOWED ON LUG WHEN PLATFORM IS INSTALLED ON BOOM TRUCK CRANE.

NO OTHER JIB TO BE ERECTED WITH THIS MPJ31 JIB STOWED.

ASME B30.23–1998 COMPLIANT

ALTEC INDUSTRIES
325 SOUTH CENTER DRIVE
DALEVILLE, VA 24085

MFG. DATE __mo/20__yr

970153057 B

Altec Industries, Inc.
325 South Center Drive
Daleville, VA 24083
1-540-992-5300

Manufacturing and Service Facilities Located Throughout the United States and Canada

www.altec.com

ALT1019263
AC38-127S-HL-3M-10/09

AH 75A/85A/100A

FOR NEW EQUIPMENT SALES, CALL
800.958.2555
TO SPEAK WITH AN ALTEC REPRESENTATIVE
or visit us online at **altec.com**

Altec®
ARTICULATING TELESCOPIC
AERIAL DEVICE

AH 75A/85A/100A

FEATURES

- Articulating and Telescoping Lower and Upper Booms
- 2100 lbs (953 kg) Total Boom-tip Capacity in Any Boom Position
- Fiberglass Platform
  - » 1300 lbs (590 kg) 40 x 60 x 42 in
  - » End Mounted Rotating Platform
- Material Handling System
  - » Up to 1500 lbs (680 kg) jib capacity in any boom position
  - » Hydraulically Extending and Articulating Jib
  - » 180 Degree Horizontal Rotation of Jib Assembly (Manual)
- Automatic Hydraulic Platform Leveling
- Single Handle Upper Control with Safety Interlock
- Category A Upgradeable - Standard Configuration Includes:
  - » Corona Ring
  - » Lower Conductive Shield
  - » Pre-wired for Leakage Monitor
- Two Hydraulic Tool Circuits at Boom Tip
- Radial Outriggers
- Outrigger Boom Interlock System
- Secondary Stowage System
- Outrigger Motion Alarm
- Back-Up Alarm

OPTIONS

- Platforms
  - » Aluminum 2100 lbs capacity 36 x 70 x 42 in
  - » Aluminum 2100 lbs capacity 36 x 96 x 42 in
  - » Fiberglass 800 lbs capacity 24 x 48 x 42 in
- Current Leakage Monitor
- Hydraulic Intensifier at Platform
- Fall Protection System
- Platform Liner
- Outrigger Pads
- Wheel Chocks

Recommended safety equipment, available through Altec Supply, include a platform liner, fall protection system, wheel chocks and outrigger pads.

LIGHTWEIGHT DESIGN

The lightweight boom design maximizes payload and increases options for chassis configuration.

PLATFORM CAPACITY

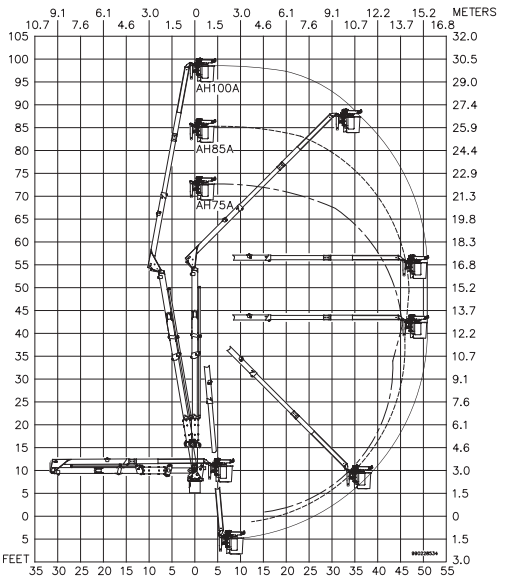
The fiberglass platform capacity holds up to 1300 lbs for increased productivity.

SPECIFICATIONS

CONFIGURATIONS

	AH75A	AH85A	AH100A
Ground to Bottom of Platform*	68.0 ft (20.7 m)	80.0 ft (24.4 m)	95.0 ft (29.0 m)
Working Height*	73.0 ft (22.3 m)	85.0 ft (26.0 m)	100.0 ft (30.5 m)
Maximum Side Reach	46.0 ft (13.7 m)	48.0 ft (14.6 m)	50.5 ft (15.4 m)
Stowed Travel Height*	12.2 ft (3.8 m)	12.5 ft (3.8 m)	12.5 ft (3.8 m)
Upper Boom Isolation	14.0 ft (4.3 m)	14.0 ft (4.3 m)	14.0 ft (4.3 m)
Lower Boom Isolation	18.0 in (457 mm)	18.0 in (457 mm)	18.0 in (457 mm)
Platform Capacity	1300 lb (590 kg)	1300 lb (590 kg)	1300 lb (590 kg)
Max. Boom Dielectric Rating	500 kV	500 kV	500 kV
Upper Boom Articulation	0° - 160°	0° - 160°	0° - 160°
Upper Boom Telescopic Extension	5.0 ft (1.5 m)	7.0 ft (2.1 m)	11.0 ft (3.4 m)
Lower Boom Articulation	0° - 105°	0° - 95°	0° - 90°
Lower Boom Telescopic Extension	8.0 ft (2.4 m)	11.0 ft (3.4 m)	14.0 ft (4.3 m)
Rotation	Continuous	Continuous	Continuous

*Based on a 40 in (1016 mm) chassis frame height in compliance with ANSI A92.2.



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4174/EXTENDABLE POLE TRLR /WT 17,000LBS/HT 64"/W 95"/ L 55' TO 83'
255/75R/22.5
4'-3"
55' TO 83' OVERALL

3392/AH 100/ALTEC 100' BUCKET ON A GT3000 TRACK BOMBER W/ROTATING BASKET/WT 60,000LBS/HT 12' W 10'-2"/L-40'/18'-9"TRACK LENGTH
TRACK LENGTH 18'-9"
vehicle pressure 2.5 psi

MJ15059/AC38127S/38T ALTEC CRANE W/127' OF MAIN BOOM ON A PINOTH TRACK BOMBER/ WT 76,000LBS L 33'/ W 11'-6"/HT 12'-1"

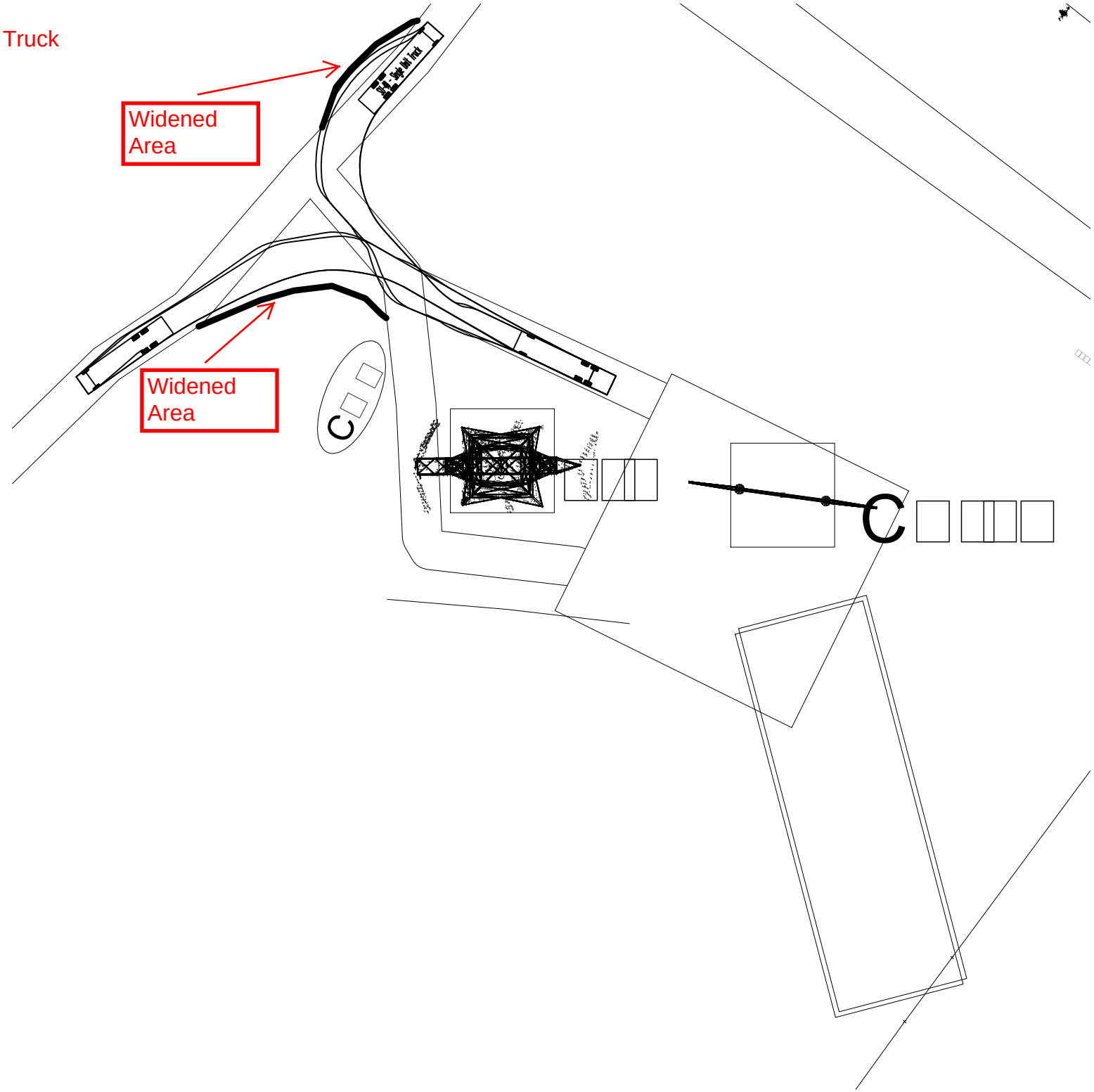
Attachment B

SU-40 Bucket Truck Vehicle Parameters and Turning Analyses

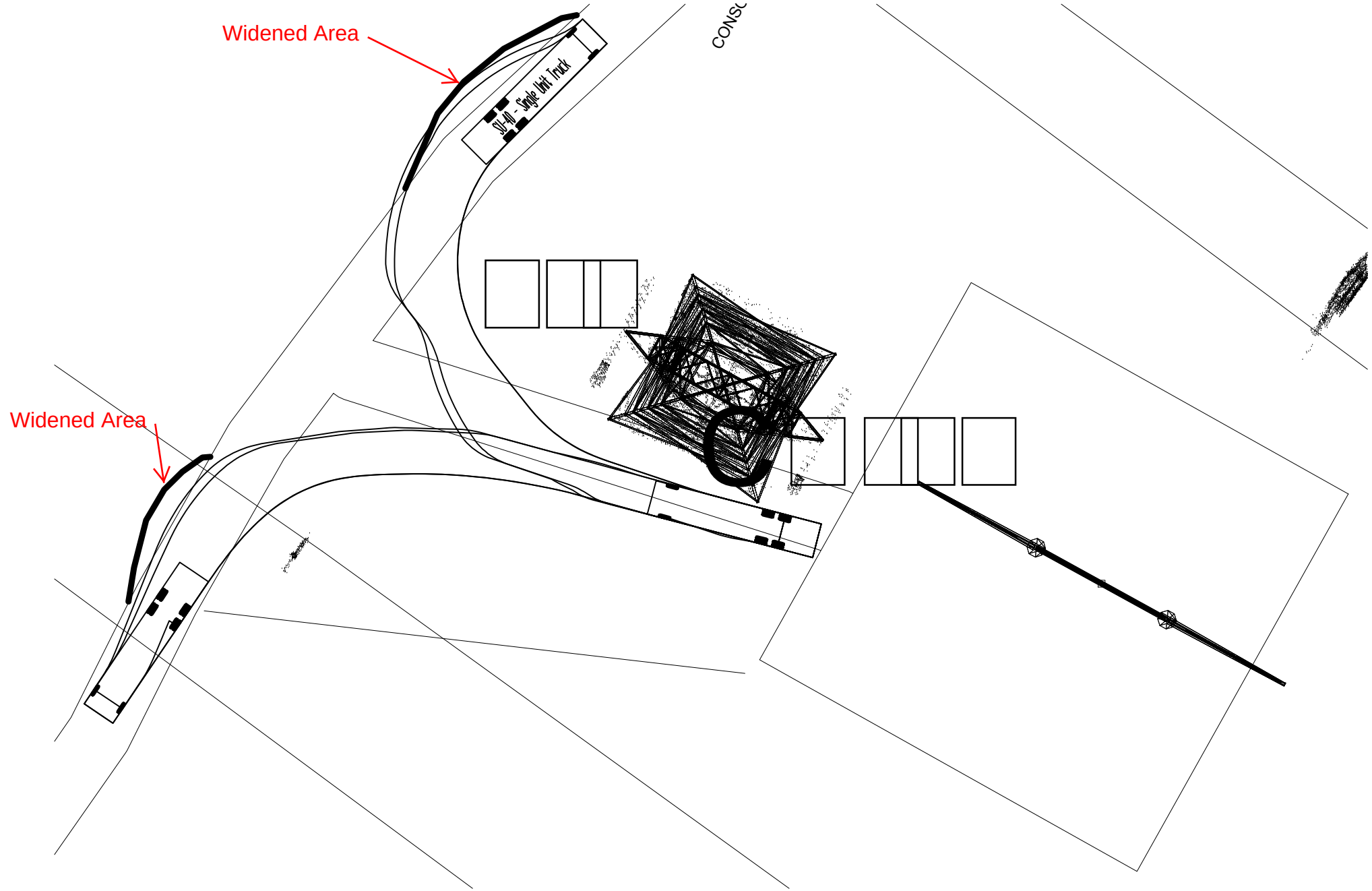
Unit Name:	**SU-40 - Single Unit Truck Tractor**
Type:	Tractor (with driver controlled steering)
Body style:	Rigid Truck (Small)
Classification	Autodesk
Source:	AASHTO handbook 2001
Description:	Design vehicle
Notes:	
Datum:	Front Primary Axle
Front Axle(s):	1 Ackerman (axles fixed, wheels turn)
Primary Front Axle Offset:	0.000ft
Effective Front Axle Offset:	0.000ft (Auto Calculated)
Maximum Wheel Angle:	31.800deg (Any Front Wheel)
Status:	Active Non Self-Steered
Track Width:	8.000ft
Total Wheels:	2 (positioned at the ends of the axle)
Tire Width:	0.800ft (Auto Calculated - proportion of Track Width)
Tire Diameter:	2.800ft (Auto Calculated - proportion of Track Width)
Rear Axle(s):	2 Fixed (All axles identical)
Primary Rear Axle Offset:	23.000ft (Innermost Axle behind Front Primary Axle)
Effective Rear Axle Offset:	25.000ft (Auto Calculated)
Maximum Wheel Angle:	Unlimited
Rear Axle Spacing:	4.000ft
Status:	Active Non Self-Steered
Track Width:	8.000ft
Total Wheels:	4 (positioned at the ends of the axle)
Tire Width:	0.800ft (Auto Calculated - proportion of Track Width)
Tire Diameter:	2.800ft (Auto Calculated - proportion of Track Width)
Steering:	Front Axle(s):
Maximum Virtual Steering Angle:	32deg
Lock-to-Lock Time (Fwd/Rev):	5.0sec / 5.0sec
Driver / Pilot	
Driver Offset Longitudinally:	-0.921ft (in front of Front Primary Axle)
Driver / Pilot Offset Laterally:	-1.969ft (Right of Centerline)
Driver Height:	7.218ft (Above ground level)
Front Coupling:	None
Rear Coupling:	None
Body outline (plan):	
Outline Type:	Rectangle
Offset (X,Y):	-4.000ft, 0.000ft
Length / Width:	39.500ft / 8.000ft

Every Effort Has Been Made To Ensure The Accuracy Of This Information
Please Check Data From Your Own Sources

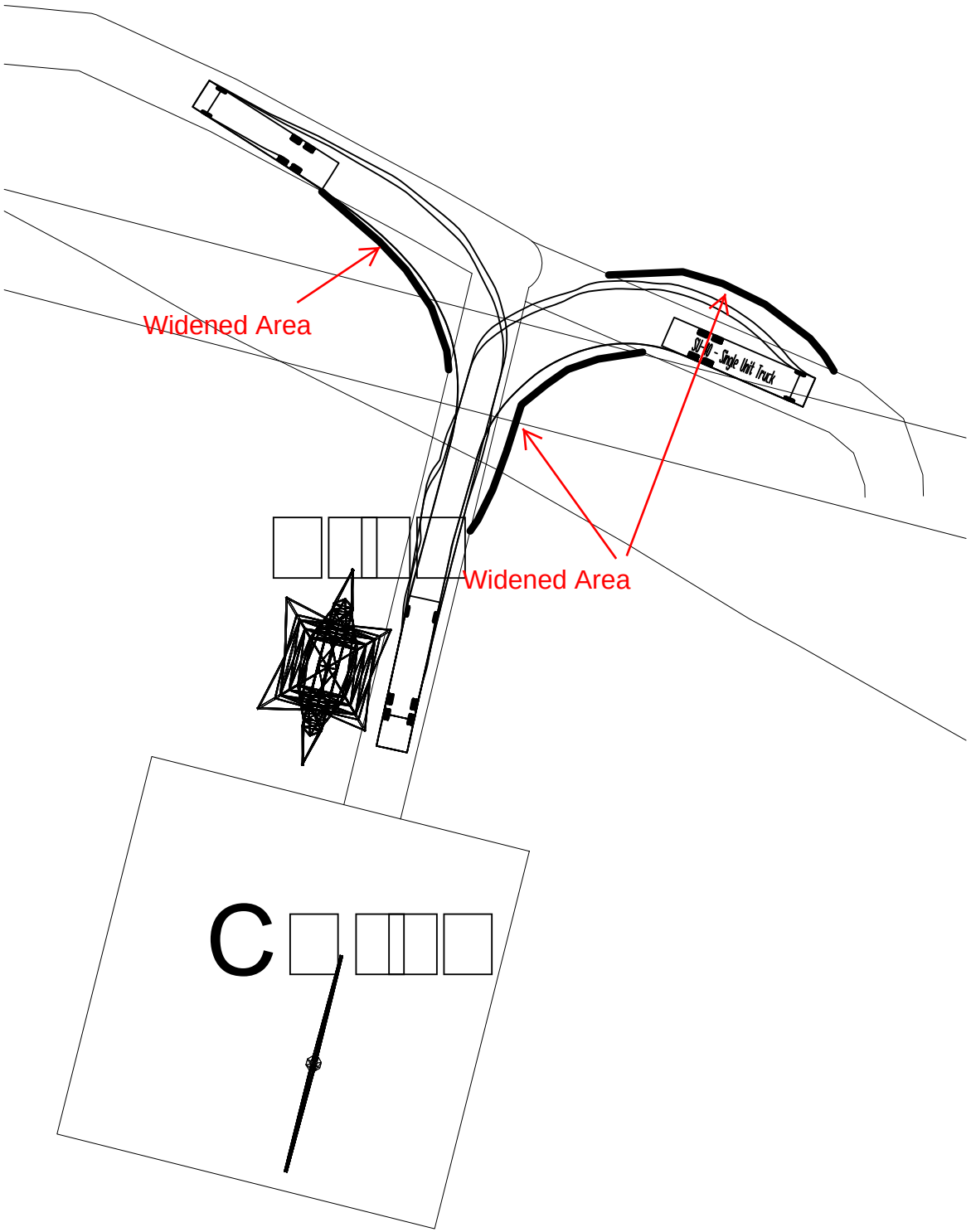
SU40 Bucket Truck



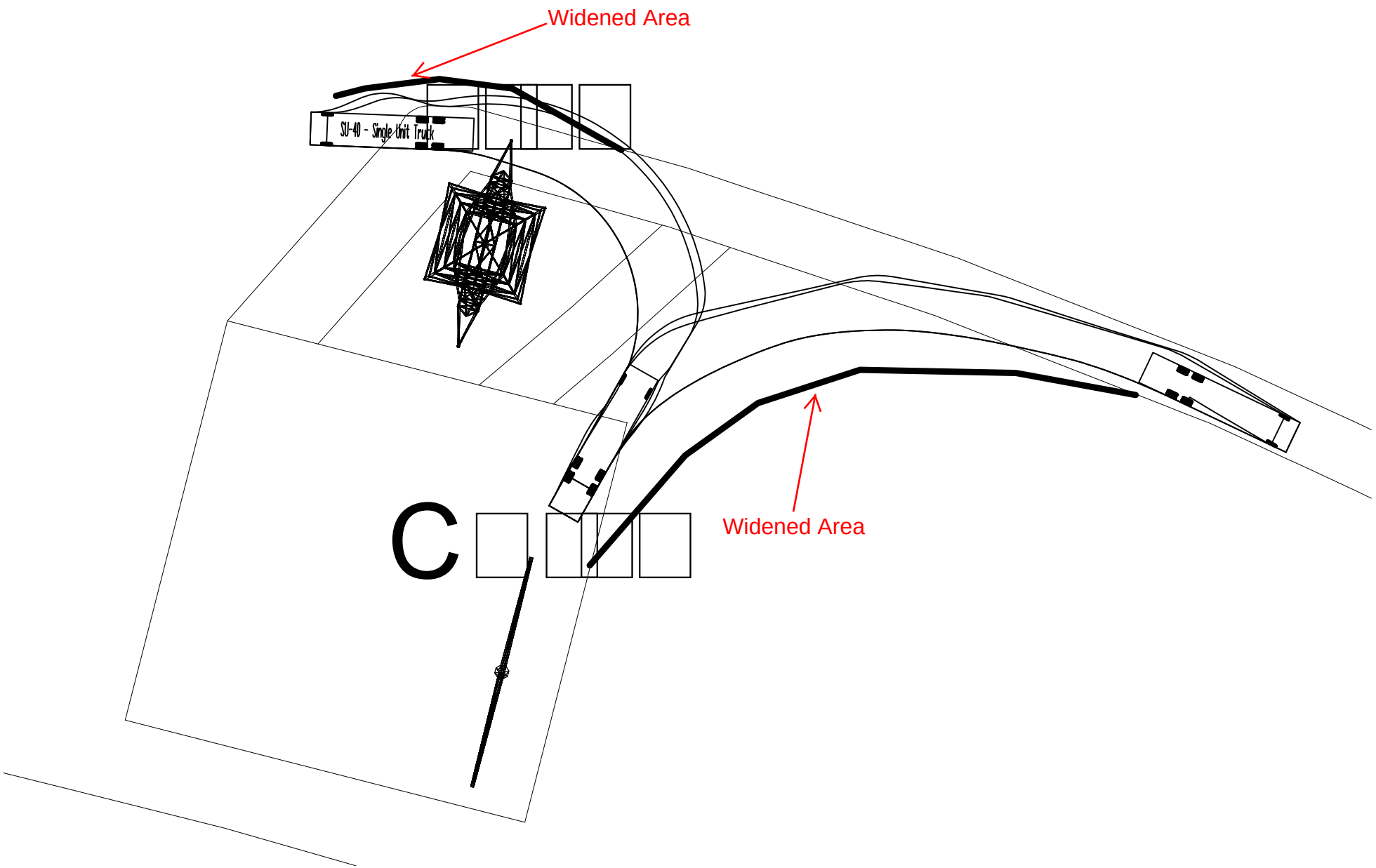
SU40 Bucket Truck



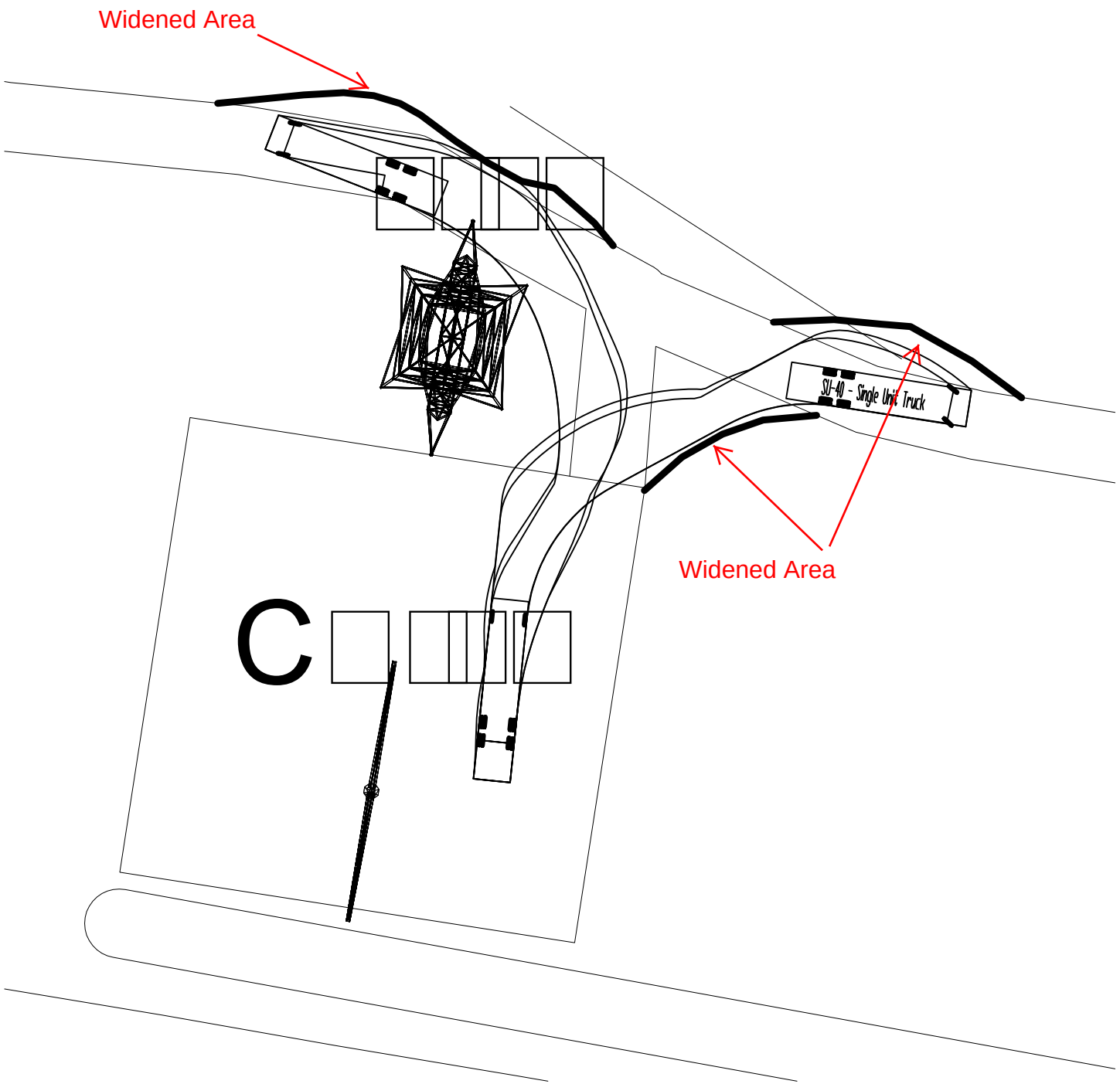
SU40 Bucket Truck



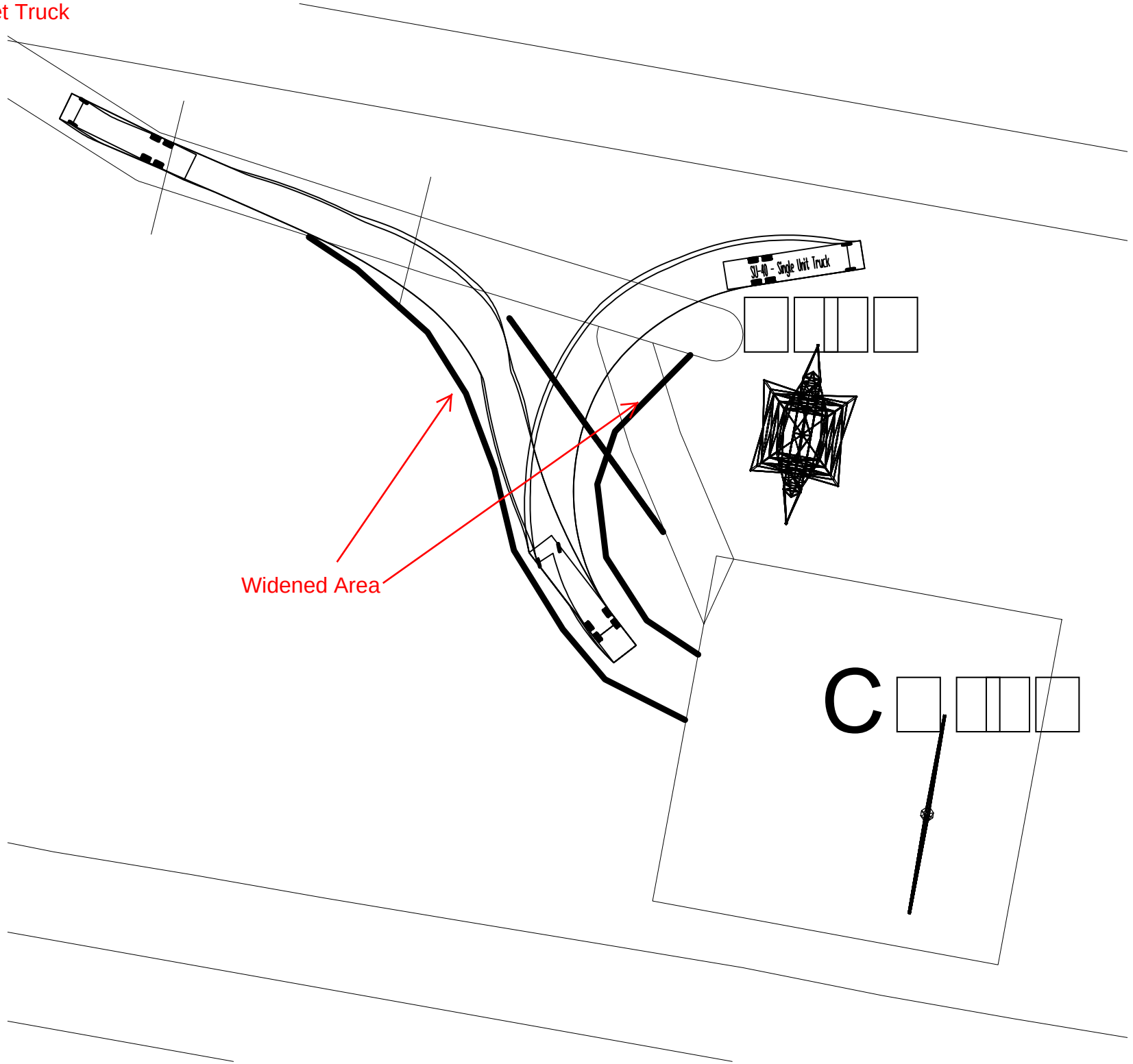
SU40 Bucket Truck



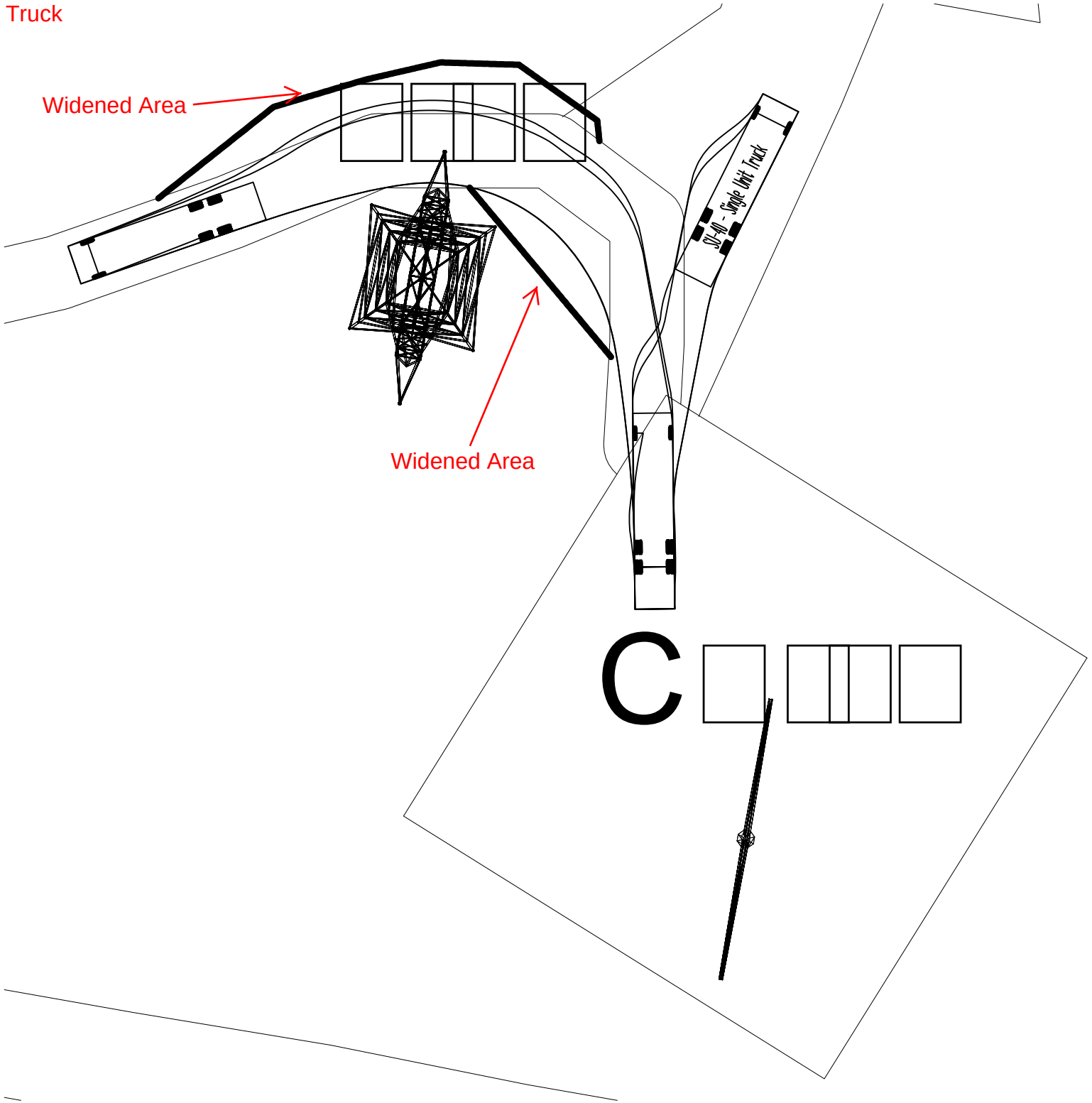
SU40 Bucket Truck



SU40 Bucket Truck



SU40 Bucket Truck

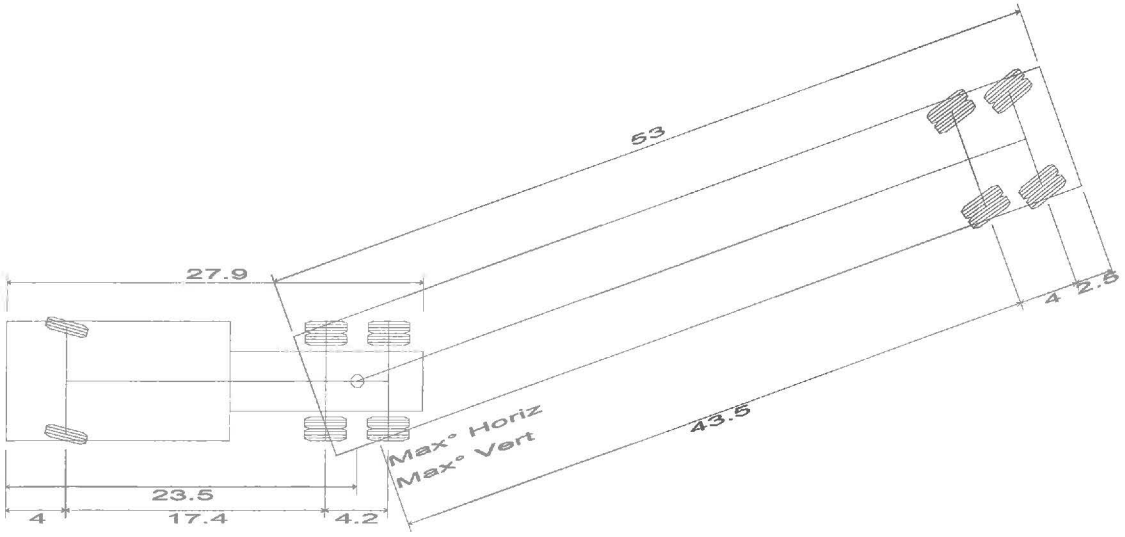


Attachment C

Wb67 Double Steering Tractor Trailer Parameters and Turning Analyses

Vehicle Tracking Vehicle Details **Ref: 100033**

Vehicle Name:	**WB-67 Double Steering**
Type:	Articulated vehicle
Category	Autodesk
Classification	Autodesk
Source:	AASHTO handbook 2011
Description:	Design vehicle
Notes:	
Unit 1 Name:	WB-67 Double Steering Tractor
Unit 2 Name:	WB-67 Double Steering Trailer 1



WB-67 Double Steering	
Overall Length	73.501ft
Overall Width	8.500ft
Overall Body Height	13.500ft
Min Body Ground Clearance	1.334ft
Max Track Width	8.500ft
Lock-to-lock time	6.00s
Max Steering Angle (Virtual)	28.40°

Every Effort Has Been Made To Ensure The Accuracy Of This Information
Please Check Data From Your Own Sources

Vehicle Tracking Vehicle Details Ref: 100033

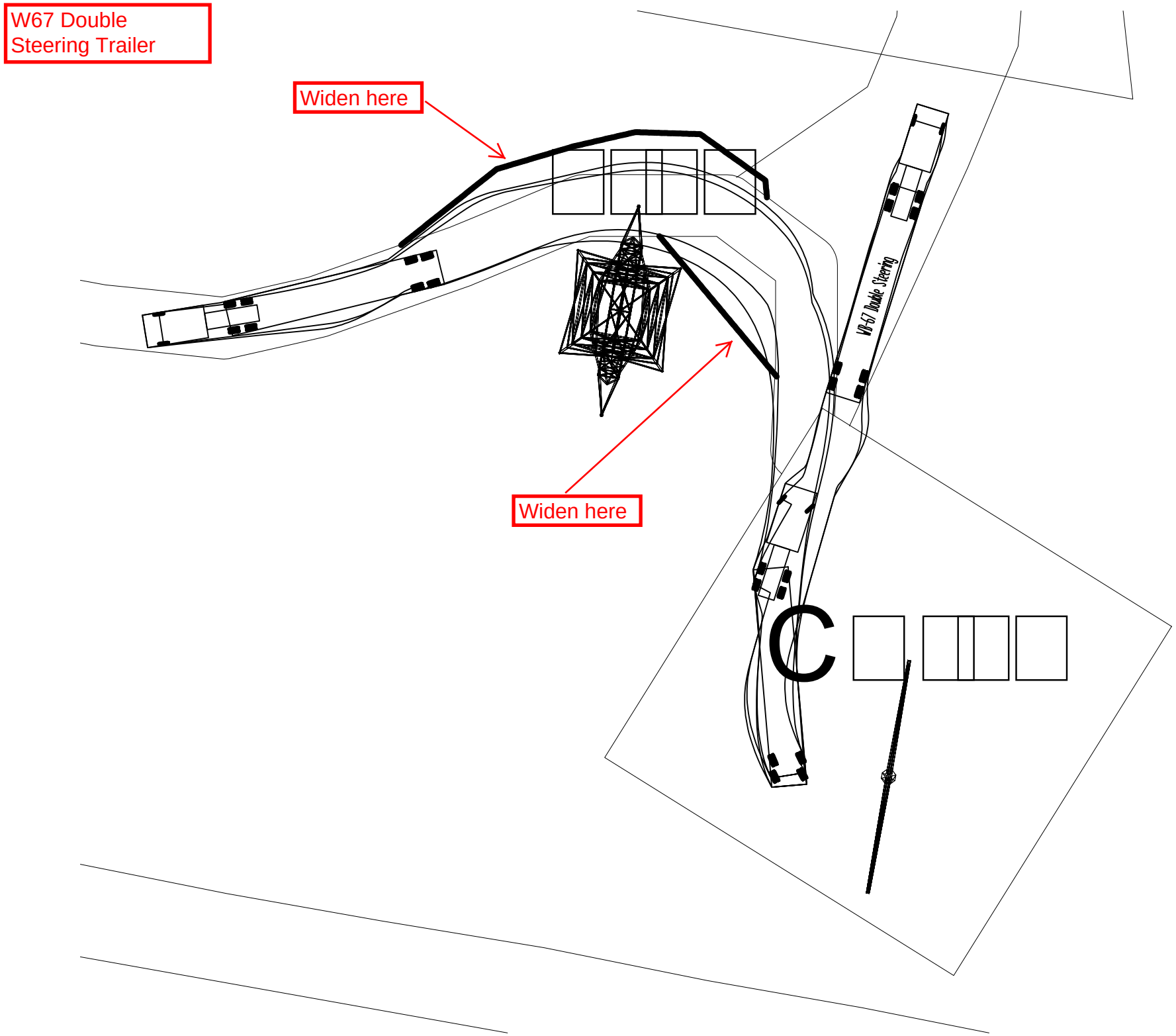
Unit Name:	**WB-67 Double Steering Tractor**
Type:	Tractor (with driver controlled steering)
Body style:	Articulated Vehicle Tractor (Large Sleeper Cab)
Classification	Autodesk
Source:	AASHTO handbook 2001
Description:	Design vehicle
Notes:	
Datum:	Front Primary Axle
Front Axle(s):	1 Ackerman (axles fixed, wheels turn)
Primary Front Axle Offset:	0.000ft
Effective Front Axle Offset:	0.000ft (Auto Calculated)
Maximum Wheel Angle:	28.400deg (Any Front Wheel)
Status:	Active Non Self-Steered
Track Width:	8.000ft
Total Wheels:	2 (positioned at the ends of the axle)
Tire Width:	0.800ft (Auto Calculated - proportion of Track Width)
Tire Diameter:	2.800ft (Auto Calculated - proportion of Track Width)
Rear Axle(s):	2 Fixed (All axles identical)
Primary Rear Axle Offset:	17.400ft (Innermost Axle behind Front Primary Axle)
Effective Rear Axle Offset:	19.500ft (Auto Calculated)
Maximum Wheel Angle:	Unlimited
Rear Axle Spacing:	4.200ft
Status:	Active Non Self-Steered
Track Width:	8.000ft
Total Wheels:	4 (positioned at the ends of the axle)
Tire Width:	0.800ft (Auto Calculated - proportion of Track Width)
Tire Diameter:	2.800ft (Auto Calculated - proportion of Track Width)
Steering:	Front Axle(s):
Maximum Virtual Steering Angle:	28deg
Lock-to-Lock Time (Fwd/Rev):	6.0sec / 6.0sec
Driver / Pilot	
Driver Offset Longitudinally:	-0.921ft (in front of Front Primary Axle)
Driver / Pilot Offset Laterally:	-1.969ft (Right of Centerline)
Driver Height:	7.382ft (Above ground level)
Front Coupling:	None
Rear Coupling:	Generic
Coupling Offset:	19.500ft (behind Front Primary Axle)
Coupling Height:	2.800ft (Auto Calculated - proportion of Tire Diameter)
Capability:	Can Tow or be Towed
Max. Horizontal Articulation Angle:	68.500deg
Max. Vertical Articulation Angle:	10.000deg
Body outline (plan):	
Outline Type:	Tractor Body

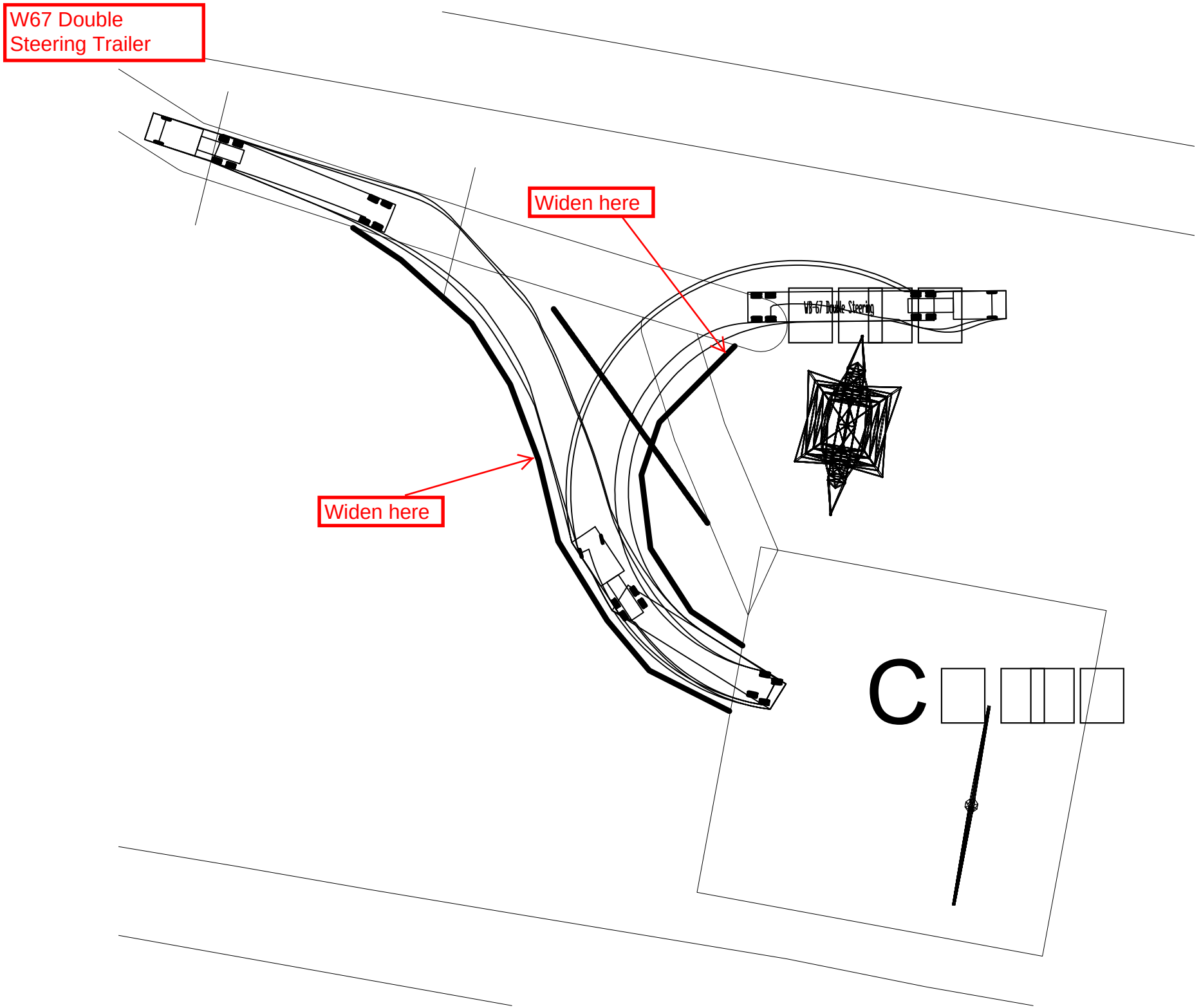
Every Effort Has Been Made To Ensure The Accuracy Of This Information
Please Check Data From Your Own Sources

Vehicle Tracking Vehicle Details **Ref: 100033**

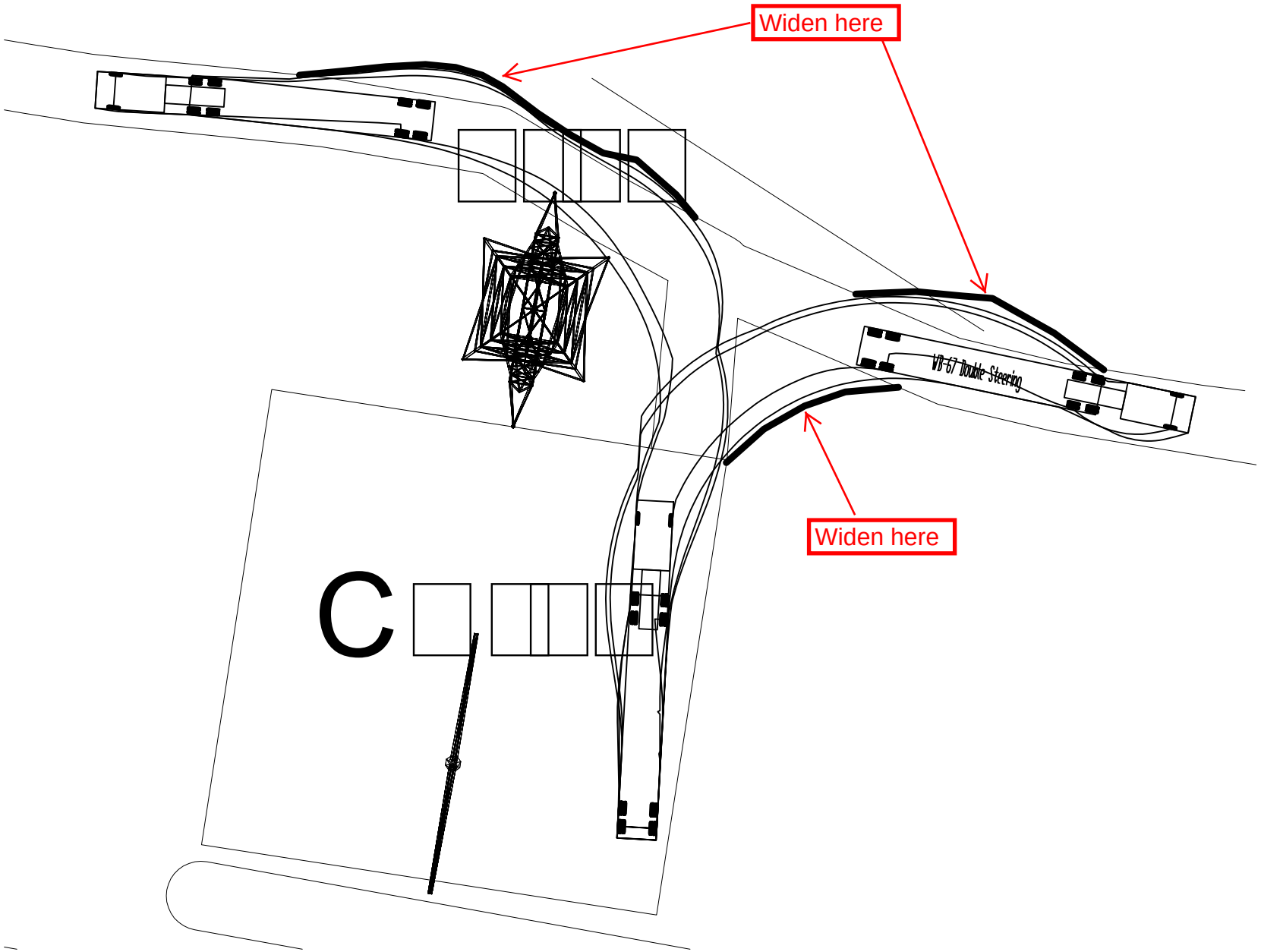
Unit Name:	**WB-67 Double Steering Trailer 1**
Type:	Trailer (no driver-controlled steering)
Body style:	Articulated Vehicle Semi-Trailer
Classification	Autodesk
Source:	AASHTO handbook 2001
Description:	Design vehicle
Notes:	
Datum:	Front Coupling
Maximum Articulation Angle:	69deg (to previous unit)
Front Axle(s):	None
Rear Axle(s):	2 Ackerman (axles fixed, wheels turn) (All axles identical)
Primary Rear Axle Offset:	43.500ft (Innermost Axle behind Front Coupling)
Effective Rear Axle Offset:	47.500ft (Auto Calculated)
Maximum Wheel Angle:	Unlimited
Rear Axle Spacing:	4.000ft
Linkage:	Rear axles linked to front axles
Basis	Angle of rear wheels
Rule 1:	Forward and reverse from 0.00deg, 100.00 based upon Tangents
Status:	Active Non Self-Steered
Track Width:	8.500ft
Total Wheels:	4 (positioned at the ends of the axle)
Tire Width:	0.850ft (Auto Calculated - proportion of Track Width)
Tire Diameter:	2.975ft (Auto Calculated - proportion of Track Width)
Front Coupling:	Generic
Coupling Offset:	0.000ft (in front of Front Coupling)
Coupling Height:	1.488ft (Auto Calculated - proportion of Tire Diameter)
Capability:	Can Tow or be Towed
Max. Horizontal Articulation Angle:	68.500deg
Max. Vertical Articulation Angle:	10.000deg
Rear Coupling:	None
Body outline (plan):	
Outline Type:	Rectangle
Offset (X,Y):	-3.000ft, 0.000ft
Length / Width:	53.000ft / 8.500ft

Every Effort Has Been Made To Ensure The Accuracy Of This Information
Please Check Data From Your Own Sources

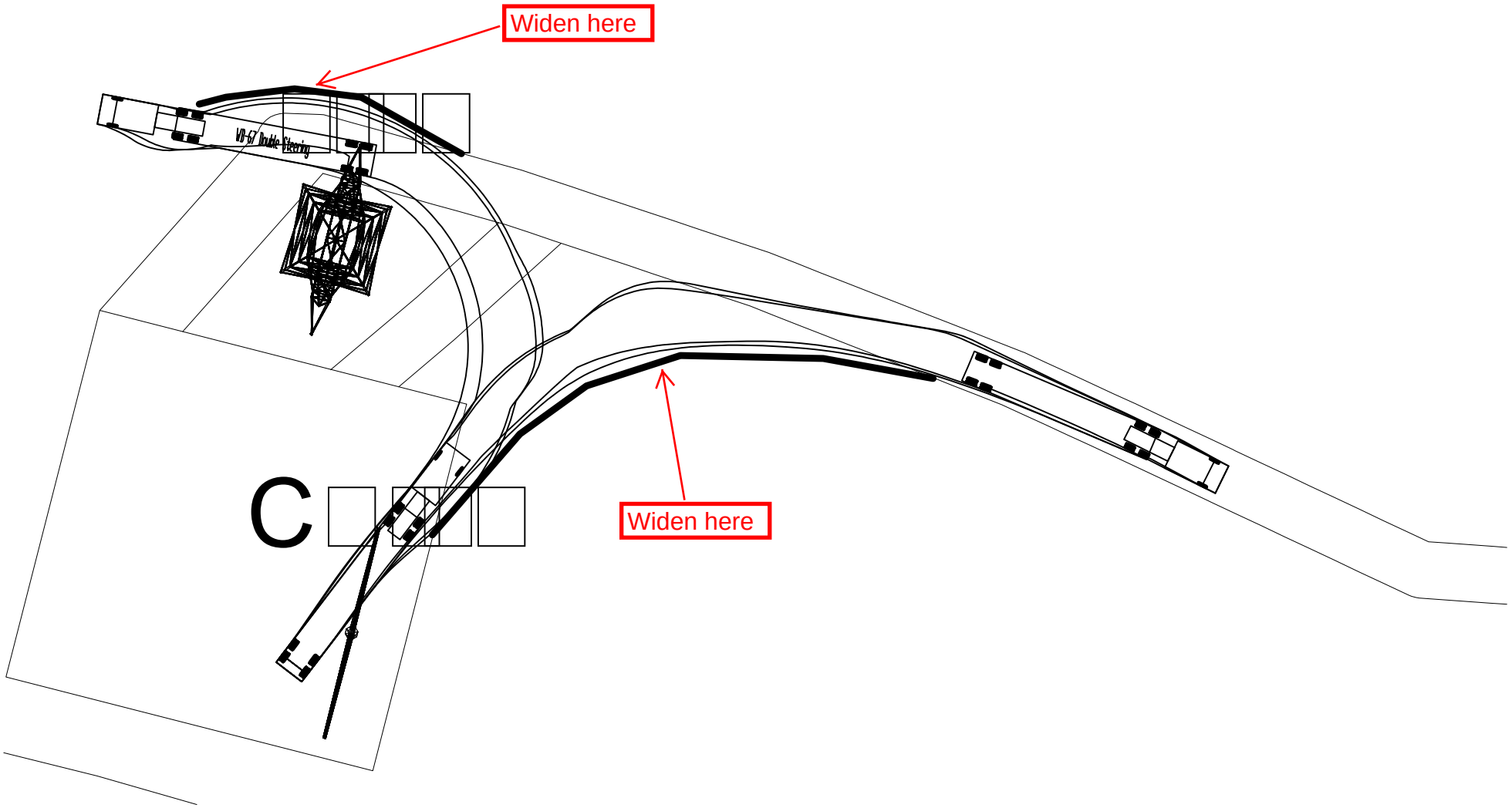

W67 Double
Steering Trailer
Widen here
Widen here
VR-67 Double Steering


W67 Double
Steering Trailer
Widen here
Widen here
W67 Double Steering

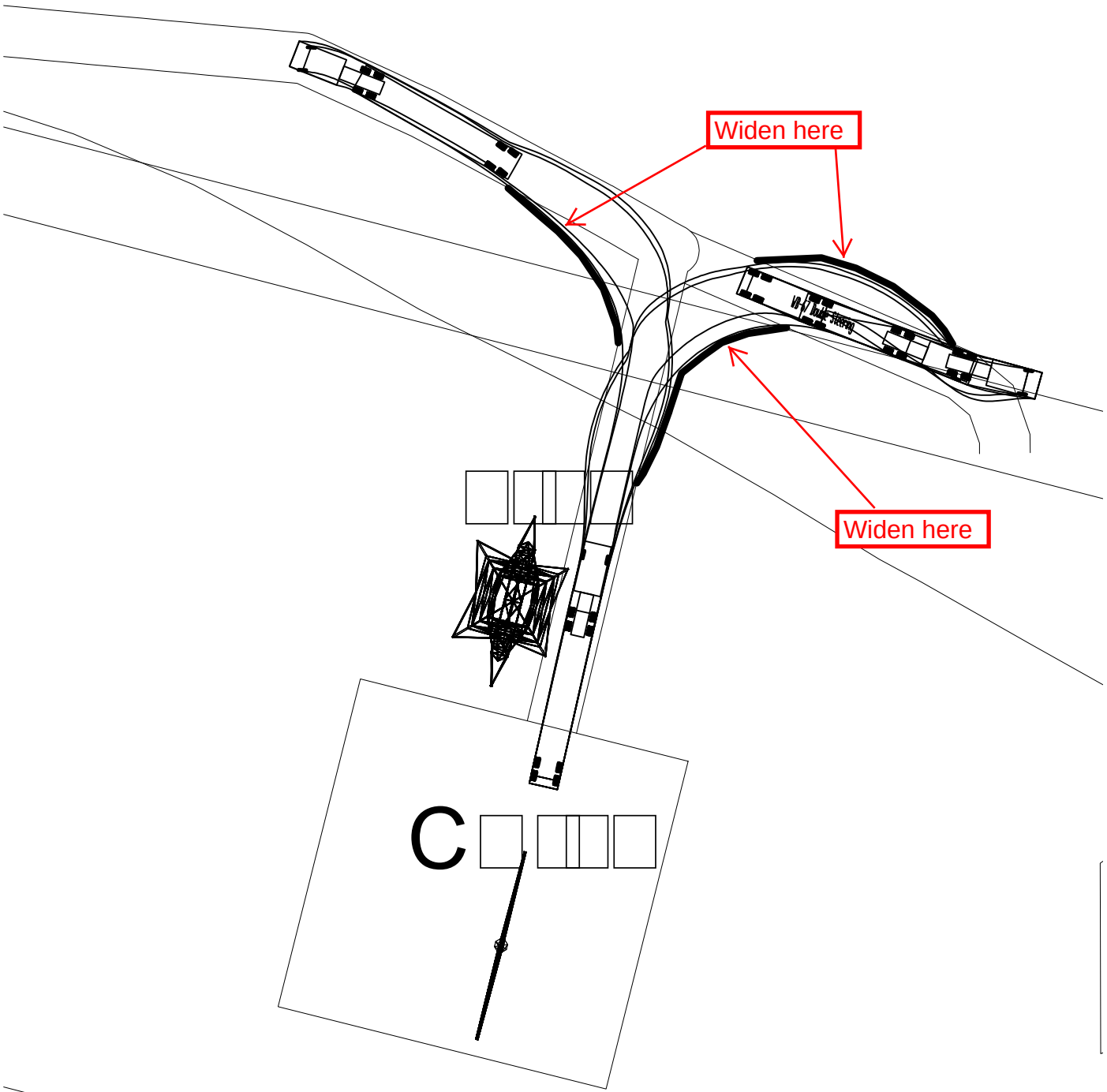
W67 Double
Steering Trailer

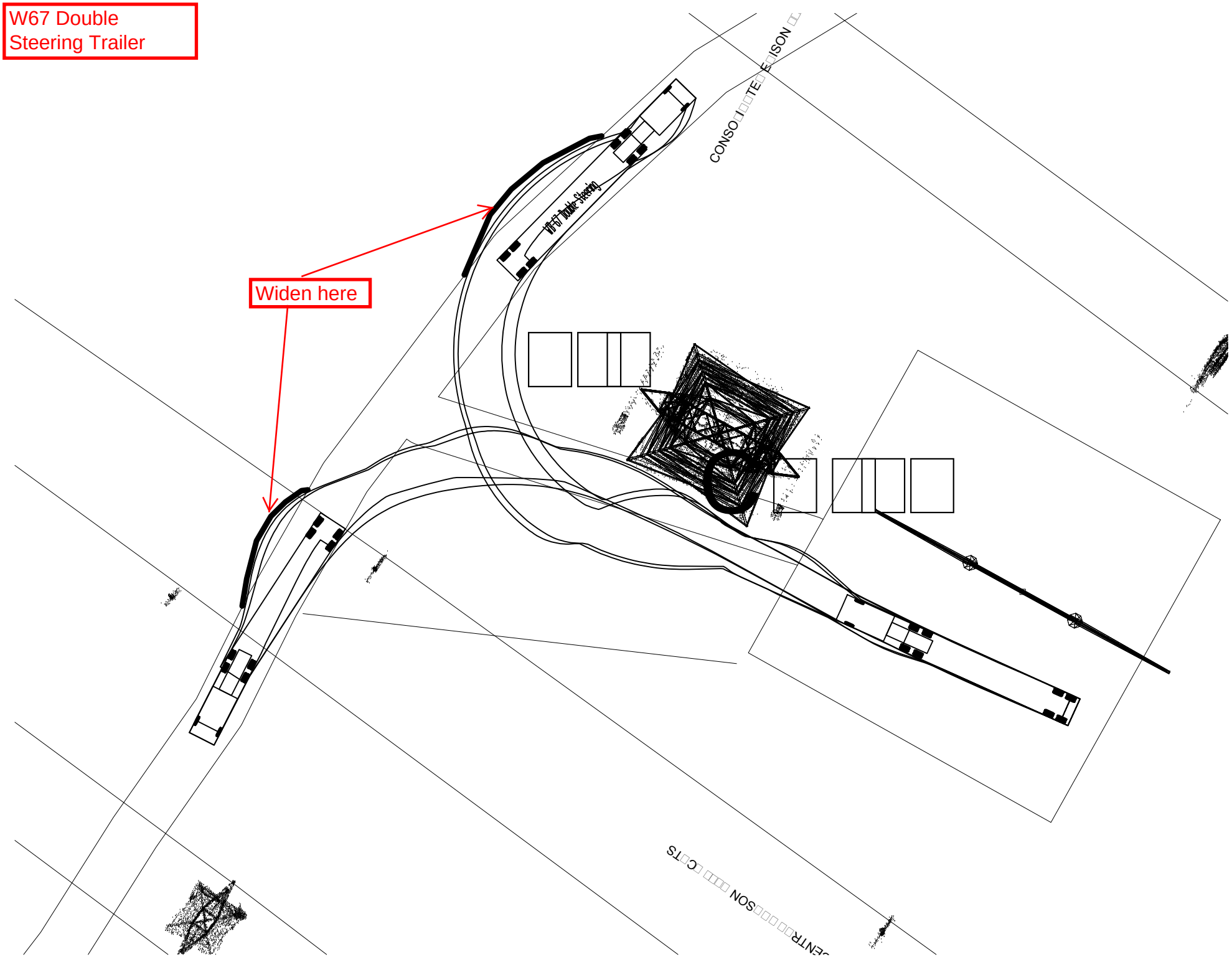


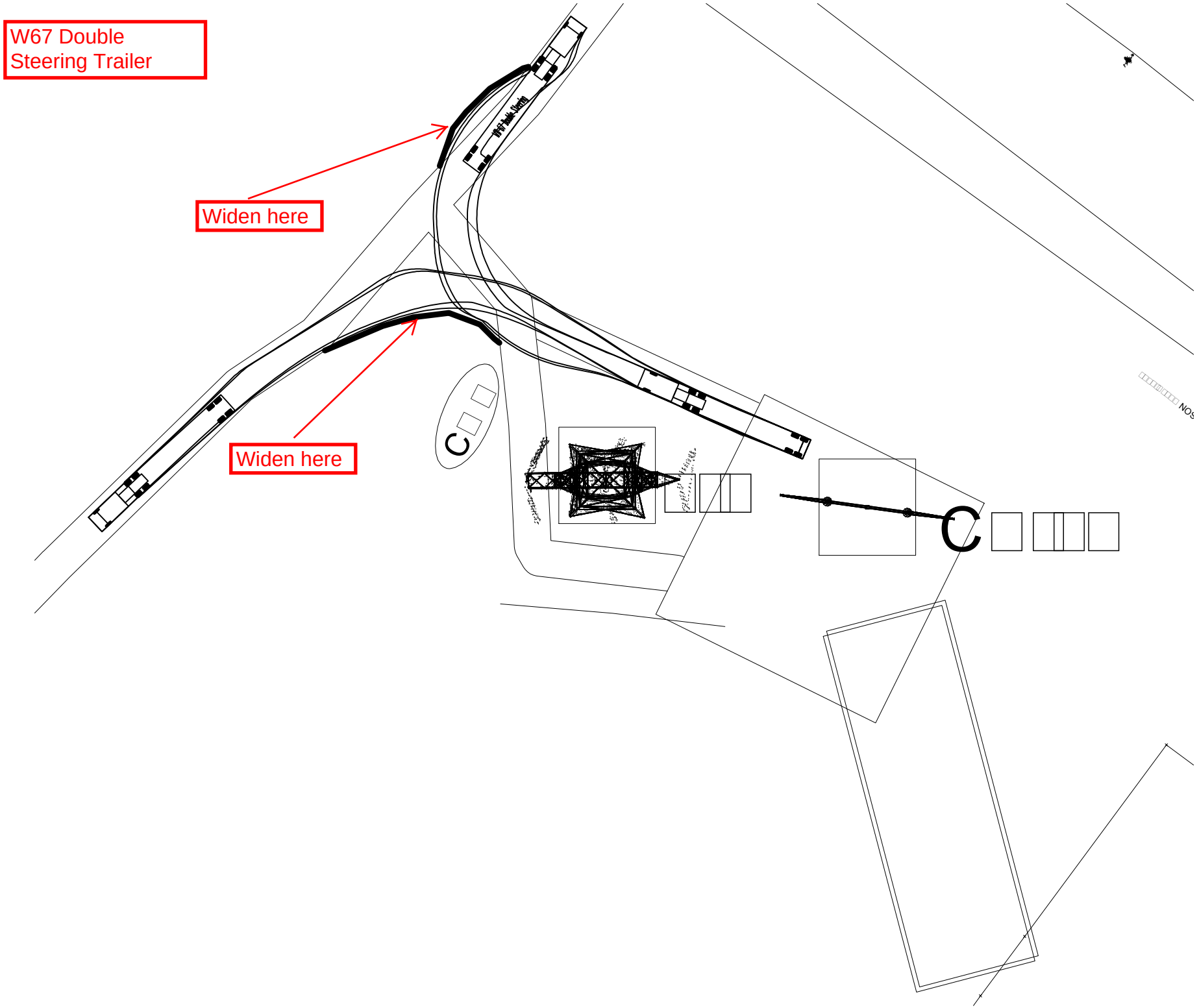
W67 Double
Steering Trailer



W67 Double
Steering Trailer




W67 Double
Steering Trailer
Widen here


W67 Double
Steering Trailer
Widen here
Widen here

Grading Plans

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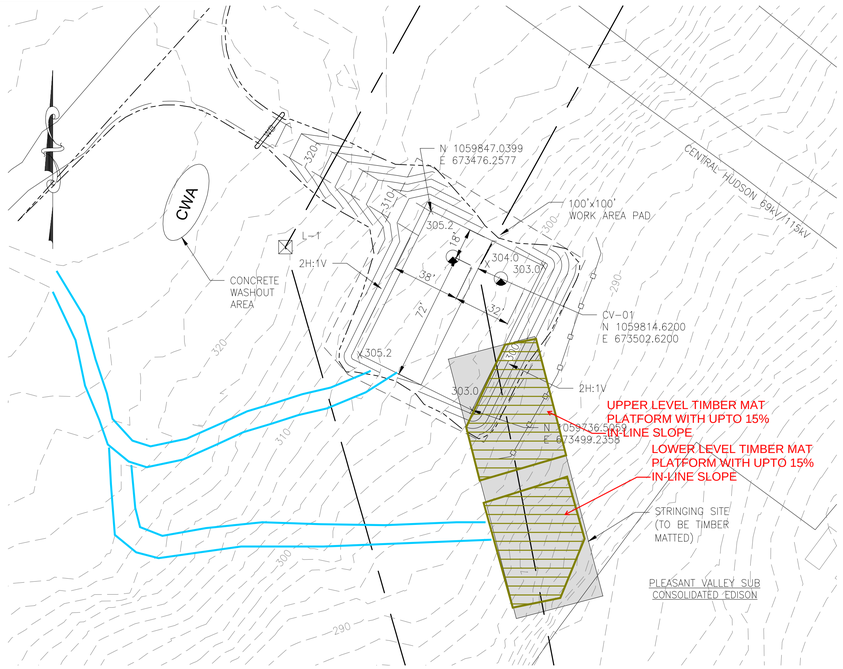
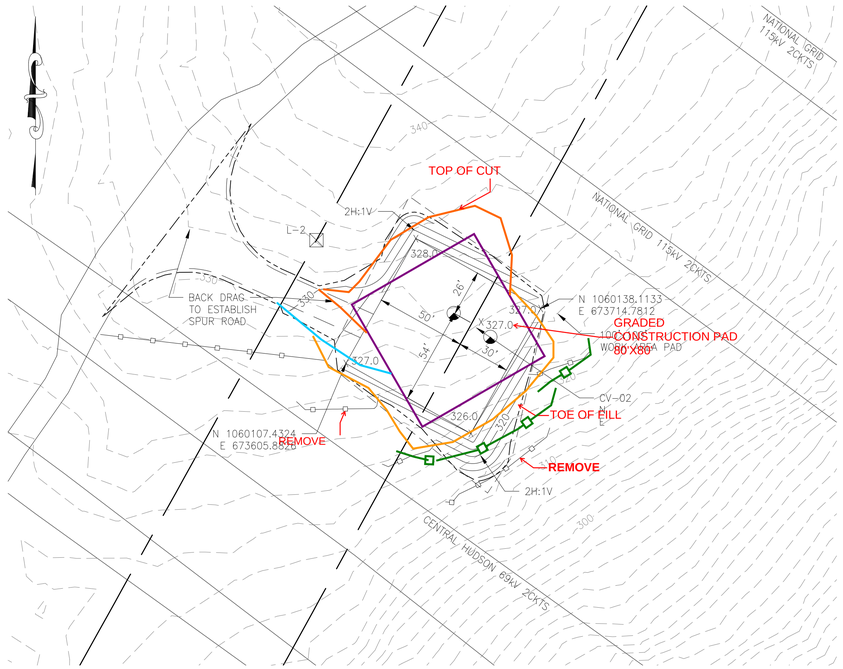
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-18



TOWER NO. CV-01



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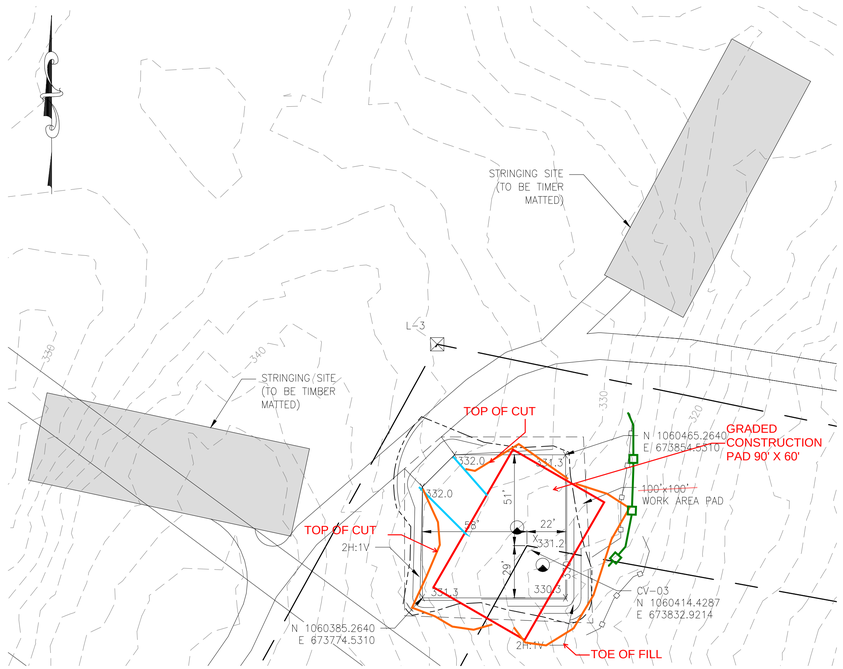

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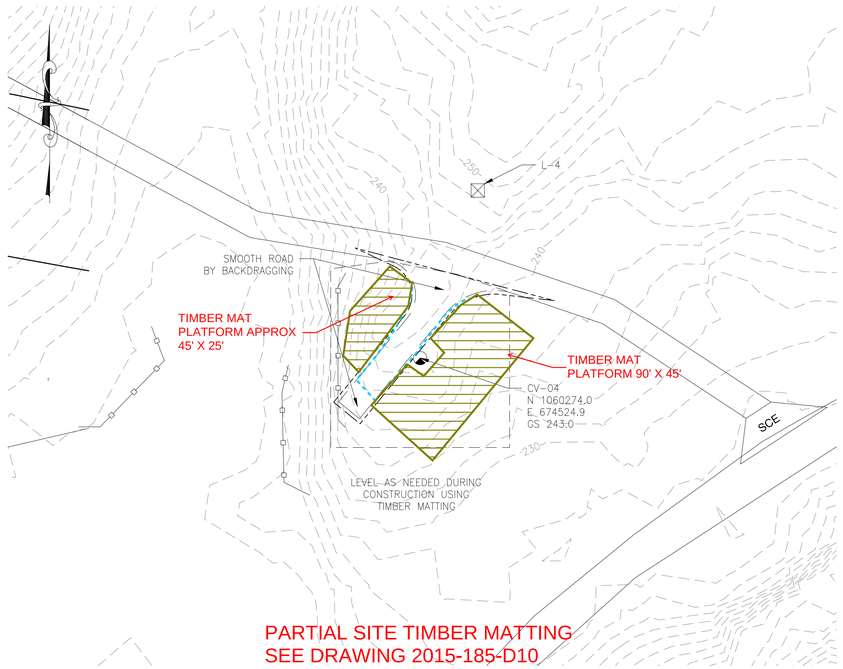

PARTIAL SITE TIMBER MATTING
SEE DRAWING 2015-185-D10

POLE NO. CV-4

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

GRADED PADS FOR CONSTRUCTION ARE LEVEL AND APPROXIMATELY 80 FT. BY 80 FT. IN PLAN.

GRADED PADS FOR MAINTENANCE ARE LEVEL AND APPROXIMATELY 25 FT. BY 25 FT. IN PLAN. THE MAINTENANCE PADS MAY BE BUILT FOLLOWING LINE CONSTRUCTION. TIMBER MATTING WILL BE UTILIZED AS A LEVEL SURFACE FOR CONSTRUCTION AT SITES WITH MAINTENANCE PADS.

SPUR ROADS WILL EITHER BE SMOOTHED PATHS OVER EXISTING TOPOGRAPHY OR GRADED. SINCE THE USE OF SPUR ROADS WILL BE INFREQUENT, IT IS PREFERRED THAT SPUR ROAD CONSTRUCTION BY BACKDRAGGING A DOZER BLADE BE USED TO CREATE A SMOOTH SURFACE. GRADING IS REQUIRED ONLY TO EITHER PREPARE A CONSTRUCTION OR MAINTENANCE PAD OR TO LIMIT SIDE SLOPES ON THE ROAD. GRADING PLANS FOR SITE CONSTRUCTION AND ACCESS ARE SHOWN ON DRAWINGS SWPPP-18 THROUGH SWPPP-26.

CUTS AND FILL SLOPES 2H:1V GENERALLY WITH 1H:1V SLOPES IN ROCK CUTS AND 1.5H:1V SLOPES IN ROCK FILLS.

ALL CUT AND FILL SLOPES WITH AN EXPOSED SLOPE OF 6 FT. OR GREATER RECEIVES AND EROSION CONTROL BLANKET

EACH SITE INCLUDING SPUR ROAD ARE DESIGNED FOR BALANCED CUT AND FILLS. MINOR ADJUSTMENTS IN GRADING MAY BE REQUIRED TO ACHIEVE THIS BALANCE.

PRIOR TO PLACING ANY FILL THE SUBGRADE SHALL BE INSPECTED FOR SUITABILITY AS A SUBGRADE, IF IMPROVEMENT IS NEEDED THEN THE SUBGRADE SHALL BE PROOF-ROLLED WITH A VIBRATORY ROLLER.

IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

UPON COMPLETION OF GRADING, PROVIDE TEMPORARY SOIL STABILIZATION WITHIN 48 HRS DISTURBANCE (MULCH, TEMPORARY SEEDING OR EROSION CONTROL BLANKET).

MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

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REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING	MJG	PGC	PGC	PGC

DIGIOIA GRAY & ASSOCIATES
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15146
(412) 372-4500
www.digioiagray.com

DRWN:	MJG
DSGN:	MJG
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS - PLEASANT VALLEY

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-18

REV. 1

STATE OF NEW YORK · PAUL G CASS · 036902 · LICENSED PROFESSIONAL ENGINEER · 12/22/16

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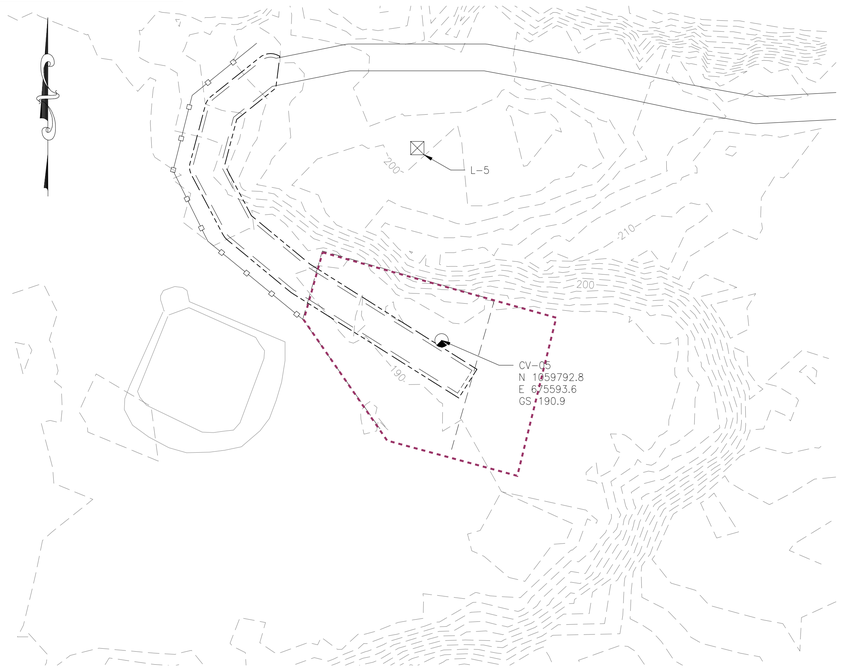
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-19



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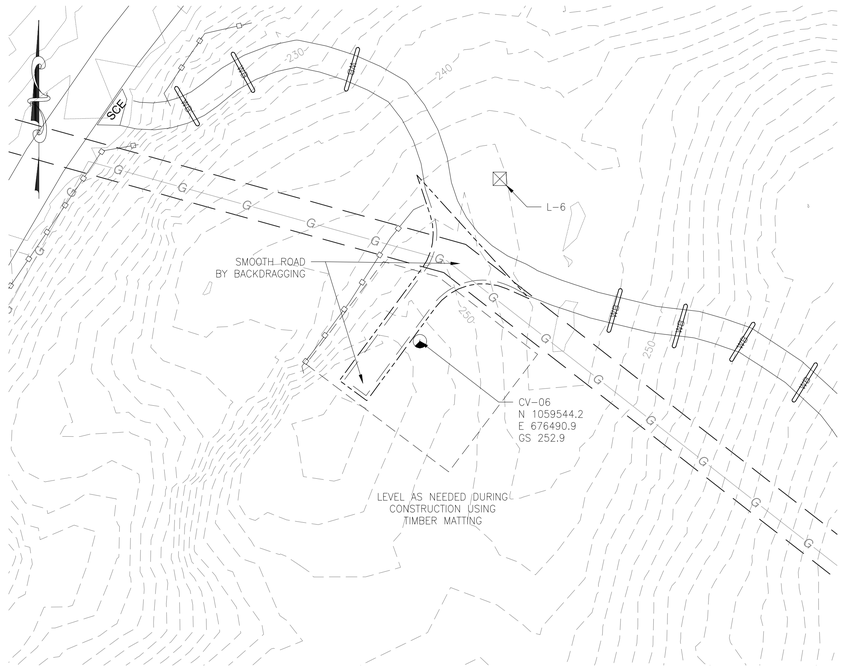

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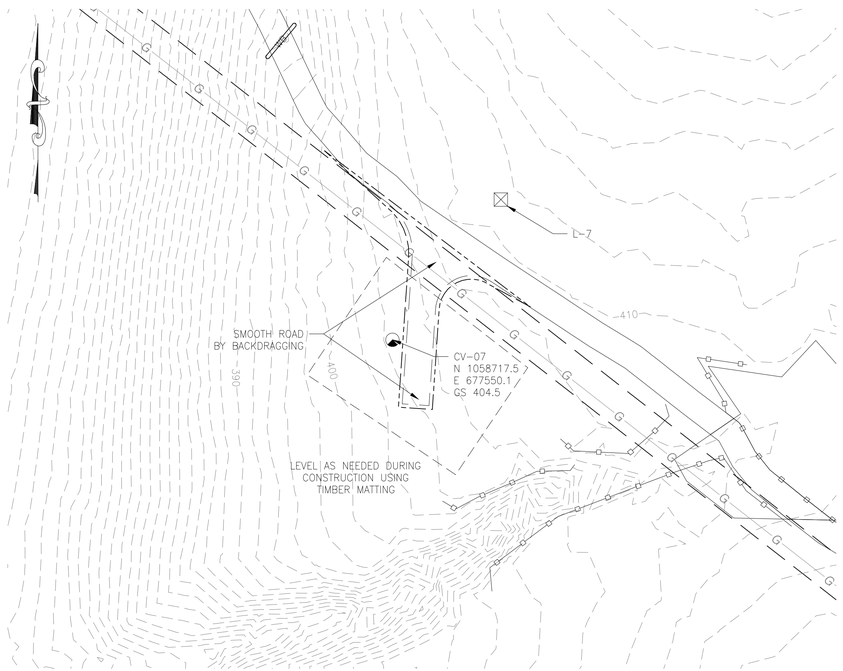

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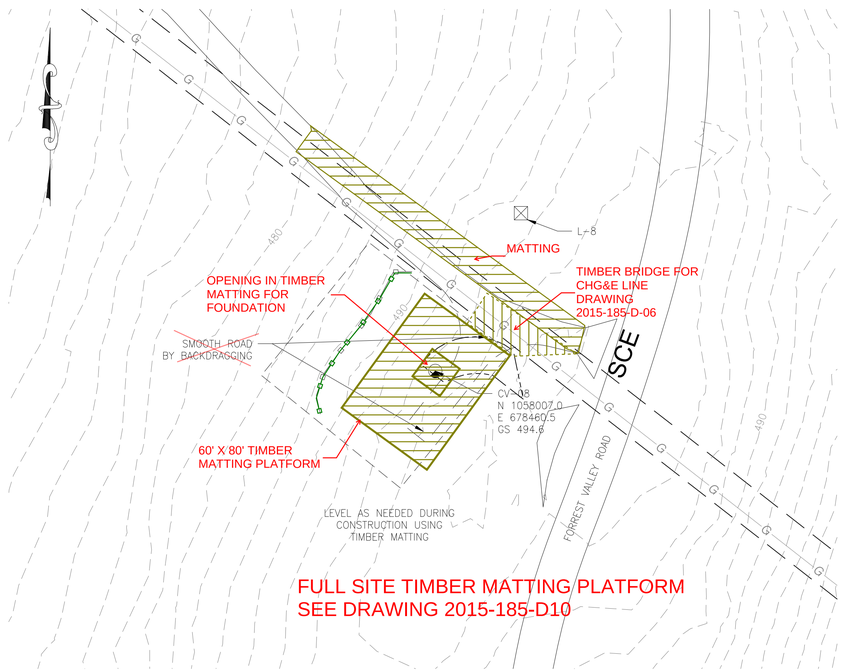

FULL SITE TIMBER MATTING PLATFORM
SEE DRAWING 2015-185-D10

POLE NO. CV-8

GRADING PLAN NOTES

GENERAL

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IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

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SEQUENCE

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MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

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- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING	MJG	PGC	PGC	PGC

DIGIOIA GRAY & ASSOCIATES
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MONROEVILLE, PA
15146
(412) 372-4500
www.digioiagray.com

DRWN:	MJG
DSGN:	MJG
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS
PLEASANT VALLEY, LAGRANGE AND UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-19

REV. 1

STATE OF NEW YORK * PAUL G CASS * LICENSED PROFESSIONAL ENGINEER 036902 12/22/16

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DIGIOIA GRAY PROJECT NO. 2015-184

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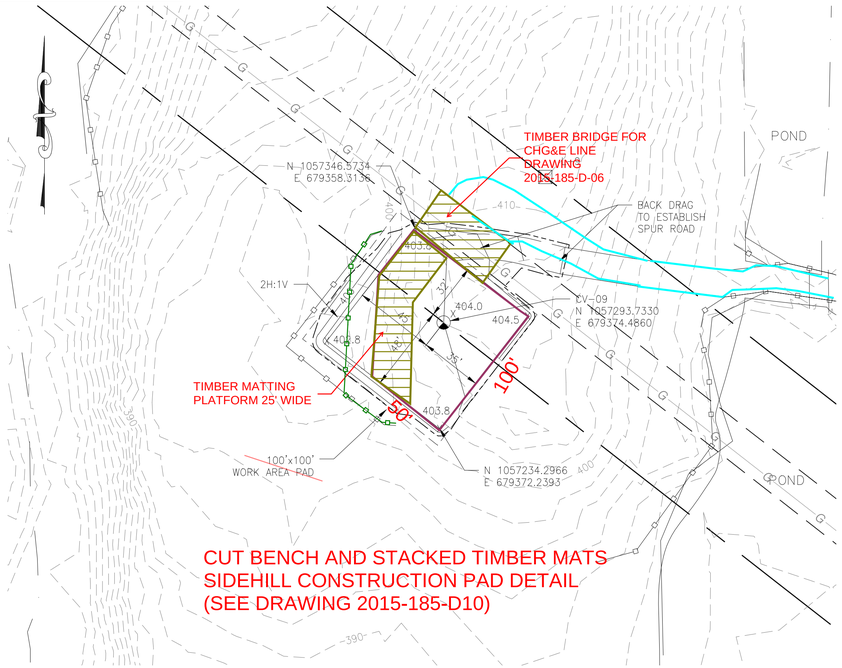

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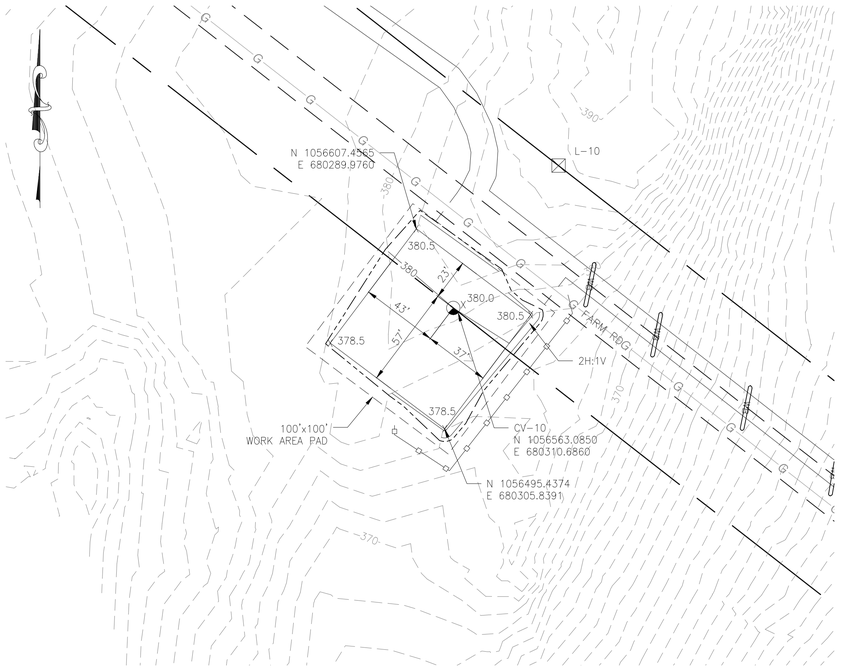
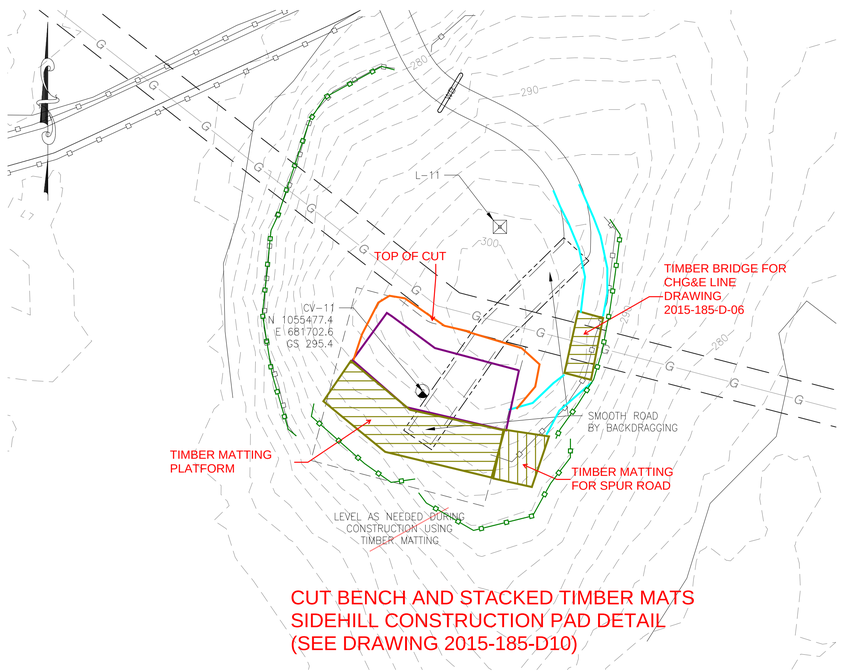


POLE NO. CV-10
PLEASANT VALLEY



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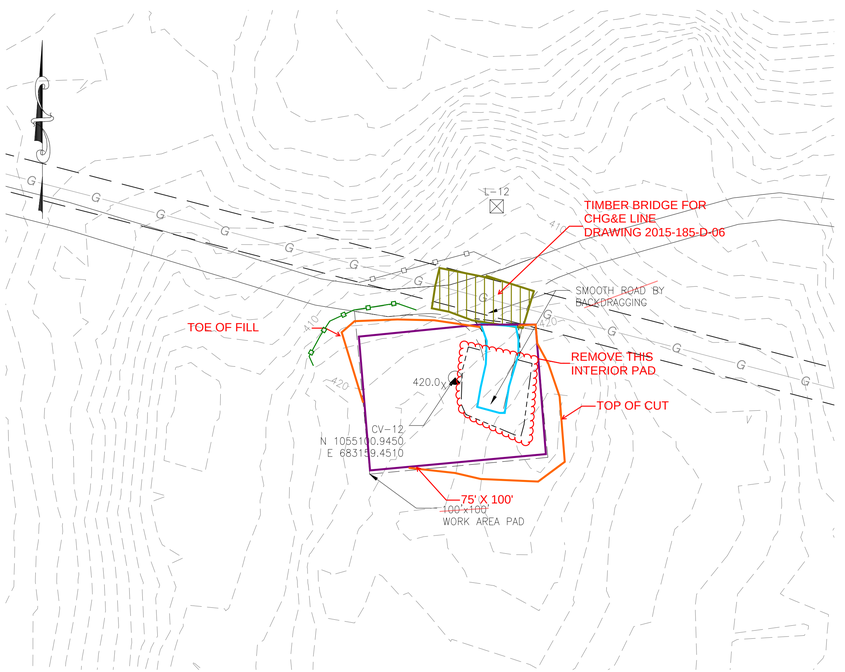

POLE NO. CV-12
PLEASANT VALLEY

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

GRADED PADS FOR CONSTRUCTION ARE LEVEL AND APPROXIMATELY 80 FT. BY 80 FT. IN PLAN.

GRADED PADS FOR MAINTENANCE ARE LEVEL AND APPROXIMATELY 25 FT. BY 25 FT. IN PLAN. THE MAINTENANCE PADS MAY BE BUILT FOLLOWING LINE CONSTRUCTION. TIMBER MATTING WILL BE UTILIZED AS A LEVEL SURFACE FOR CONSTRUCTION AT SITES WITH MAINTENANCE PADS.

SPUR ROADS WILL EITHER BE SMOOTHED PATHS OVER EXISTING TOPOGRAPHY OR GRADED. SINCE THE USE OF SPUR ROADS WILL BE INFREQUENT, IT IS PREFERRED THAT SPUR ROAD CONSTRUCTION BY BACKDRAGGING A DOZER BLADE BE USED TO CREATE A SMOOTH SURFACE. GRADING IS REQUIRED ONLY TO EITHER PREPARE A CONSTRUCTION OR MAINTENANCE PAD OR TO LIMIT SIDE SLOPES ON THE ROAD. GRADING PLANS FOR SITE CONSTRUCTION AND ACCESS ARE SHOWN ON DRAWINGS SWPPP-18 THROUGH SWPPP-26.

CUTS AND FILL SLOPES 2H:1V GENERALLY WITH 1H:1V SLOPES IN ROCK CUTS AND 1.5H:1V SLOPES IN ROCK FILLS.

ALL CUT AND FILL SLOPES WITH AN EXPOSED SLOPE OF 6 FT. OR GREATER RECEIVES AND EROSION CONTROL BLANKET

EACH SITE INCLUDING SPUR ROAD ARE DESIGNED FOR BALANCED CUT AND FILLS. MINOR ADJUSTMENTS IN GRADING MAY BE REQUIRED TO ACHIEVE THIS BALANCE.

PRIOR TO PLACING ANY FILL THE SUBGRADE SHALL BE INSPECTED FOR SUITABILITY AS A SUBGRADE, IF IMPROVEMENT IS NEEDED THEN THE SUBGRADE SHALL BE PROOF-ROLLED WITH A VIBRATORY ROLLER.

IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

UPON COMPLETION OF GRADING, PROVIDE TEMPORARY SOIL STABILIZATION WITHIN 48 HRS DISTURBANCE (MULCH, TEMPORARY SEEDING OR EROSION CONTROL BLANKET).

MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING	MJG	PGC	PGC	PGC

DIGIOIA GRAY & ASSOCIATES
570 BEATTY ROAD
MONROEVILLE, PA
15146
(412) 372-4500
www.digioiagray.com

DRWN:	MJG
DSGN:	MJG
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS
UNION VALE, LAGRANGE AND DOVER

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-20

REV. 1

STATE OF NEW YORK · PAUL G CASS · LICENSED PROFESSIONAL ENGINEER · 036902 · 12/22/16

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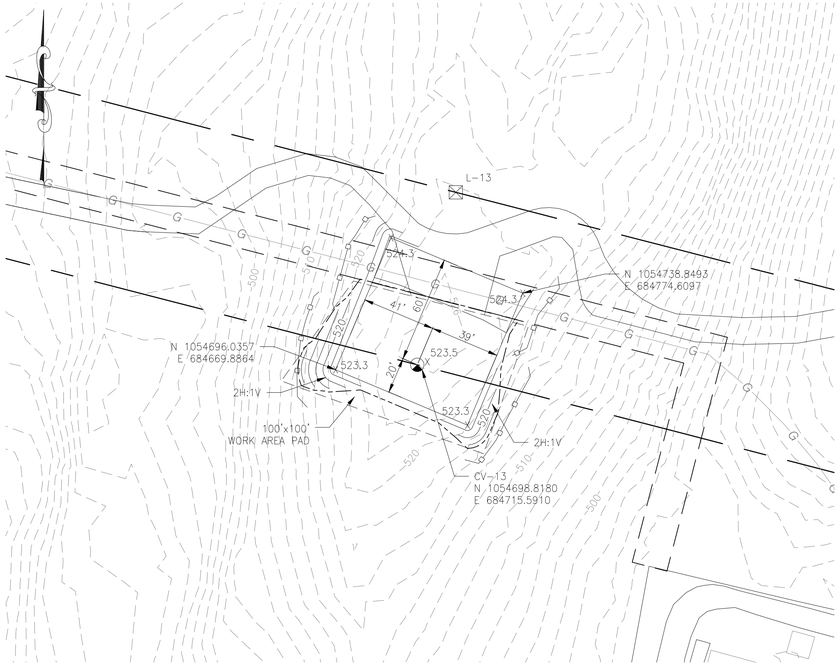
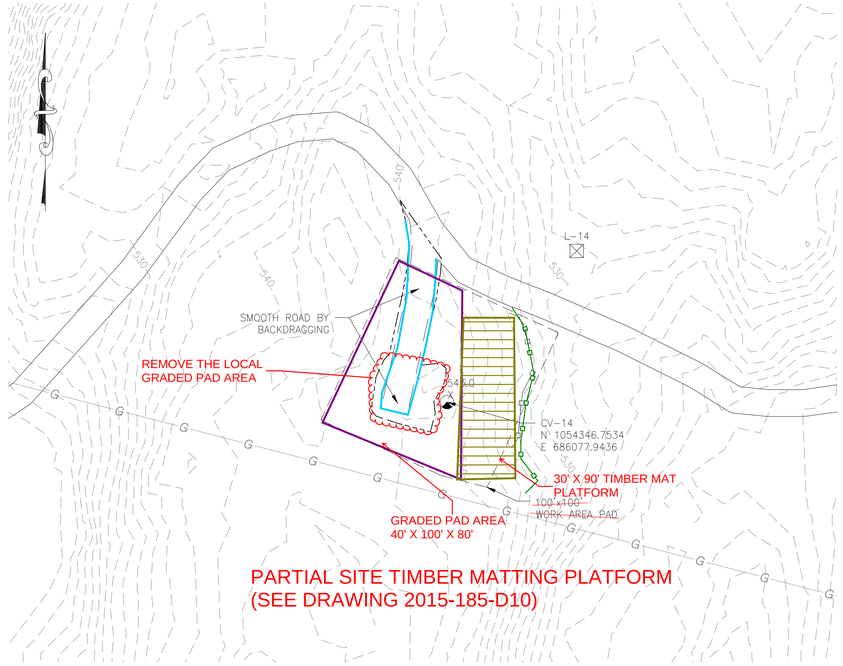
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-21



POLE NO. CV-13
PLEASANT VALLEY



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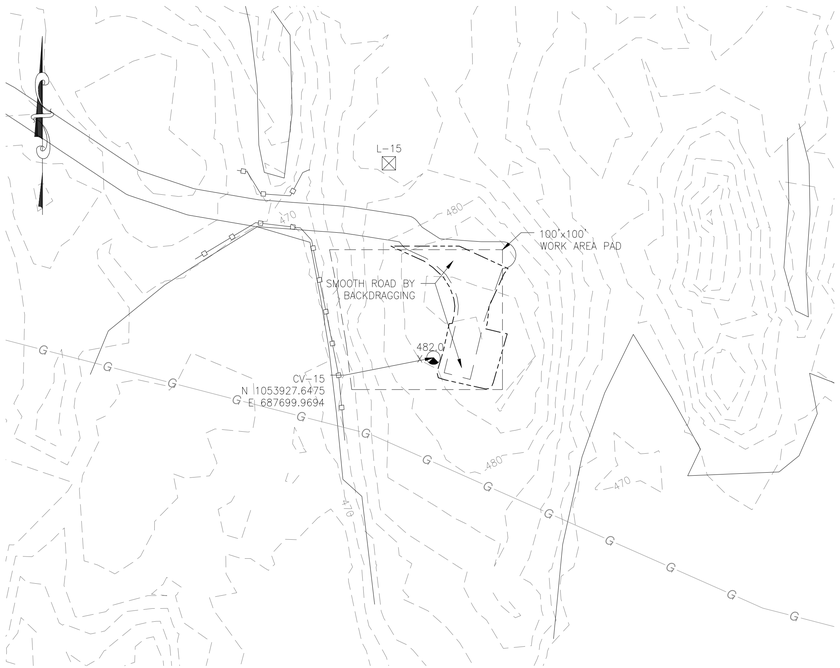

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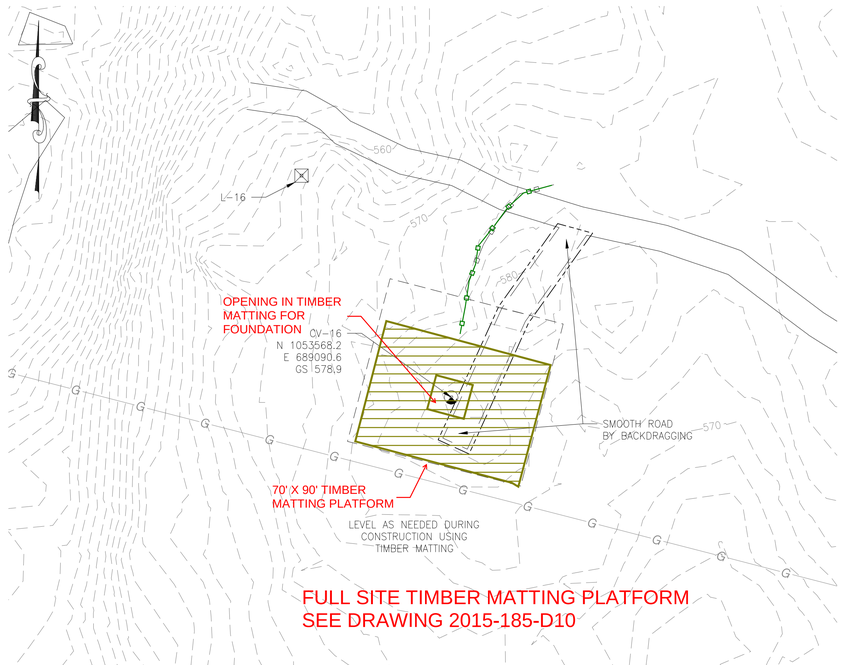

POLE NO. CV-16

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

GRADED PADS FOR CONSTRUCTION ARE LEVEL AND APPROXIMATELY 80 FT. BY 80 FT. IN PLAN.

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CUTS AND FILL SLOPES 2H:1V GENERALLY WITH 1H:1V SLOPES IN ROCK CUTS AND 1.5H:1V SLOPES IN ROCK FILLS.

ALL CUT AND FILL SLOPES WITH AN EXPOSED SLOPE OF 6 FT. OR GREATER RECEIVES AND EROSION CONTROL BLANKET

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IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

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INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

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PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
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ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING	MJG	PGC	PGC	PGC

DIGIOIA GRAY & ASSOCIATES
570 BEATTY ROAD
MONROEVILLE, PA
15146
(412) 372-4500
www.digioiagray.com

DRWN:	MJG
DSGN:	MJG
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS- UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-21

REV. 1

STATE OF NEW YORK · PAUL G CASS · LICENSED PROFESSIONAL ENGINEER · 086902 · 12/22/16

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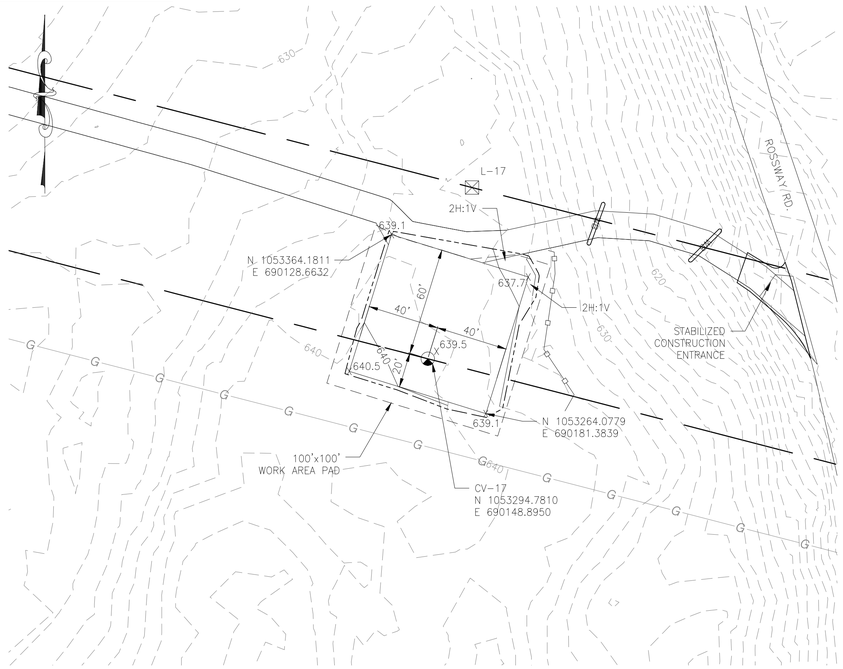
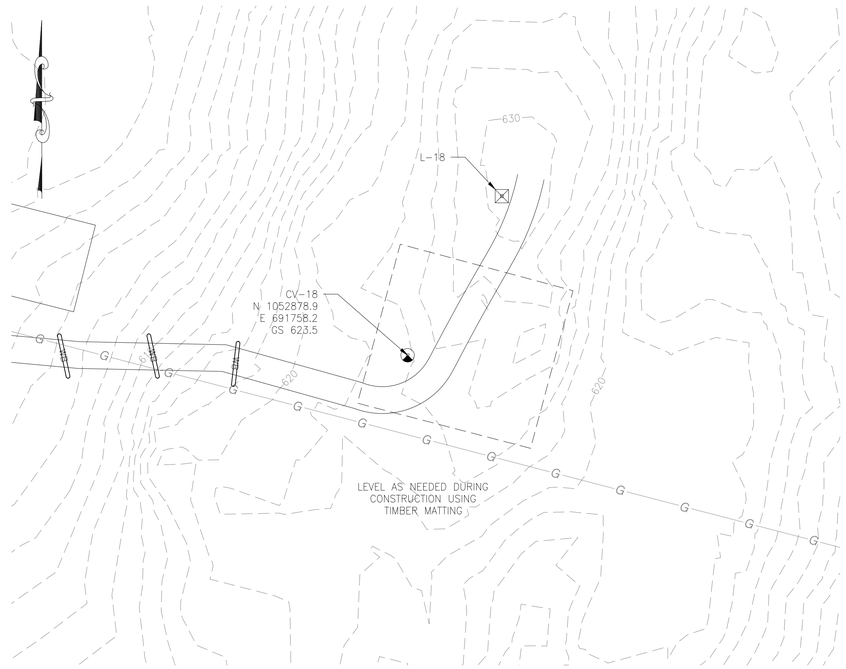
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-22



POLE NO. CV-17
LAGRANGE



POLE NO. CV-18

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

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IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

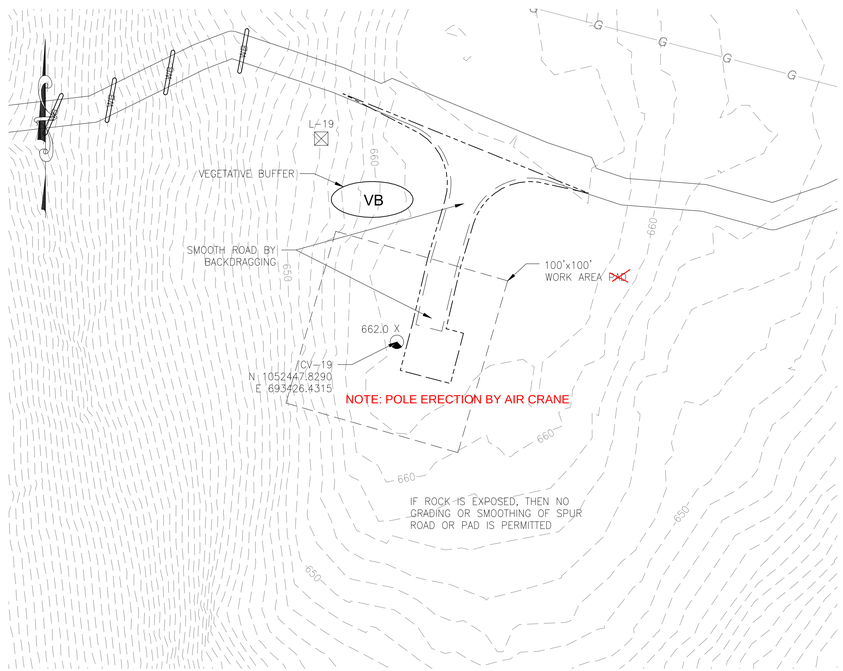
UPON COMPLETION OF GRADING, PROVIDE TEMPORARY SOIL STABILIZATION WITHIN 48 HRS DISTURBANCE (MULCH, TEMPORARY SEEDING OR EROSION CONTROL BLANKET).

MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS



POLE NO. CV-19
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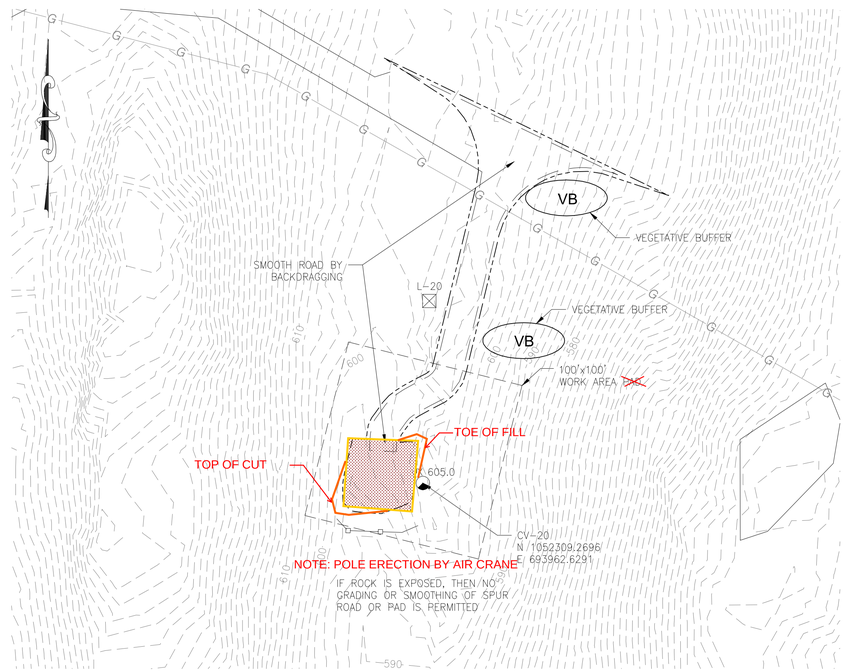

POLE NO. CV-20
LAGRANGE

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING, IDENTIFIED VEGETATIVE BUFFER AREAS	MJG	PGC	PGC	PGC

DIGIOIA GRAY & ASSOCIATES
570 BEATTY ROAD
MONROEVILLE, PA
15146
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www.digioiagray.com

DRWN:	MJG
DSGN:	MJG
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS- UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-22

REV. 1

STATE OF NEW YORK · PAUL G CASS · 086902 · LICENSED PROFESSIONAL ENGINEER · 12/22/16

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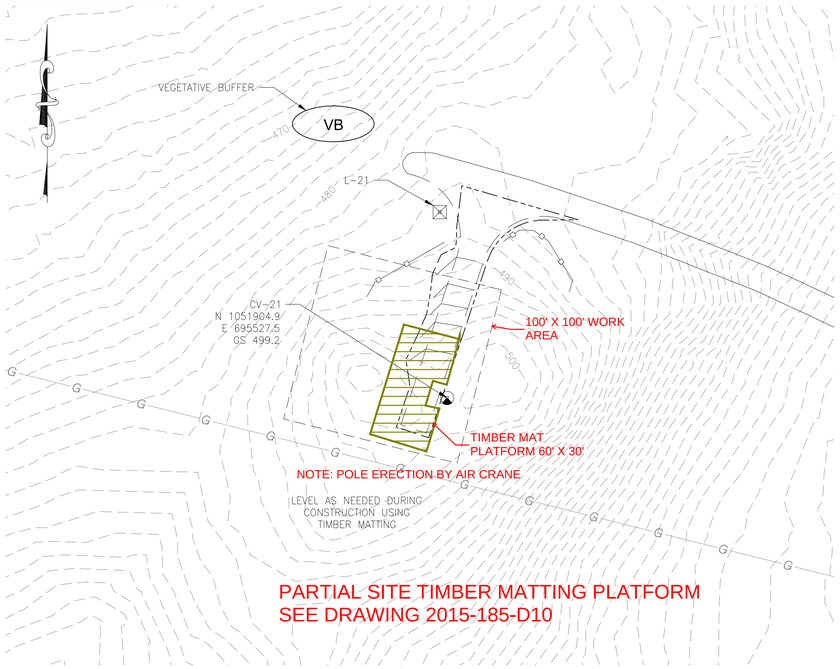
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-23



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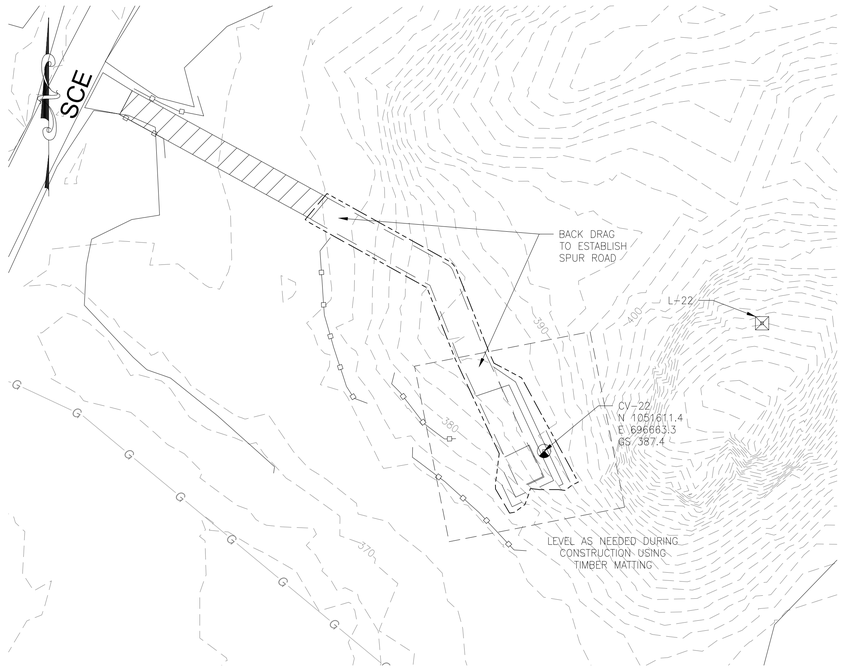

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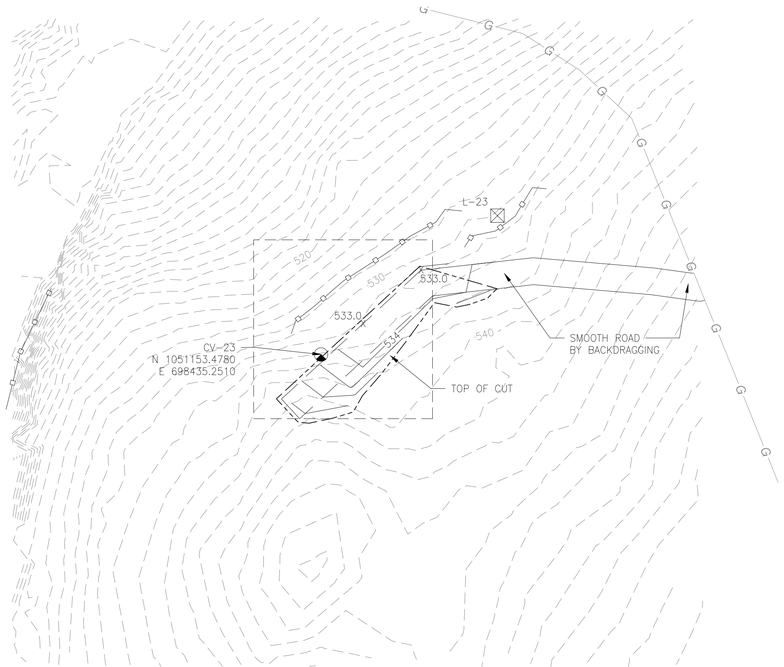
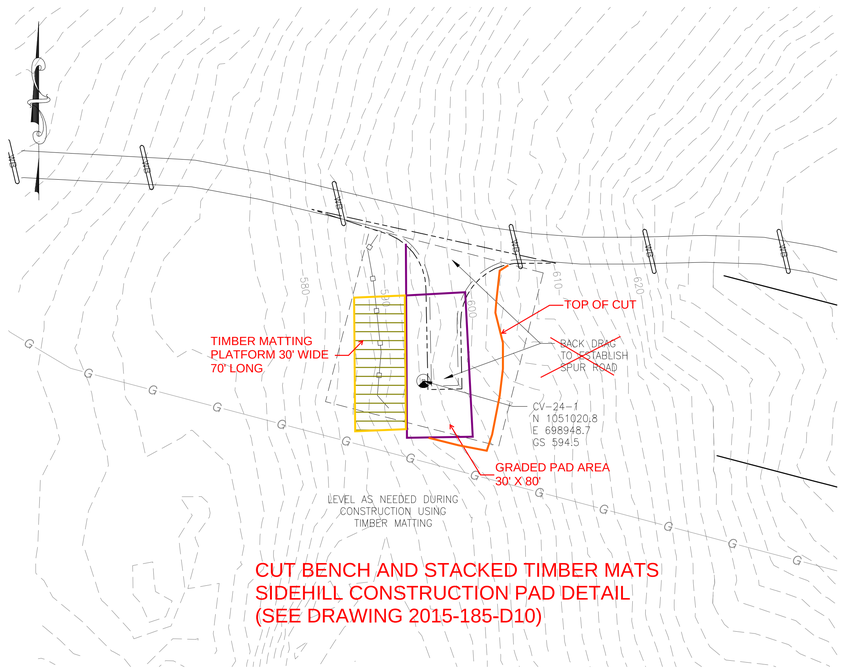


POLE NO. CV-23 SPUR ROAD
LAGRANGE



POLE NO. CV-24-1

GRADING PLAN NOTES

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RELATED DRAWINGS

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2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING	MJG	PGC	PGC	PGC

DIGIOIA GRAY & ASSOCIATES
570 BEATTY ROAD
MONROEVILLE, PA
15146
(412) 372-4500
www.digioiagray.com

DRWN:	MJG
DSGN:	MJG
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS- UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-23

REV. 1

STATE OF NEW YORK · PAUL G CASS · 036902 · LICENSED PROFESSIONAL ENGINEER · 12/22/16

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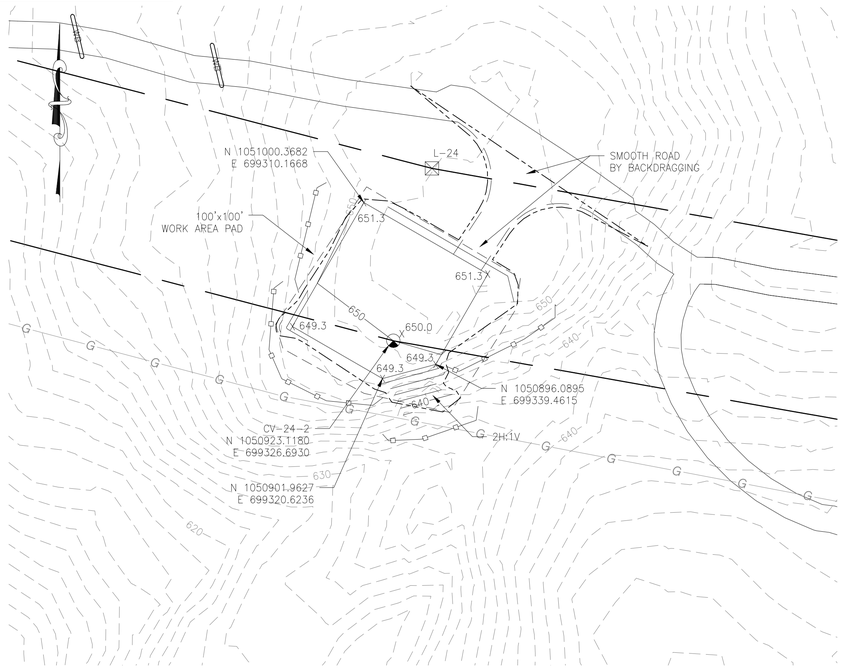
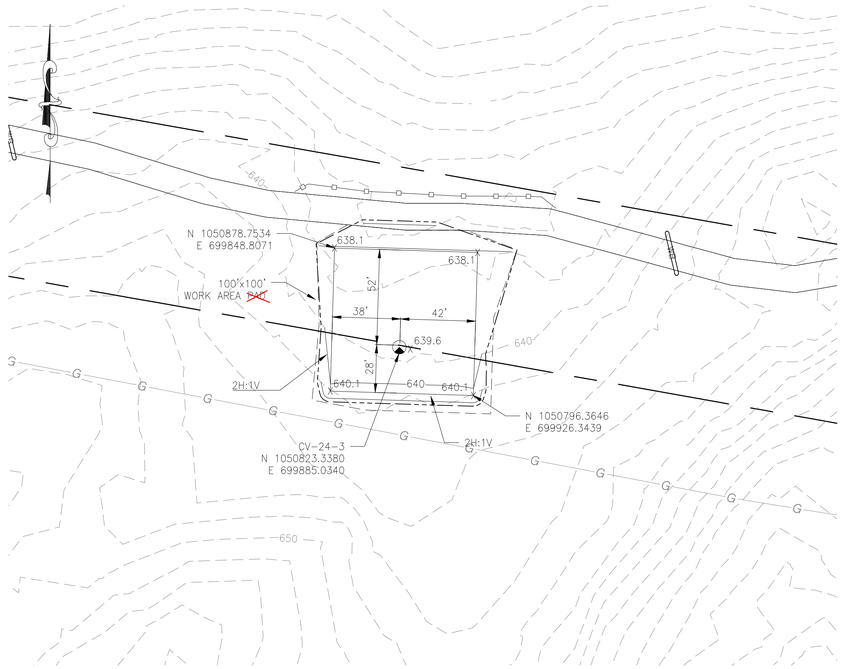
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-24



POLE NO. CV-24-2
UNION VALE



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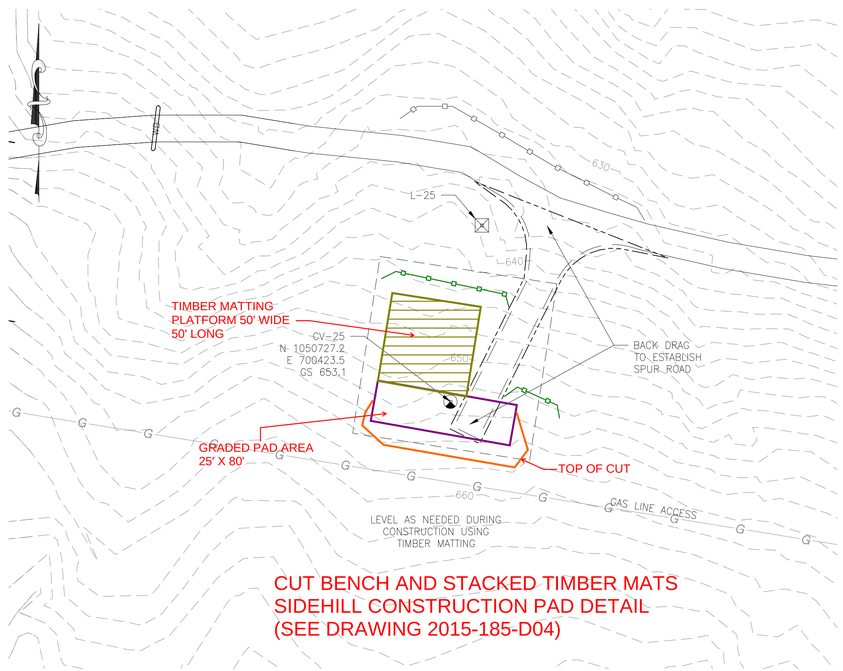
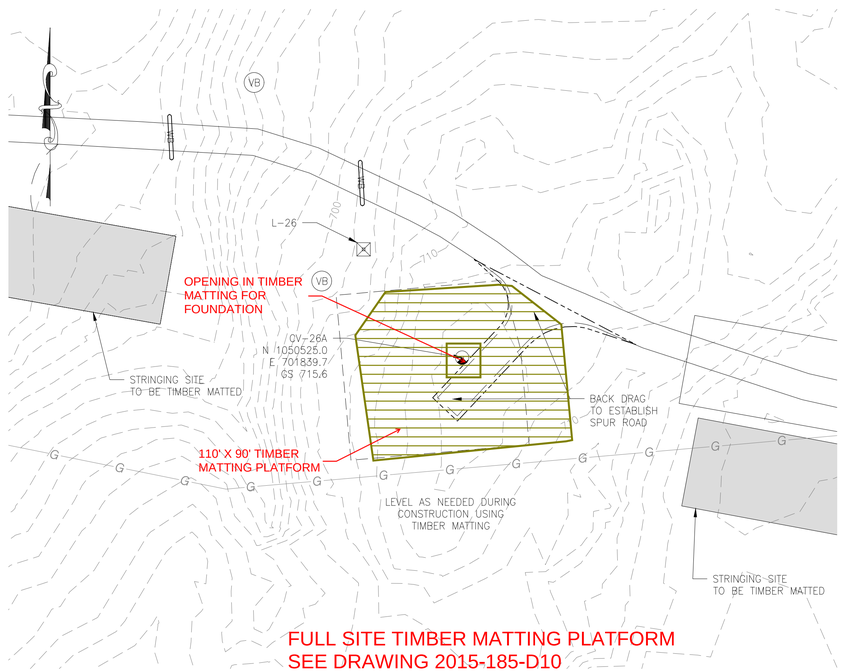
UNION VALE



CUT BENCH AND STACKED TIMBER MATS
SIDEHILL CONSTRUCTION PAD DETAIL
(SEE DRAWING 2015-185-D04)

POLE NO. CV-25



FULL SITE TIMBER MATTING PLATFORM
SEE DRAWING 2015-185-D10

POLE NO. CV-26A

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

GRADED PADS FOR CONSTRUCTION ARE LEVEL AND APPROXIMATELY 80 FT. BY 80 FT. IN PLAN.

GRADED PADS FOR MAINTENANCE ARE LEVEL AND APPROXIMATELY 25 FT. BY 25 FT. IN PLAN. THE MAINTENANCE PADS MAY BE BUILT FOLLOWING LINE CONSTRUCTION. TIMBER MATTING WILL BE UTILIZED AS A LEVEL SURFACE FOR CONSTRUCTION AT SITES WITH MAINTENANCE PADS.

SPUR ROADS WILL EITHER BE SMOOTHED PATHS OVER EXISTING TOPOGRAPHY OR GRADED. SINCE THE USE OF SPUR ROADS WILL BE INFREQUENT, IT IS PREFERRED THAT SPUR ROAD CONSTRUCTION BY BACKDRAGGING A DOZER BLADE BE USED TO CREATE A SMOOTH SURFACE. GRADING IS REQUIRED ONLY TO EITHER PREPARE A CONSTRUCTION OR MAINTENANCE PAD OR TO LIMIT SIDE SLOPES ON THE ROAD. GRADING PLANS FOR SITE CONSTRUCTION AND ACCESS ARE SHOWN ON DRAWINGS SWPPP-18 THROUGH SWPPP-26.

CUTS AND FILL SLOPES 2H:1V GENERALLY WITH 1H:1V SLOPES IN ROCK CUTS AND 1.5H:1V SLOPES IN ROCK FILLS.

ALL CUT AND FILL SLOPES WITH AN EXPOSED SLOPE OF 6 FT. OR GREATER RECEIVES AND EROSION CONTROL BLANKET

EACH SITE INCLUDING SPUR ROAD ARE DESIGNED FOR BALANCED CUT AND FILLS. MINOR ADJUSTMENTS IN GRADING MAY BE REQUIRED TO ACHIEVE THIS BALANCE.

PRIOR TO PLACING ANY FILL THE SUBGRADE SHALL BE INSPECTED FOR SUITABILITY AS A SUBGRADE, IF IMPROVEMENT IS NEEDED THEN THE SUBGRADE SHALL BE PROOF-ROLLED WITH A VIBRATORY ROLLER.

IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

UPON COMPLETION OF GRADING, PROVIDE TEMPORARY SOIL STABILIZATION WITHIN 48 HRS DISTURBANCE (MULCH, TEMPORARY SEEDING OR EROSION CONTROL BLANKET).

MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING	MJG	PGC	PGC	PGC

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DRWN:	MJG
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APP:	PGC
DATE:	12-22-16
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CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS- UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-24

REV. 1

STATE OF NEW YORK · PAUL G CASS · 11/25/16 · 096902 · LICENSED PROFESSIONAL ENGINEER

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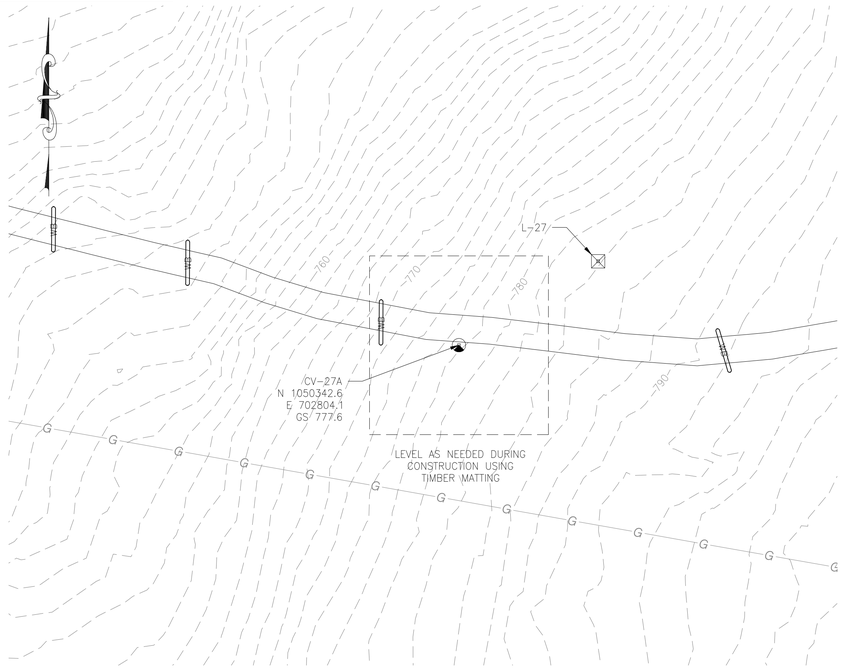
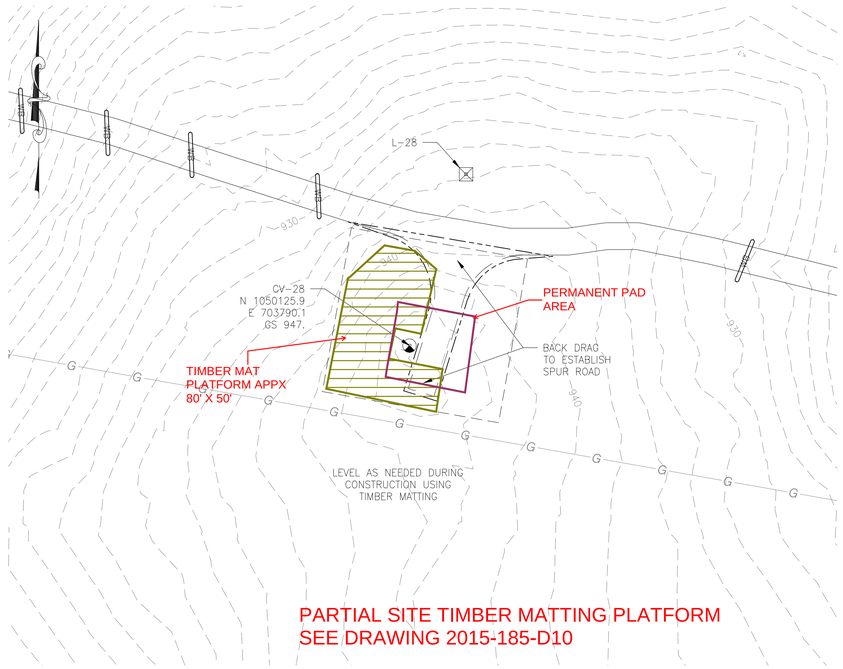
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-25



POLE NO. CV-27A



PARTIAL SITE TIMBER MATTING PLATFORM
SEE DRAWING 2015-185-D10

POLE NO. CV-28

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

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CUTS AND FILL SLOPES 2H:1V GENERALLY WITH 1H:1V SLOPES IN ROCK CUTS AND 1.5H:1V SLOPES IN ROCK FILLS.

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PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

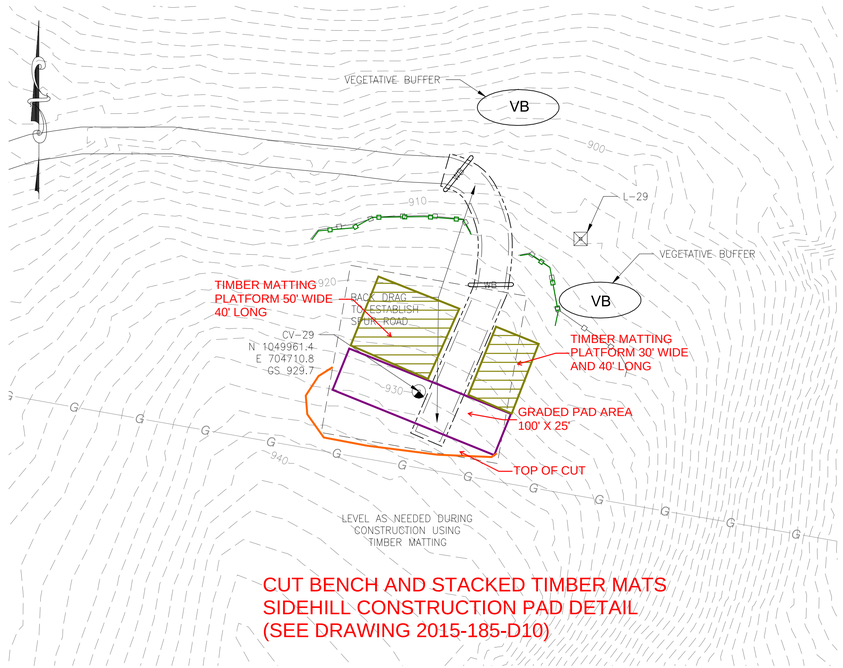
UPON COMPLETION OF GRADING, PROVIDE TEMPORARY SOIL STABILIZATION WITHIN 48 HRS DISTURBANCE (MULCH, TEMPORARY SEEDING OR EROSION CONTROL BLANKET).

MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

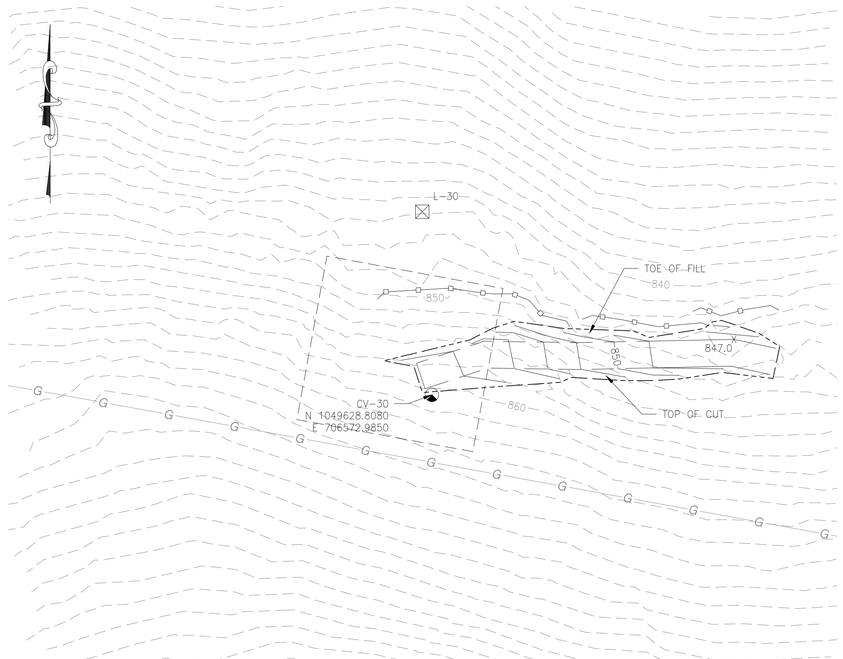
RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS



CUT BENCH AND STACKED TIMBER MATS
SIDEHILL CONSTRUCTION PAD DETAIL
(SEE DRAWING 2015-185-D10)

POLE NO. CV-29



POLE NO. CV-30 SPUR ROAD
UNION VALE

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- G — EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
1	5/5/17	ADDED LIMITS OF GRADING, IDENTIFIED VEGETATIVE BUFFER AREAS	MJG	PGC	PGC	PGC

REFERENCE TITLE | NUMBER

DIGIOIA GRAY & ASSOCIATES
570 BEATTY ROAD
MONROEVILLE, PA
15146
(412) 372-4500
www.digioiagray.com

DRWN:	MJG
DSGN:	MJG
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS- UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-25 | REV. 1

STATE OF NEW YORK · PAUL G CASS · 036902 · LICENSED PROFESSIONAL ENGINEER · 12/22/16

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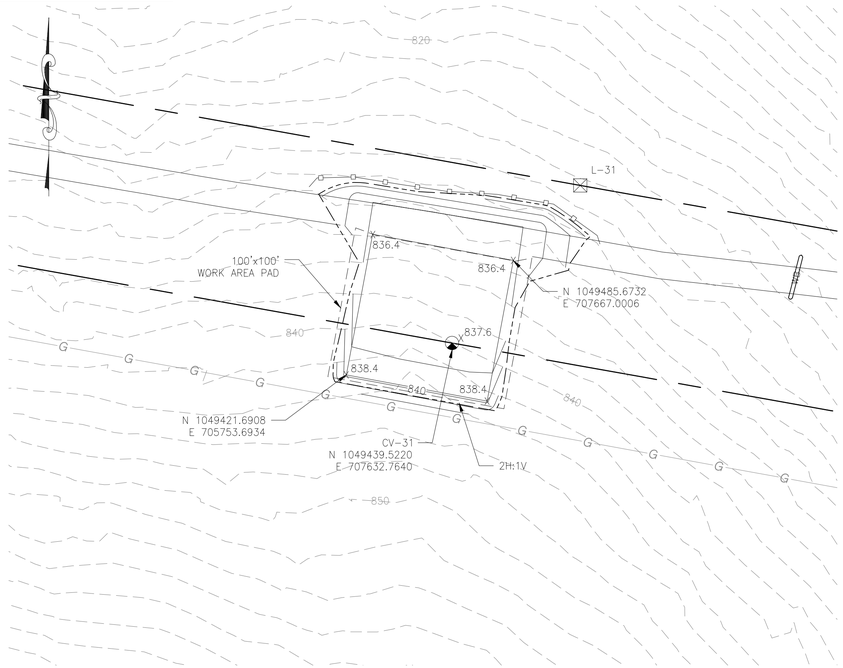
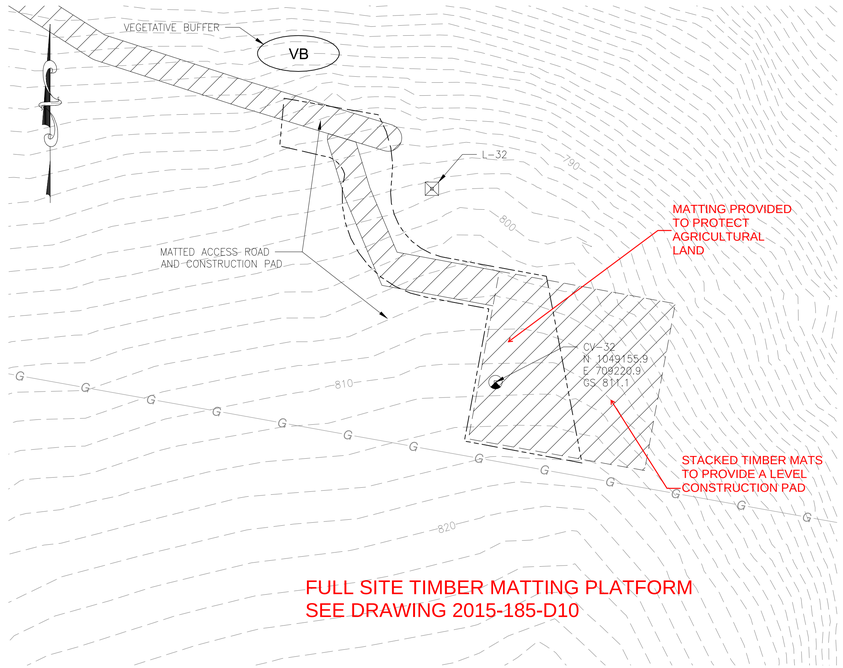
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-26



POLE NO. CV-31
UNION VALE



FULL SITE TIMBER MATTING PLATFORM
SEE DRAWING 2015-185-D10

POLE NO. CV-32

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

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CUTS AND FILL SLOPES 2H:1V GENERALLY WITH 1H:1V SLOPES IN ROCK CUTS AND 1.5H:1V SLOPES IN ROCK FILLS.

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SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

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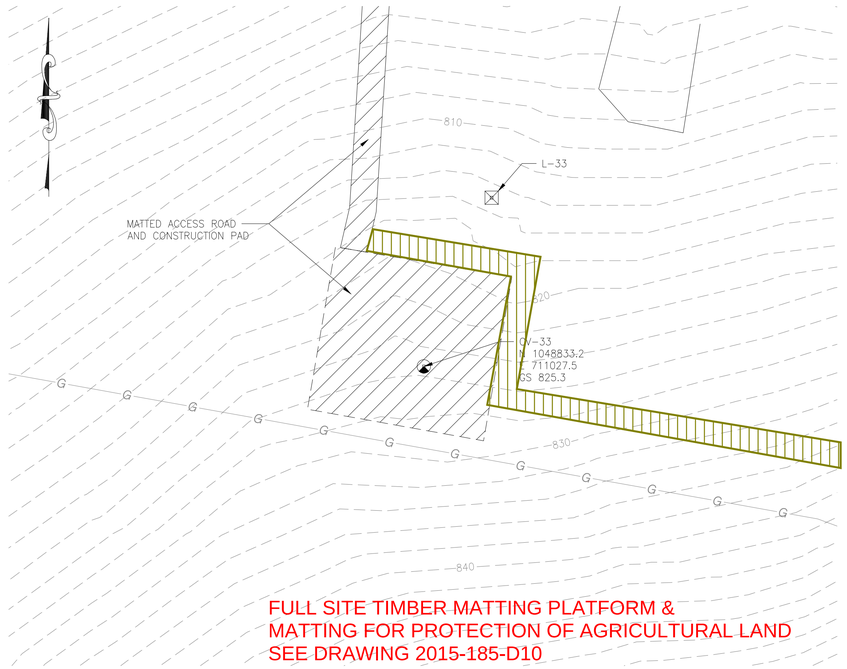
PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

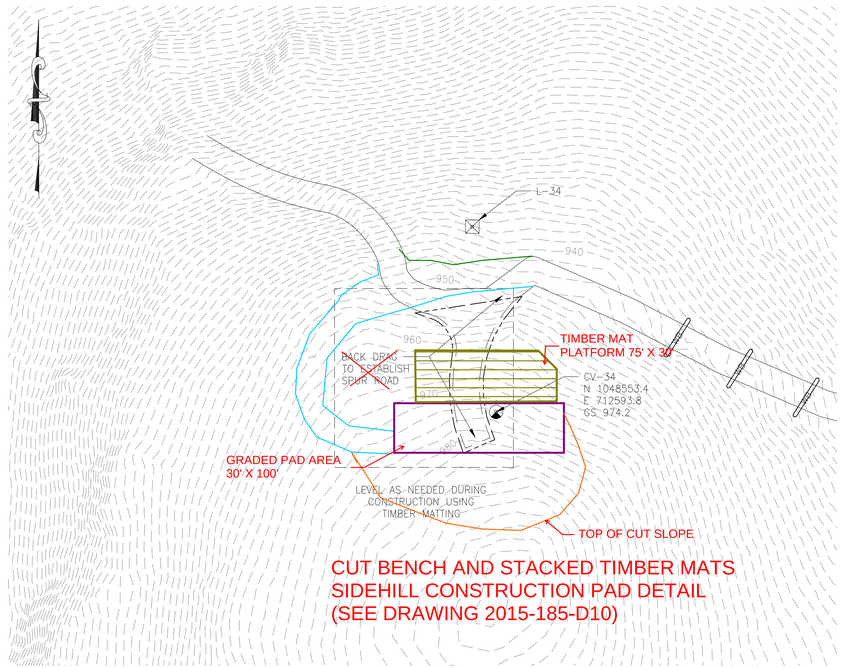
RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS



FULL SITE TIMBER MATTING PLATFORM &
MATTING FOR PROTECTION OF AGRICULTURAL LAND
SEE DRAWING 2015-185-D10

POLE NO. CV-33



CUT BENCH AND STACKED TIMBER MATS
SIDEHILL CONSTRUCTION PAD DETAIL
(SEE DRAWING 2015-185-D10)

POLE NO. CV-34

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- G — EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
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Cricket Valley Energy

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CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS- UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-26

REV. 1

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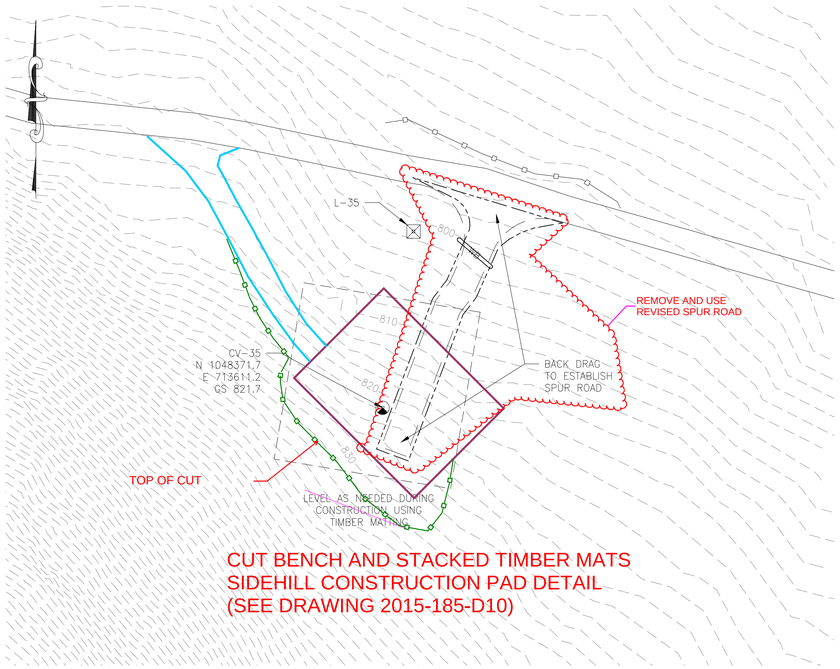
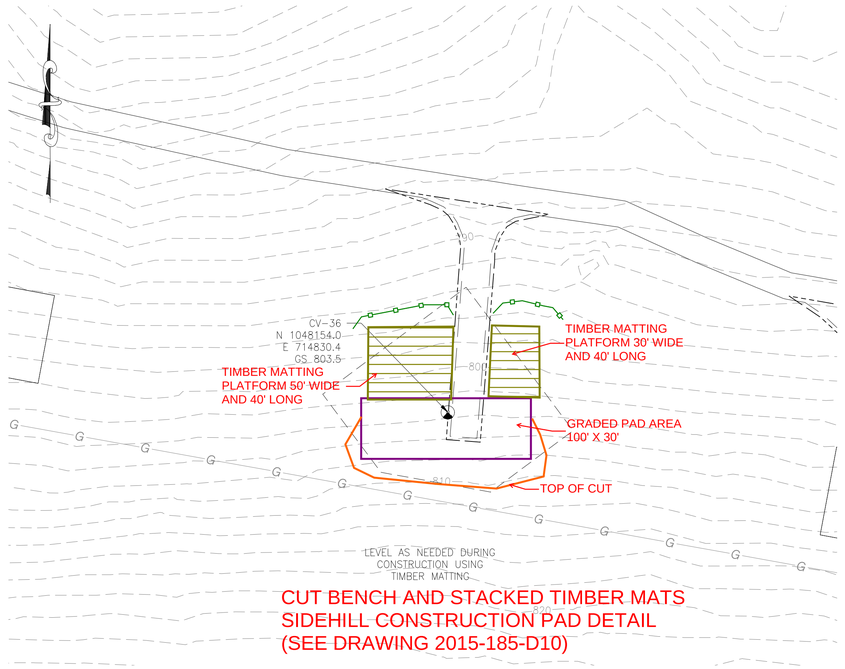
DIGIOIA GRAY PROJECT NO. 2015-184



POLE NO. CV-35



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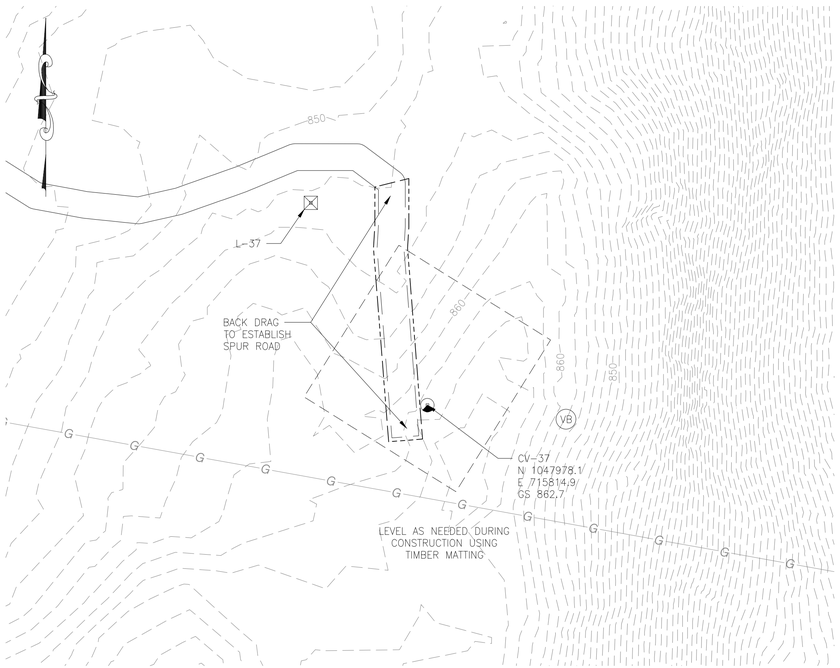

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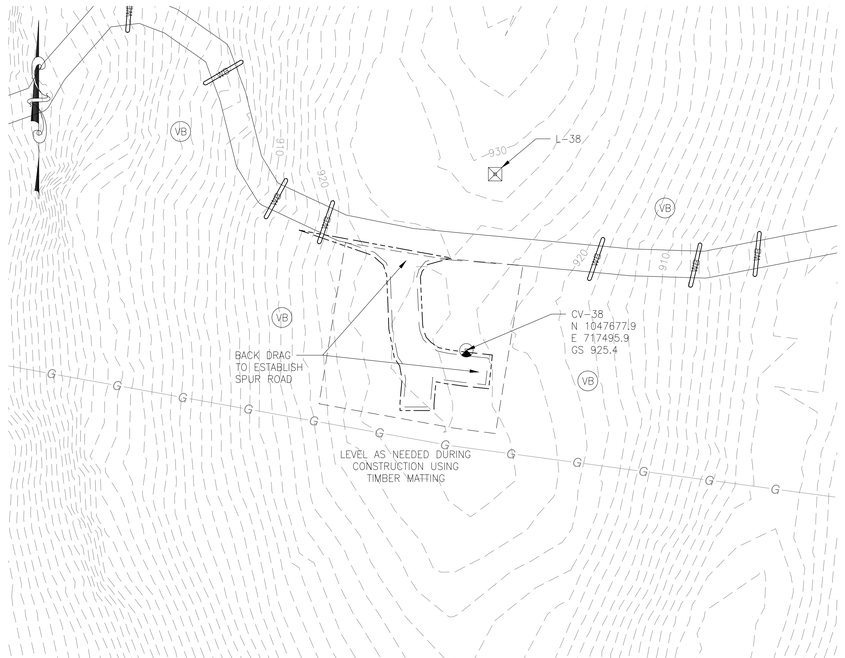

POLE NO. CV-38

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

GRADED PADS FOR CONSTRUCTION ARE LEVEL AND APPROXIMATELY 80 FT. BY 80 FT. IN PLAN.

GRADED PADS FOR MAINTENANCE ARE LEVEL AND APPROXIMATELY 25 FT. BY 25 FT. IN PLAN. THE MAINTENANCE PADS MAY BE BUILT FOLLOWING LINE CONSTRUCTION. TIMBER MATTING WILL BE UTILIZED AS A LEVEL SURFACE FOR CONSTRUCTION AT SITES WITH MAINTENANCE PADS.

SPUR ROADS WILL EITHER BE SMOOTHED PATHS OVER EXISTING TOPOGRAPHY OR GRADED. SINCE THE USE OF SPUR ROADS WILL BE INFREQUENT, IT IS PREFERRED THAT SPUR ROAD CONSTRUCTION BY BACKDRAGGING A DOZER BLADE BE USED TO CREATE A SMOOTH SURFACE. GRADING IS REQUIRED ONLY TO EITHER PREPARE A CONSTRUCTION OR MAINTENANCE PAD OR TO LIMIT SIDE SLOPES ON THE ROAD. GRADING PLANS FOR SITE CONSTRUCTION AND ACCESS ARE SHOWN ON DRAWINGS SWPPP-18 THROUGH SWPPP-26.

CUTS AND FILL SLOPES 2H:1V GENERALLY WITH 1H:1V SLOPES IN ROCK CUTS AND 1.5H:1V SLOPES IN ROCK FILLS.

ALL CUT AND FILL SLOPES WITH AN EXPOSED SLOPE OF 6 FT. OR GREATER RECEIVES AND EROSION CONTROL BLANKET

EACH SITE INCLUDING SPUR ROAD ARE DESIGNED FOR BALANCED CUT AND FILLS. MINOR ADJUSTMENTS IN GRADING MAY BE REQUIRED TO ACHIEVE THIS BALANCE.

PRIOR TO PLACING ANY FILL THE SUBGRADE SHALL BE INSPECTED FOR SUITABILITY AS A SUBGRADE, IF IMPROVEMENT IS NEEDED THEN THE SUBGRADE SHALL BE PROOF-ROLLED WITH A VIBRATORY ROLLER.

IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

UPON COMPLETION OF GRADING, PROVIDE TEMPORARY SOIL STABILIZATION WITHIN 48 HRS DISTURBANCE (MULCH, TEMPORARY SEEDING OR EROSION CONTROL BLANKET).

MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING, ADDED BARRIER FENCE	MJG	PGC	PGC	PGC

DIGIOIA GRAY & ASSOCIATES
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MONROEVILLE, PA
15146
(412) 372-4500
www.digioiagray.com

DRWN:	MJG
DSGN:	MJG
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS- UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-27

REV. 1

STATE OF NEW YORK * PAUL G CASS * 11/22/16 * 096902 * LICENSED PROFESSIONAL ENGINEER

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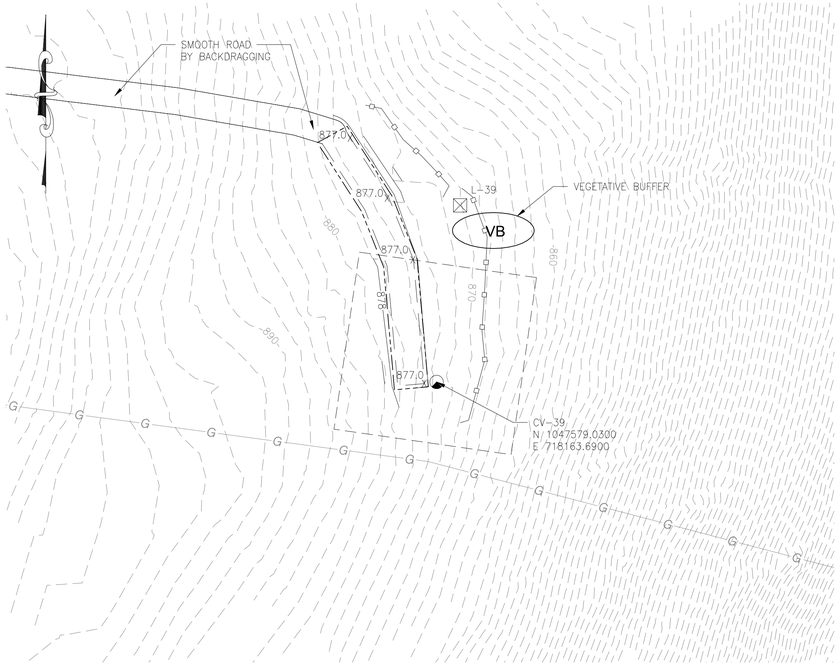
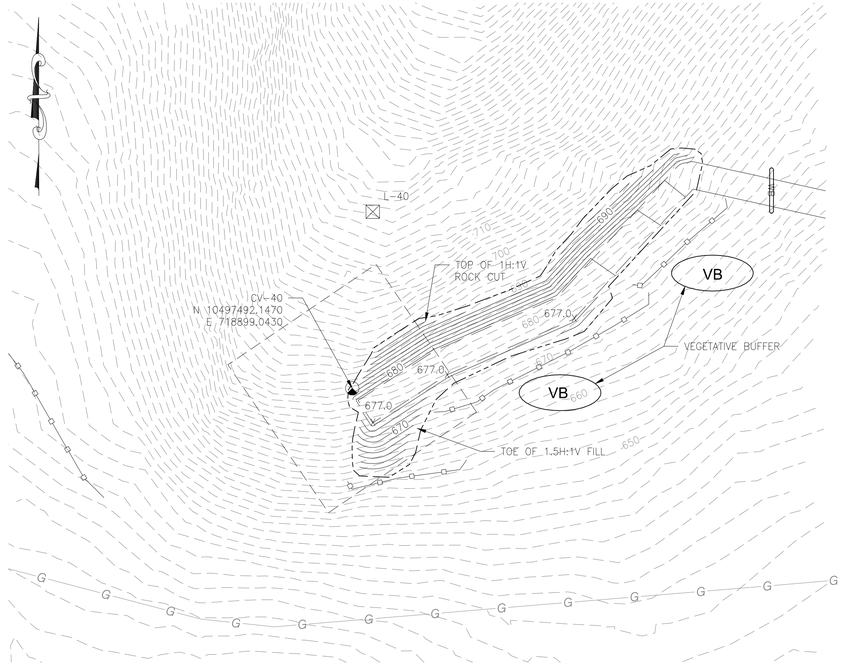
DIGIOIA GRAY PROJECT NO. 2015-184

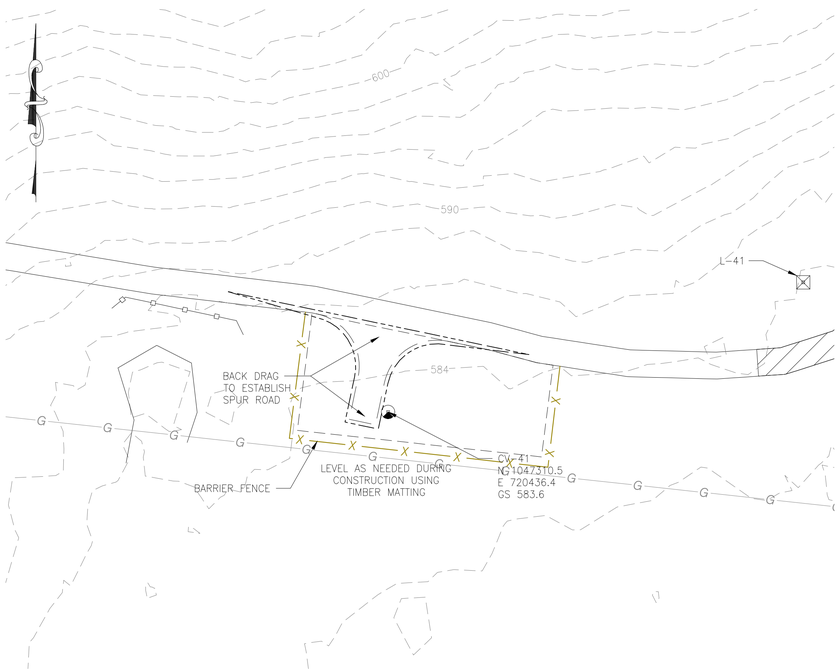
DRAWING NO. SWPPP-28



POLE NO. CV-39 SPUR ROAD
UNION VALE



POLE NO. CV-40 SPUR ROAD
UNION VALE



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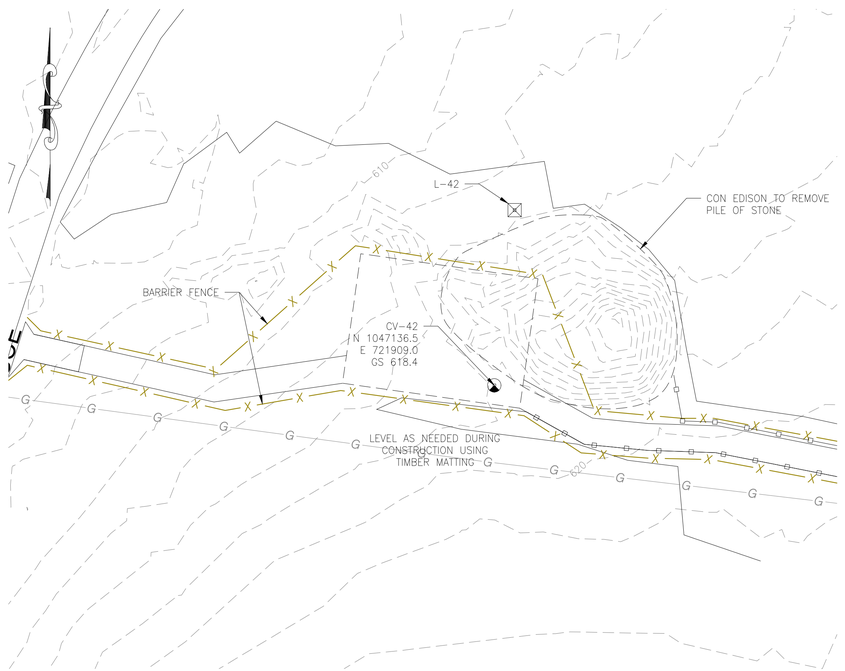

POLE NO. CV-42

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

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IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

UPON COMPLETION OF GRADING, PROVIDE TEMPORARY SOIL STABILIZATION WITHIN 48 HRS DISTURBANCE (MULCH, TEMPORARY SEEDING OR EROSION CONTROL BLANKET).

MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

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REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING, IDENTIFIED VEGETATIVE BUFFER AREAS, ADDED BARRIER FENCE	MJG	PGC	PGC	PGC

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MONROEVILLE, PA
15146
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www.digioiagray.com

DRWN: MJG
DSGN: MJG
CHK'D: PGC
APP: PGC
DATE: 12-22-16
SCALE: AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS- UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-28

REV. 1

STATE OF NEW YORK * PAUL G CASS * 11/22/16 * 096902 * LICENSED PROFESSIONAL ENGINEER

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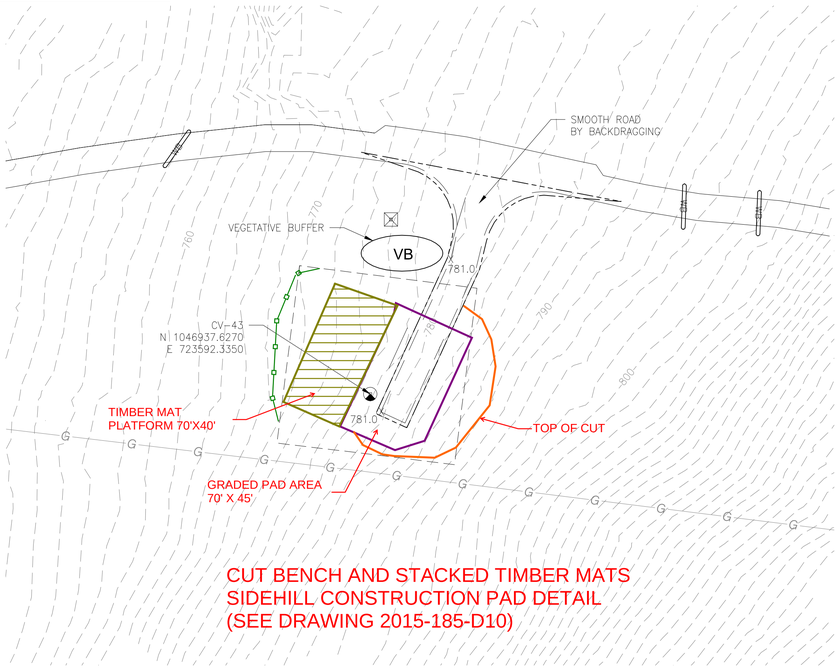
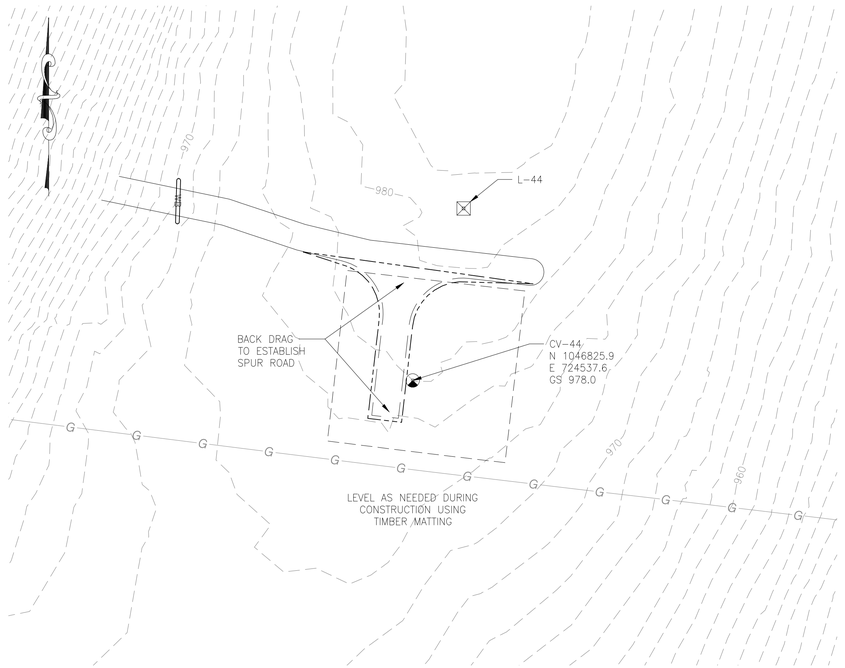
DRAWING NO. SWPPP-29

DIGIOIA GRAY PROJECT NO. 2015-184



CUT BENCH AND STACKED TIMBER MATS
SIDEHILL CONSTRUCTION PAD DETAIL
(SEE DRAWING 2015-185-D10)

POLE NO. CV-43 SPUR ROAD
UNION VALE



POLE NO. CV-44

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

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EACH SITE INCLUDING SPUR ROAD ARE DESIGNED FOR BALANCED CUT AND FILLS. MINOR ADJUSTMENTS IN GRADING MAY BE REQUIRED TO ACHIEVE THIS BALANCE.

PRIOR TO PLACING ANY FILL THE SUBGRADE SHALL BE INSPECTED FOR SUITABILITY AS A SUBGRADE, IF IMPROVEMENT IS NEEDED THEN THE SUBGRADE SHALL BE PROOF-ROLLED WITH A VIBRATORY ROLLER.

IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

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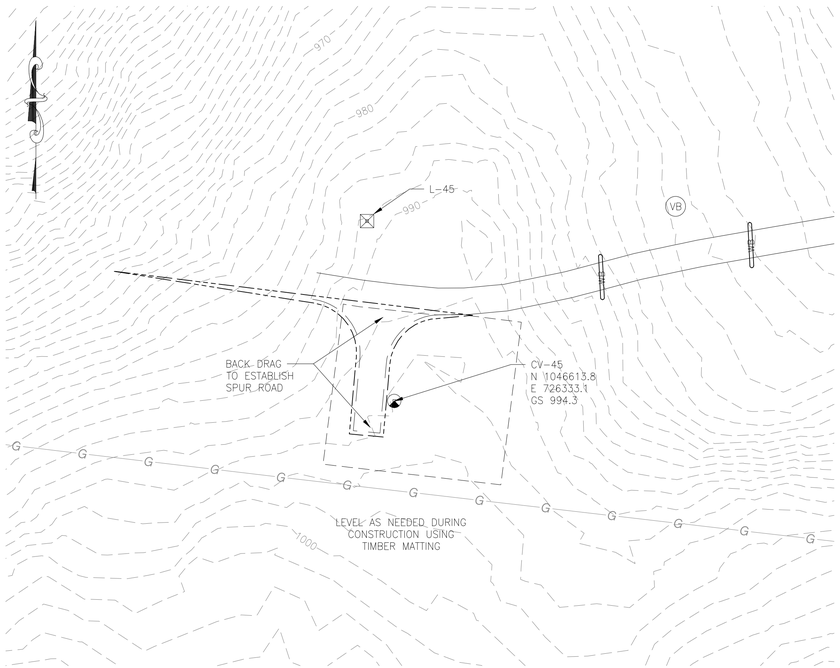
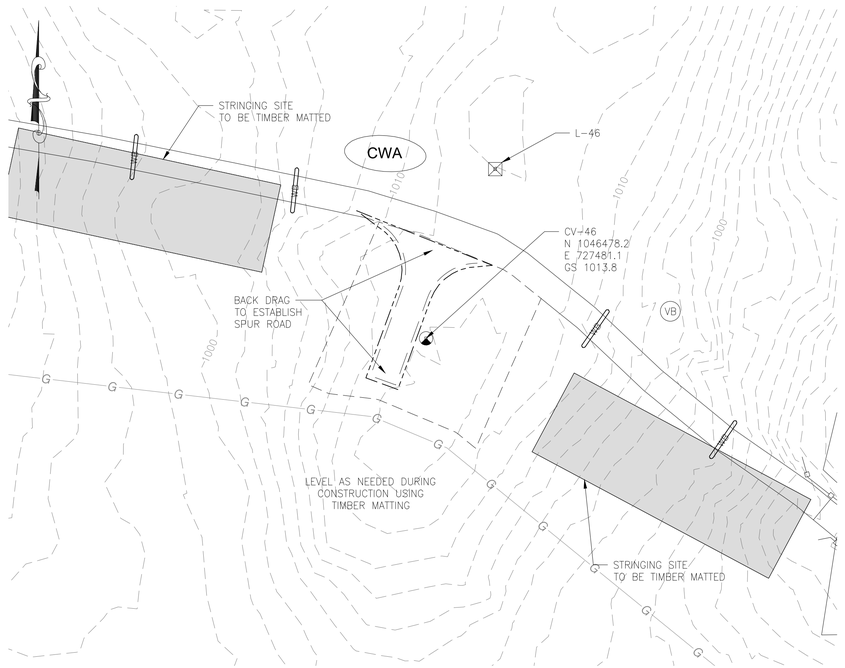
RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS



POLE NO. CV-45



POLE NO. CV-46

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1		ADDED LIMITS OF GRADING, IDENTIFIED VEGETATIVE BUFFER AREAS	MJG	PGC	PGC	PGC

DIGIOIA GRAY & ASSOCIATES
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DRWN:	MJG
DSGN:	MJG
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS- UNION VALE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-29

REV. 1

STATE OF NEW YORK * PAUL G CASS * 086902 LICENSED PROFESSIONAL ENGINEER 12/22/16

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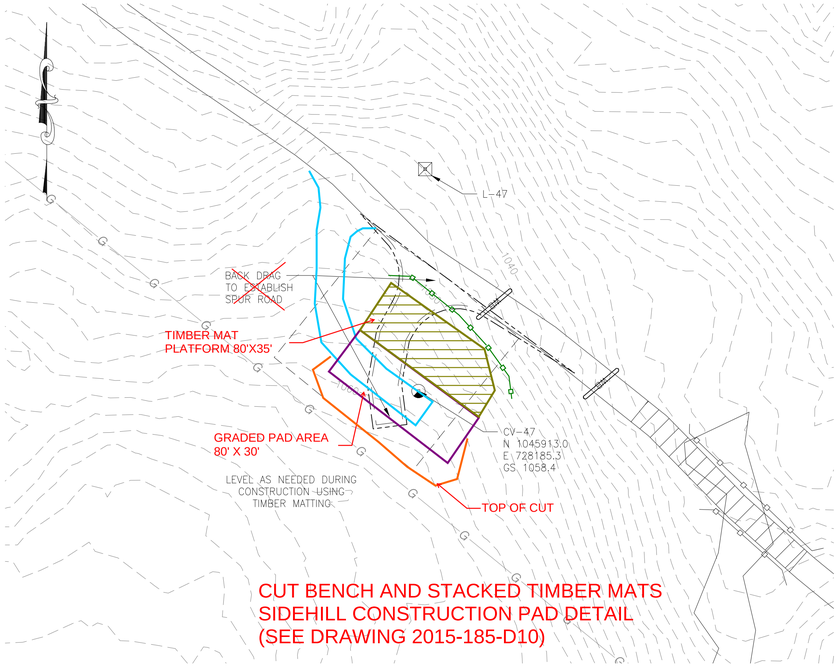
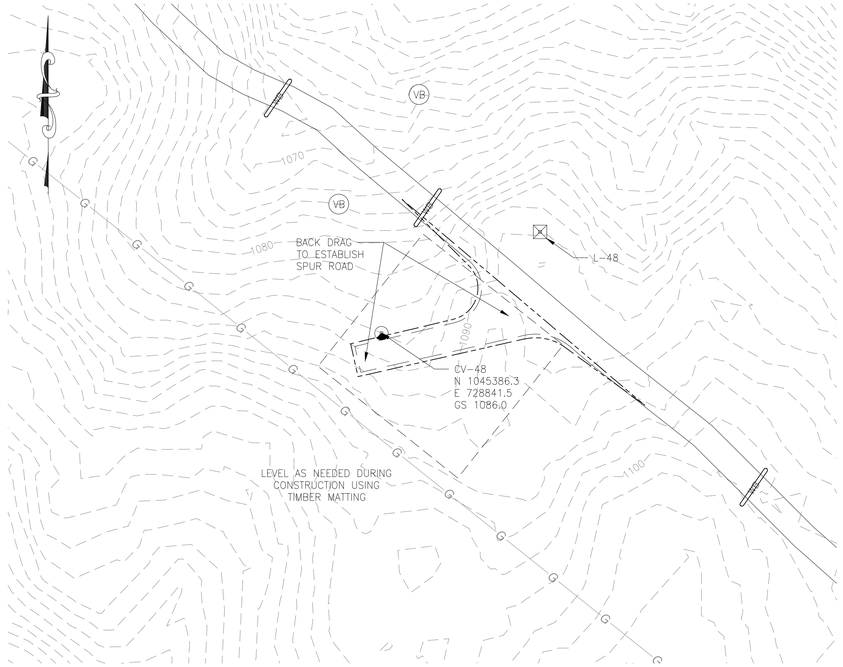
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-30



POLE NO. CV-47



POLE NO. CV-48

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

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IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

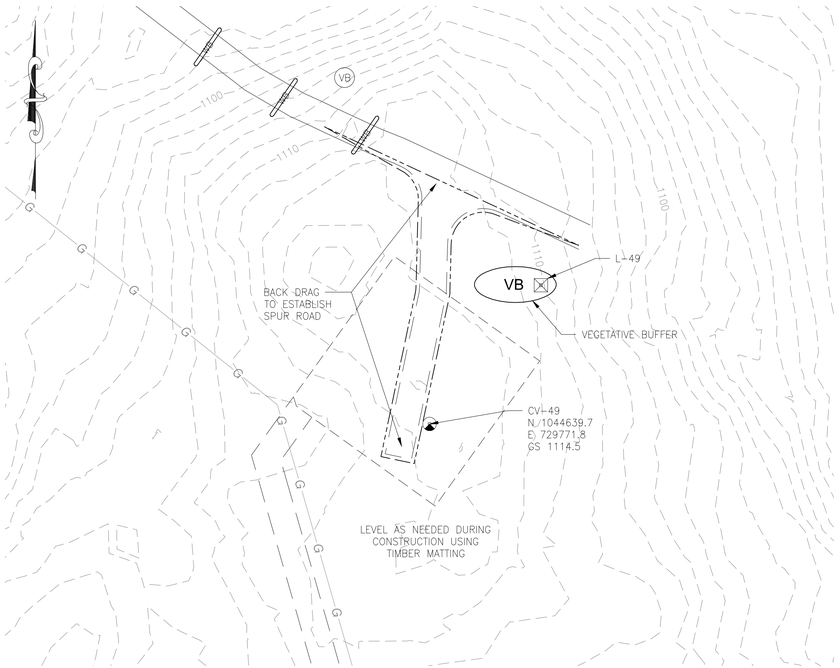
UPON COMPLETION OF GRADING, PROVIDE TEMPORARY SOIL STABILIZATION WITHIN 48 HRS DISTURBANCE (MULCH, TEMPORARY SEEDING OR EROSION CONTROL BLANKET).

MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS



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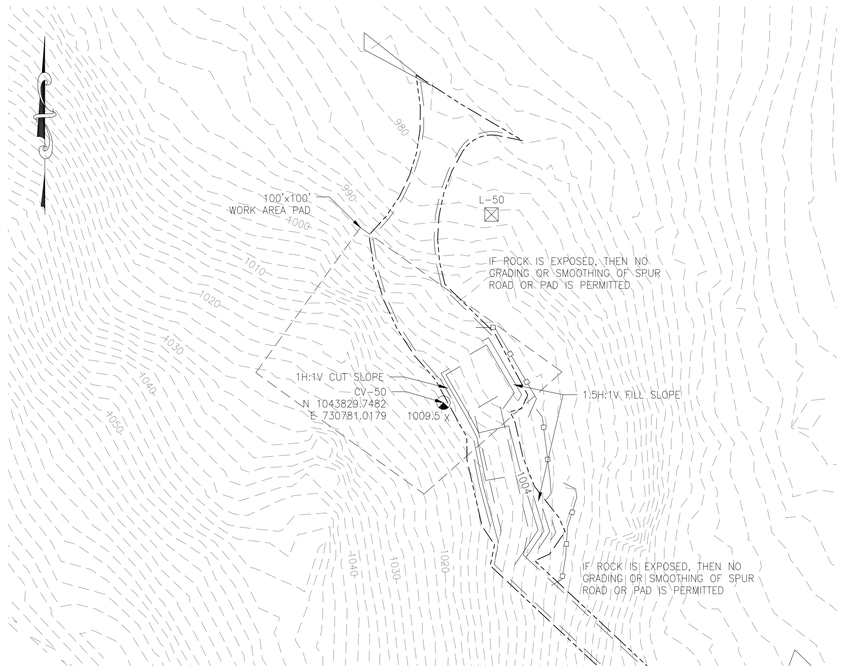

POLE NO. CV-50
DOVER

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING, IDENTIFIED VEGETATIVE BUFFER AREAS	MJG	PGC	PGC	PGC

DIGIOIA GRAY & ASSOCIATES
570 BEATTY ROAD
MONROEVILLE, PA
15146
(412) 372-4500
www.digioiagray.com

DRWN:	LEB
DSGN:	LEB
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS - PLEASANT VALLEY AND LAGRANGE

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-30

REV. 1

STATE OF NEW YORK · PAUL G CASS · 11/25/16 · 096902 · LICENSED PROFESSIONAL ENGINEER

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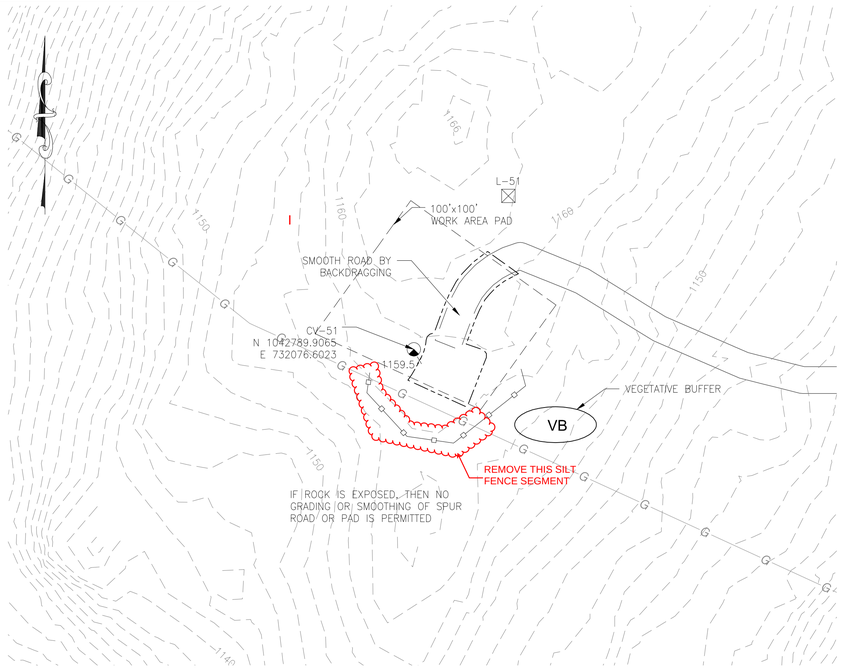
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-31



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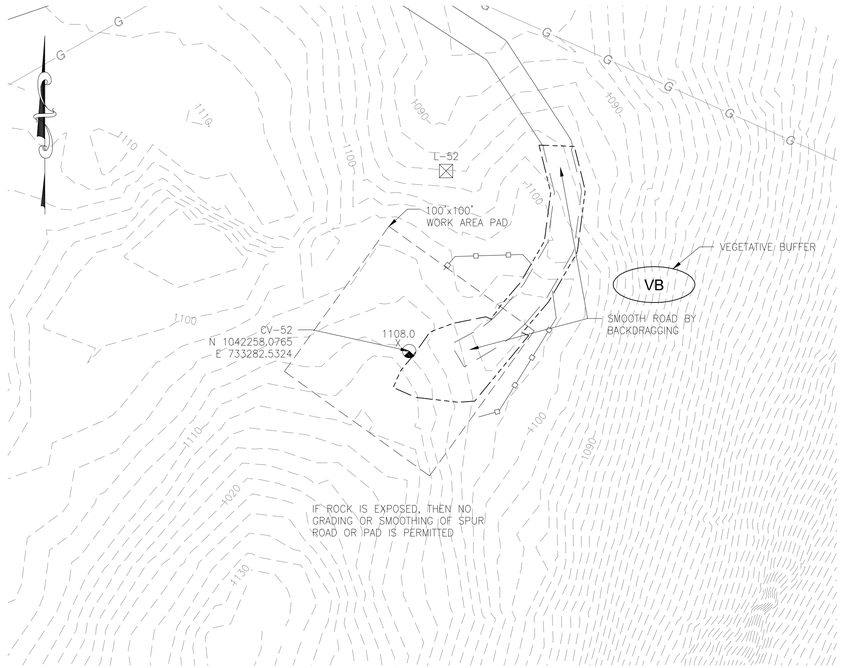

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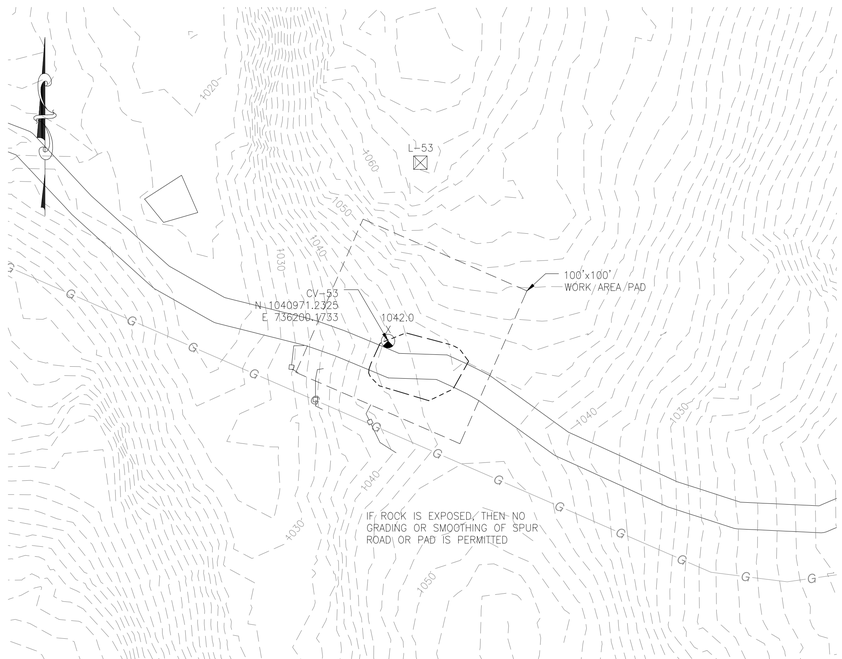

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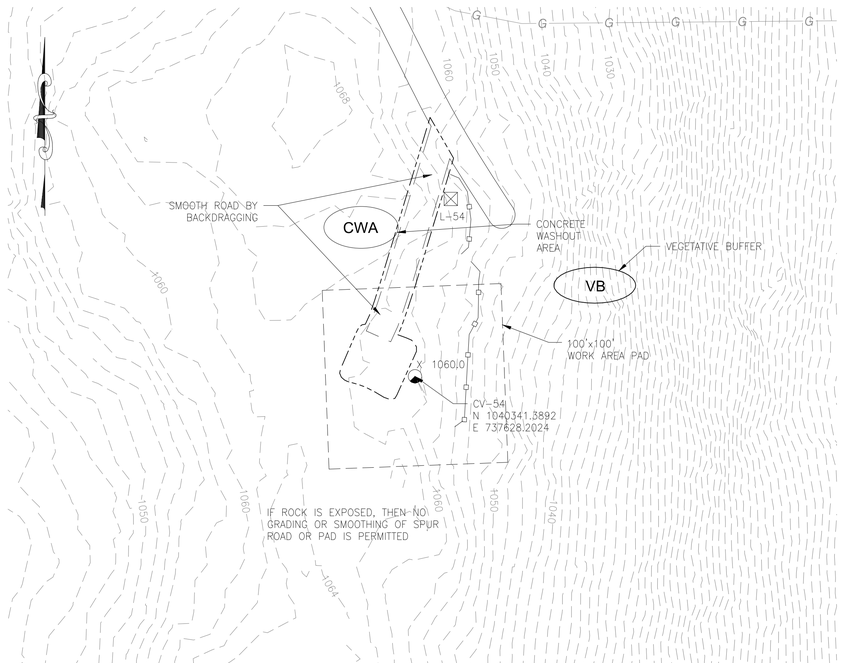

POLE NO. CV-54

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

GRADED PADS FOR CONSTRUCTION ARE LEVEL AND APPROXIMATELY 80 FT. BY 80 FT. IN PLAN.

GRADED PADS FOR MAINTENANCE ARE LEVEL AND APPROXIMATELY 25 FT. BY 25 FT. IN PLAN. THE MAINTENANCE PADS MAY BE BUILT FOLLOWING LINE CONSTRUCTION. TIMBER MATTING WILL BE UTILIZED AS A LEVEL SURFACE FOR CONSTRUCTION AT SITES WITH MAINTENANCE PADS.

SPUR ROADS WILL EITHER BE SMOOTHED PATHS OVER EXISTING TOPOGRAPHY OR GRADED. SINCE THE USE OF SPUR ROADS WILL BE INFREQUENT, IT IS PREFERRED THAT SPUR ROAD CONSTRUCTION BY BACKDRAGGING A DOZER BLADE BE USED TO CREATE A SMOOTH SURFACE. GRADING IS REQUIRED ONLY TO EITHER PREPARE A CONSTRUCTION OR MAINTENANCE PAD OR TO LIMIT SIDE SLOPES ON THE ROAD. GRADING PLANS FOR SITE CONSTRUCTION AND ACCESS ARE SHOWN ON DRAWINGS SWPPP-18 THROUGH SWPPP-26.

CUTS AND FILL SLOPES 2H:1V GENERALLY WITH 1H:1V SLOPES IN ROCK CUTS AND 1.5H:1V SLOPES IN ROCK FILLS.

ALL CUT AND FILL SLOPES WITH AN EXPOSED SLOPE OF 6 FT. OR GREATER RECEIVES AND EROSION CONTROL BLANKET

EACH SITE INCLUDING SPUR ROAD ARE DESIGNED FOR BALANCED CUT AND FILLS. MINOR ADJUSTMENTS IN GRADING MAY BE REQUIRED TO ACHIEVE THIS BALANCE.

PRIOR TO PLACING ANY FILL THE SUBGRADE SHALL BE INSPECTED FOR SUITABILITY AS A SUBGRADE, IF IMPROVEMENT IS NEEDED THEN THE SUBGRADE SHALL BE PROOF-ROLLED WITH A VIBRATORY ROLLER.

IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

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RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
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- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
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ROADS, ACCESS ROADS AND SPUR ROADS
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EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
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DRWN:	LEB
DSGN:	LEB
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS - LAGRANGE AND DOVER

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-31

REV. 1

STATE OF NEW YORK * PAUL G CASS * 036902 LICENSED PROFESSIONAL ENGINEER 12/22/16

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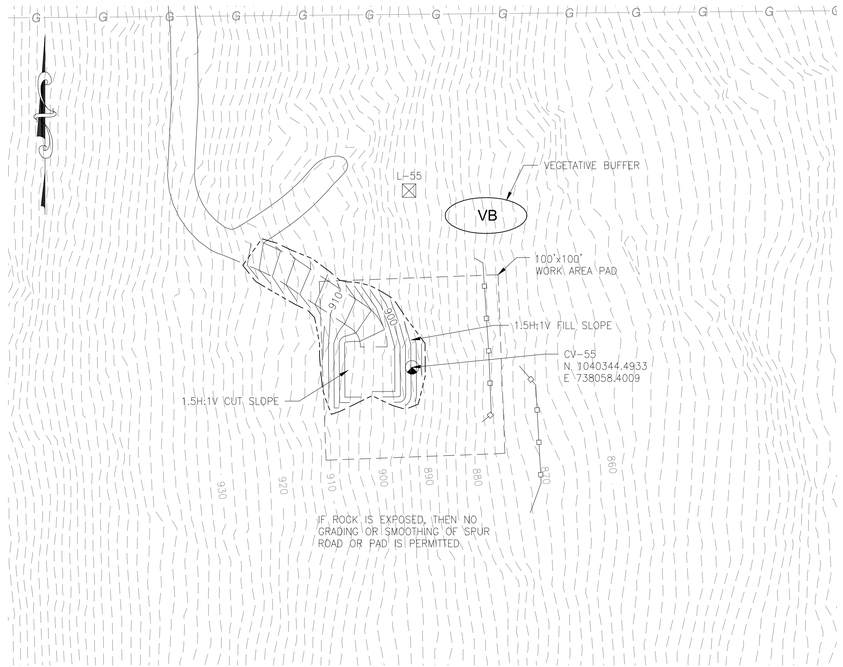
DRAWING NO. SWPPP-32

DIGIOIA GRAY PROJECT NO. 2015-184



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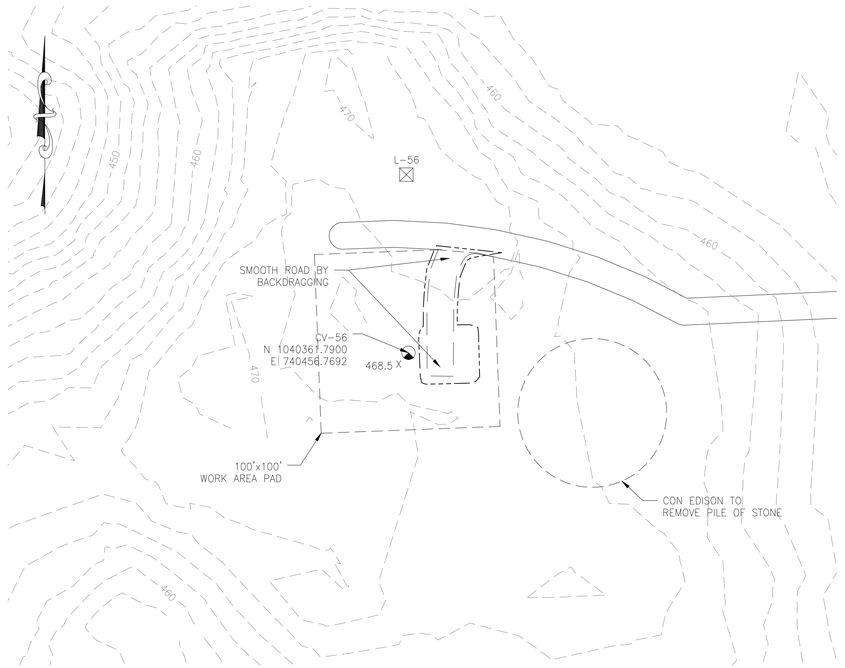

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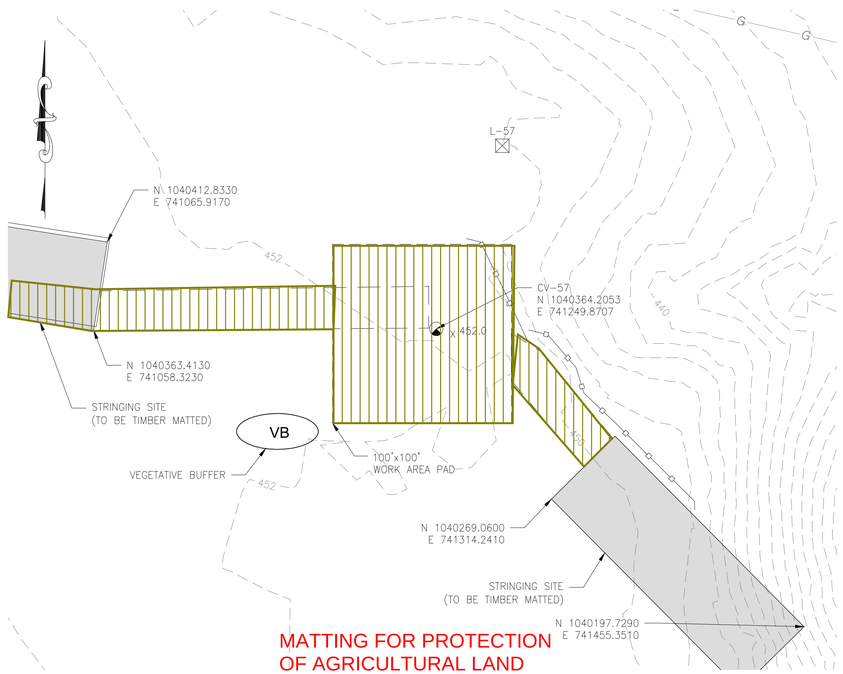

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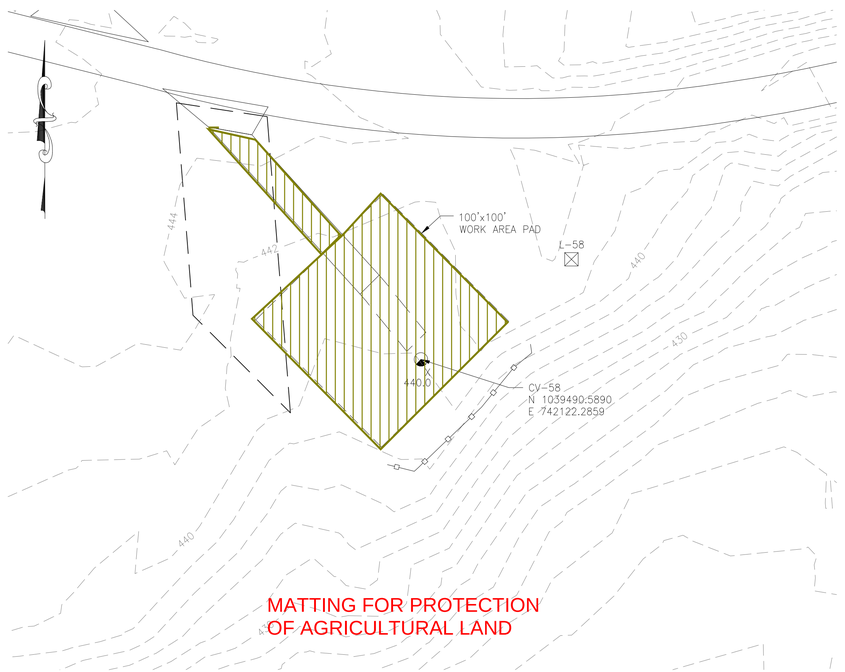

POLE NO. CV-58

GRADING PLAN NOTES

GENERAL

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SEQUENCE

DEFINE AND MARK LIMITS OF WORK

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IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

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RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
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TRANSMISSION WORK AREAS
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- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
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EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

ft 0 40 80 160

REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING, IDENTIFIED VEGETATIVE BUFFER AREAS	MJG	PGC	PGC	PGC

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DRWN:	LEB
DSGN:	LEB
CHK'D:	PGC
APP:	PGC
DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS - DOVER

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-32

REV. 1

STATE OF NEW YORK * PAUL G CASS * LICENSED PROFESSIONAL ENGINEER 036902 12/22/16

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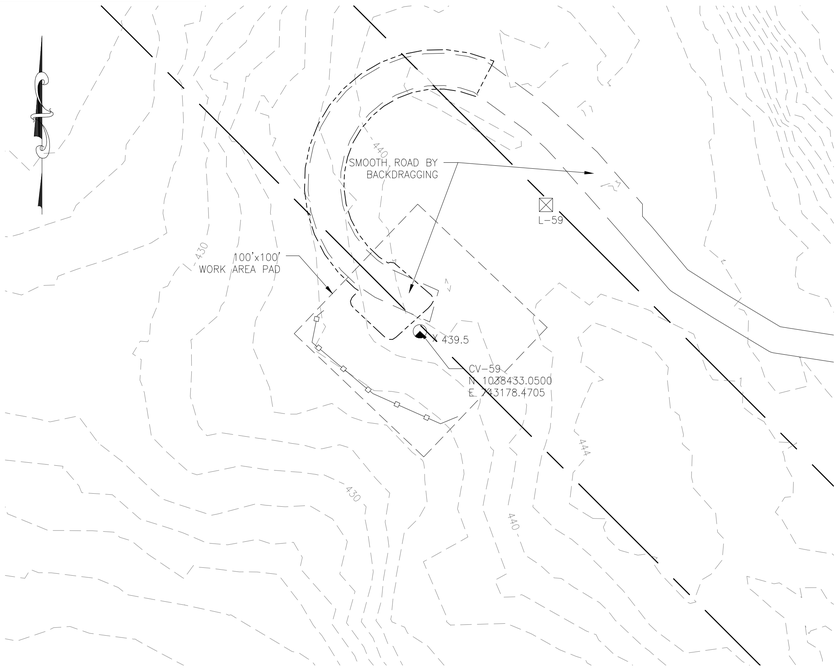
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-33



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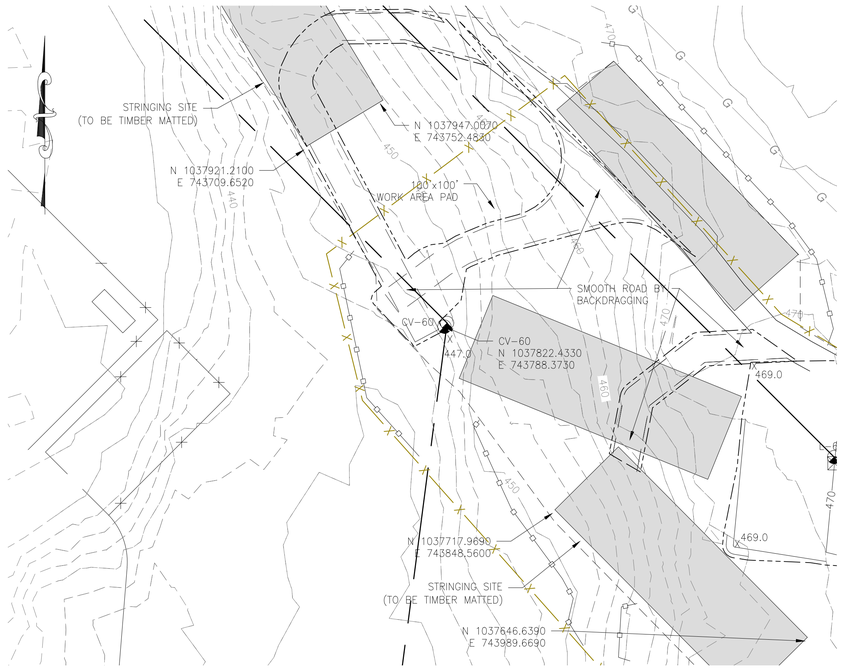

POLE NO. CV-60

GRADING PLAN NOTES

GENERAL

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PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

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RELATED DRAWINGS

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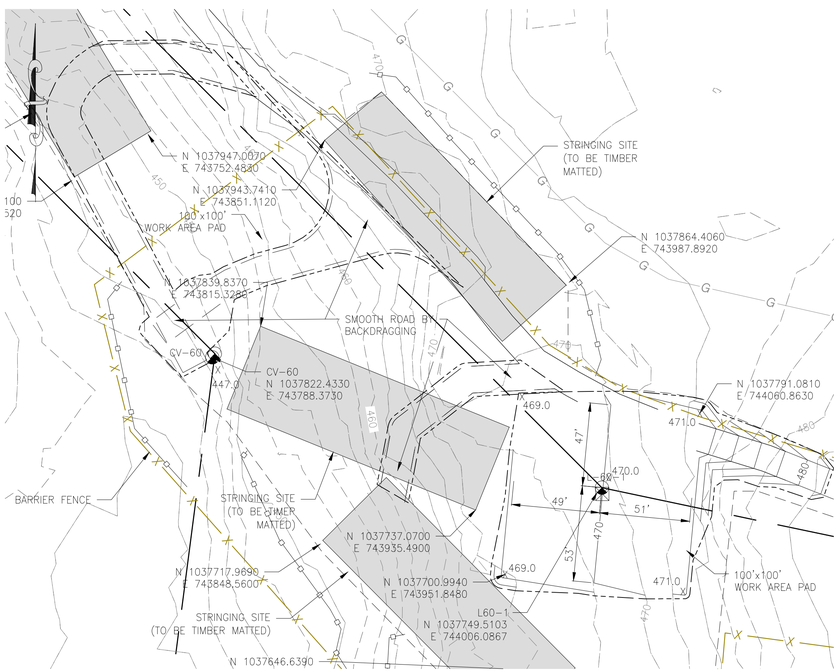
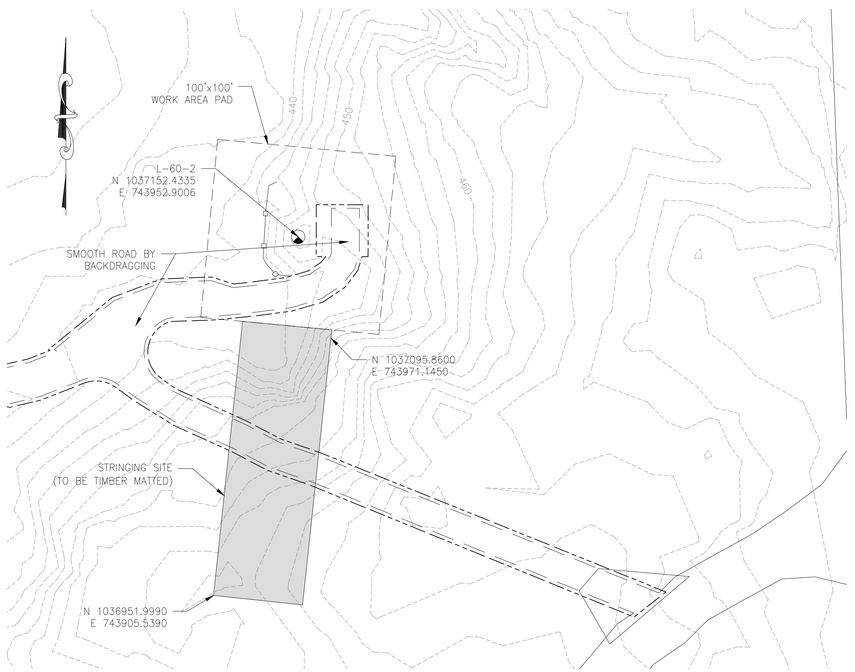
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS



POLE NO. L-60-1
DOVER



POLE NO. L-60-2

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION POLE
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA
- LIMITS OF GRADING

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
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Cricket Valley Energy

ft 0 40 80 160

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REFERENCE TITLE	NUMBER	NO.	DATE	REVISION	DRWN	DSGN	CHK'D	APP
		1	5/5/17	ADDED LIMITS OF GRADING, ADDED BARRIER FENCE	MJG	PGC	PGC	PGC

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CHK'D:	PGC
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DATE:	12-22-16
SCALE:	AS NOTED

CRICKET VALLEY 345kV TRANSMISSION LINES
GRADING PLANS - DOVER

CRICKET VALLEY ENERGY CENTER
DOVER PLAINS, NEW YORK

DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-33

REV. 1

STATE OF NEW YORK · PAUL G CASS · 11/22/16 · 096902 · LICENSED PROFESSIONAL ENGINEER

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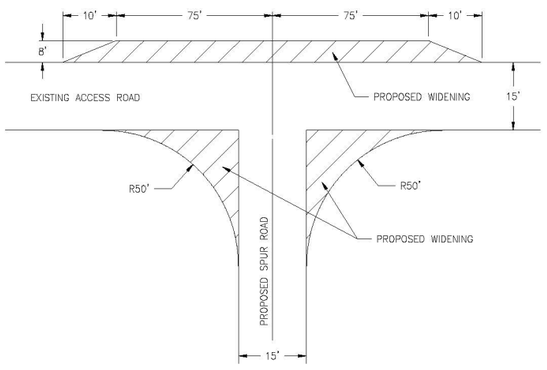
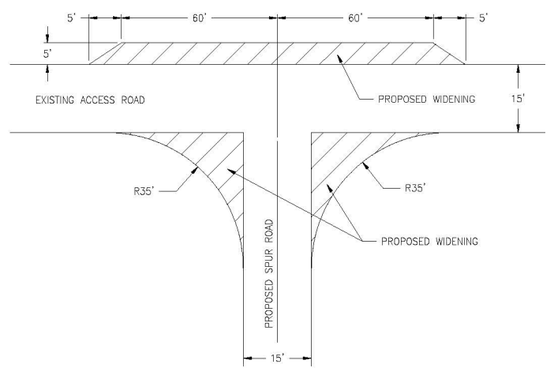
DIGIOIA GRAY PROJECT NO. 2015-184

DRAWING NO. SWPPP-34



TYPICAL ACCESS ROAD TO SPUR ROAD CORNER DETAIL 1 (N.T.S.)

(INCLUDES TRACTOR WITH STEERABLE POLE TRAILER)



TYPICAL ACCESS ROAD TO SPUR ROAD CORNER DETAIL 2 (N.T.S.)

(INCLUDES ALL VEHICLES UNDER 40', EXCLUDES TRACTOR WITH STEERABLE POLE TRAILER)

GRADING PLAN NOTES

GENERAL

CONFINE CONSTRUCTION EQUIPMENT AND VEHICLES TO DEFINED ACCESS ROADS, SPUR ROADS, PADS (CONSTRUCTION, MAINTENANCE, STRINGING AND TIMBER).

GRADED PADS FOR CONSTRUCTION ARE LEVEL AND APPROXIMATELY 80 FT. BY 80 FT. IN PLAN.

GRADED PADS FOR MAINTENANCE ARE LEVEL AND APPROXIMATELY 25 FT. BY 25 FT. IN PLAN. THE MAINTENANCE PADS MAY BE BUILT FOLLOWING LINE CONSTRUCTION. TIMBER MATTING WILL BE UTILIZED AS A LEVEL SURFACE FOR CONSTRUCTION AT SITES WITH MAINTENANCE PADS.

SPUR ROADS WILL EITHER BE SMOOTHED PATHS OVER EXISTING TOPOGRAPHY OR GRADED. SINCE THE USE OF SPUR ROADS WILL BE INFREQUENT, IT IS PREFERRED THAT SPUR ROAD CONSTRUCTION BY BACKDRAGGING A DOZER BLADE BE USED TO CREATE A SMOOTH SURFACE. GRADING IS REQUIRED ONLY TO EITHER PREPARE A CONSTRUCTION OR MAINTENANCE PAD OR TO LIMIT SIDE SLOPES ON THE ROAD. GRADING PLANS FOR SITE CONSTRUCTION AND ACCESS ARE SHOWN ON DRAWINGS SWPPP-18 THROUGH SWPPP-26.

CUTS AND FILL SLOPES 2H:1V GENERALLY WITH 1H:1V SLOPES IN ROCK CUTS AND 1.5H:1V SLOPES IN ROCK FILLS.

ALL CUT AND FILL SLOPES WITH AN EXPOSED SLOPE OF 6 FT. OR GREATER RECEIVES AND EROSION CONTROL BLANKET

EACH SITE INCLUDING SPUR ROAD ARE DESIGNED FOR BALANCED CUT AND FILLS. MINOR ADJUSTMENTS IN GRADING MAY BE REQUIRED TO ACHIEVE THIS BALANCE.

PRIOR TO PLACING ANY FILL THE SUBGRADE SHALL BE INSPECTED FOR SUITABILITY AS A SUBGRADE, IF IMPROVEMENT IS NEEDED THEN THE SUBGRADE SHALL BE PROOF-ROLLED WITH A VIBRATORY ROLLER.

IF STONE IS REQUIRED DUE TO A SOFT SUBGRADE, PLACE STONE OVER GEOTEXTILE.

IF STONE IS REQUIRED DUE TO ROCKY SURFACE CONDITIONS, UTILIZE NYSDOT TYPE 4 SUBGRADE STONE. IF THE SLOPE OF THE ROAD IS STEEPER THAN 20% THEN A GEOGRID SHALL BE PLACED BETWEEN THE SUBGRADE AND THE STONE SUBBASE.

PLACE FILL IN LIFTS NOT EXCEEDING 8 INCHES. IF MATERIAL IS GRANULAR OR ROCK, COMPACT BY TRACKING EQUIPMENT ACROSS THE LIFT (COVER ALL AREAS) BY AT LEAST 4 PASSES. IF MATERIAL IS COHESIVE, COMPACT AS SPECIFIED BY THE ENGINEER ON THE SITE TO ACHIEVE A DRY DENSITY OF AT LEAST 100% STANDARD PROCTOR DENSITY (ASTM D698).

SEQUENCE

DEFINE AND MARK LIMITS OF WORK

CLEARING AND GRUBBING AS NECESSARY FOR SPUR ROAD, PAD AREAS AND INSTALLATION OF PERIMETER EROSION AND SEDIMENTATION CONTROLS

IDENTIFY AND PROTECT BURIED UTILITIES

INSTALLATION OF REMAINING EROSION AND SEDIMENTATION CONTROL FEATURES

SMOOTHING OR GRADING AS SPECIFIED

UPON COMPLETION OF GRADING, PROVIDE TEMPORARY SOIL STABILIZATION WITHIN 48 HRS DISTURBANCE (MULCH, TEMPORARY SEEDING OR EROSION CONTROL BLANKET).

MOW AS REQUIRED TO IDENTIFY DESIGNATED SPUR ROADS AND PAD AREAS.

PROVIDE PERMANENT SOIL STABILIZATION AND SEEDING WITH 7 DAYS FOLLOWING COMPLETION OF CONSTRUCTION.

RELATED DRAWINGS

SWPPP-01 THROUGH -017 PLANS	EROSION AND SEDIMENTATION CONTROL
2051-185-D-01 THROUGH -04 DETAILS (SEE APPENDIX S	EROSION AND SEDIMENTATION CONTROL OF THE EM&CP FOR DRAWINGS

LEGEND:

TRANSMISSION FACILITIES
- PROPOSED TRANSMISSION POLE
- EXISTING TRANSMISSION TOWER
- TRANSMISSION CENTERLINE ALIGNMENT
- EXISTING DISTRIBUTION POLES

TRANSMISSION WORK AREAS
- WORK PAD AREA
- STRINGING EQUIPMENT AREA

GAS PIPELINE
- EXISTING GAS PIPELINE

BASEMAP FEATURES
- CHG&E 20' GASLINE EASEMENT
- EXISTING 2 FT CONTOURS

ROADS, ACCESS ROADS AND SPUR ROADS
- EXISTING ACCESS ROAD
- PROPOSED SPUR ROAD

EROSION AND SEDIMENT CONTROL
- PROPOSED SILT FENCE
- WB PROPOSED WATER BAR

Cricket Valley Energy

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DRAWING NO. SWPPP-34

REV. 0

STATE OF NEW YORK PAUL G CASS 086902 LICENSED PROFESSIONAL ENGINEER

Job# 2517005700

CV Access Points

Name	Address	Type guard	Update	Sheet	Crossing
Existing N-2	5 Niagara Rd & 1482 Main St (Rt.44) Pleasant Valley NY 12569				
CV-01	30 Kovacs Ln Pleasant Valley NY 12569	H-frame	Additional	P-1	T-line
		H-frame	Additional	P-1	
		H-frame	Additional	P-1	T-line
		H-frame	Additional	P-1	
CV-02		H-frame	Additional	P-1	T-line
		H-frame	Additional	P-1	
		H-frame	Additional	P-1	T-line
		H-frame	Additional	P-1	
CV-03					
CV-04	35 Niagara Rd Pleasant Valley NY 12569	H-frame		P-2	3-phase Dist crossing, Comunication crossing
	1533 Main St (Rt.44) Pleasant Valley NY 12569	H-frame		P-2	Road crossing , sec & comunication crossing
CV-05	1098 Freedom Rd Pleasant Valley NY 12569	H-frame		P-2	3-phase Dist crossing, Comunication crossing
CV-05	31 South Ave Pleasant Valley NY 12569				
CV-06					
CV-07	199 Forest Valley Rd Pleasant Valley NY 12569	Boom truck		P-2	Road crossing
CV-08					
CV-09	555 (GPS) Traver Rd Pleasant Valley NY 12569	H-frame		P-3	Road crossing, single phase crossing
CV-10					
CV-11	515 Traver Rd Pleasant Valley NY 12569				
CV-12					
CV-13	68 Medaugh Rd Pleasant Valley NY 12569 (Water Dept Gate)	H-frame		P-4	3 phase sub T
		H-frame		P-4	
CV-14					
CV-15					
	Tanconic PKY	H-frame		P-5	Road crossing (T-PKY)
		H-frame		P-5	

Coordinates are approx.

Field Verify acc points numbers for house address

Job# 2517005700

CV Access Points

Name	Address	Type guard	Update	Sheet	Crossing
CV-16	4427 (GPS) Rossway Rd Pleasant Valley NY 12569				
CV-17		H-frame	Move	P-5	Road crossing, single phase crossing
CV-18	262 Downing Rd Pleasant Valley NY 12569	Boom truck	Move	P-5	Road crossing, single phase crossing (riser)
CV-19	95 Downing Rd Pleasant Valley NY 12569	H-frame	Move	P-6	Road crossing Single phase
CV-20					
CV-21	215 Gidley Rd Pleasant Valley NY 12569				
CV-22	235 Gidley Rd Pleasant Valley NY 12569	Boom truck	Road	P-7	Road crossing, single phase crossing
	402 (GPS) Skidmore rd Pleasant Valley NY 12569	H-frame		P-7	Road crossing
	2977 (GPS) Hwy 82 Pleasant Valley NY 12569	H-frame		P-7	Road crossing 3 Phase crossing (Rt82)
		H-frame	Additional	P-7	
CV-23	183 N Smith Rd Lagrange, NY 12540 (Cawley Way)				
CV-24-1					
CV-24-2					Access throug tree line at end of ROW near bone yard
CV-24-3					
CV-25					
	164 Oswego Rd Pleasant Valley NY 12569				
CV-26A	206 N Smith Rd Lagrange, NY 12540				
CV-27A		H-frame		P-8	Road crossing, single phase crossing
CV-28		Boom truck	Additional	P-8	
CV-29					
CV-30	797 Waterbury Hill Rd Lagrangeville, NY 12540				
CV-31					
CV-32		H-frame		P-9	Road crossing, single phase crossing
CV-33	291 N Clove Rd Verbank, NY 12585				
CV-34	443 (GPS) N Clove Rd Verbank, NY 12585				
CV-35		Boom truck	Additional	P-10	Low Single phase
CV-36					
CV-37					

Field Verify acc points numbers for house address

Coordinates are approx.

Job# 2517005700

CV Access Points

Name	Address	Type guard	Update	Sheet	Crossing
CV-38	591 N Clove Rd Millbrook, NY 12545	H-frame		P-11	Road crossing, single phase crossing
CV-39					
CV-40	127 Chestnut Ridge Rd Millbrook, NY 12545				
CV-41					
CV-42	127 Chestnut Ridge Rd Millbrook, NY 12545	H-frame		P-12	Road crossing, single phase crossing
CV-43					
CV-44					
CV-45	1 McCourt Rd Dover Plains, NY 12522				
CV-46					
CV-47					
CV-48					accaess
CV-49					
CV-50					
CV-51					
CV-52		H-frame		P15	Road crossing, single phase crossing
CV-53	96E (176 GPS) Cooperstown Rd Dover Plains, NY 12522				
CV-54					
CV-55					
CV-56	186 Dover Furnace Rd 12522	Boom truck		P-16	Road crossing, 2x dead wire crossing
CV-57					
CV-58	115 Dover Furnace Rd 12522	H-frame	Move	P-17	Road crossing, single phase delta crossing
CV-59	2263 (GPS) NY-22 Dover Plains NY 12522				RxR Need MNRR permit language
CV-60					
L60-1		H-frame		P-18	Road crossing 3 Phase crossing
		Boom truck	Additional	P18	

Coordinates are approx.

Field Verify acc points numbers
for house address

Job# 2517005700

CV Access Points

Name	Address	Type guard	Update	Sheet	Crossing
CVEC take	2238 NY-22 Dover Plains NY 12522				
60-2					
60-3					
	2441 NY-22 Dover Plains NY 12522				
	31 (GPS) Cricket Hill Rd Dover Plains NY 12522				
61	43 (GPS) Cricket Hill Rd Dover Plains NY 12522				
62					
63	349 Old Route 22 Dover Plains, NY 12522				
64					
	344 (GPS) Old Route 22 Dover Plains, NY 12522	H-frame		P19	Road crossing 3 Phase crossing
		H-frame	Additional		
65	716 (GPS) Berkshire Rd Dover Plains, NY 12522				
66		H-frame	Additional	P20	Road crossing 3 Phase crossing
		H-frame	Additional	P-20	
67	716 (GPS) Berkshire Rd Dover Plains, NY 12522	H-frame			Road crossing 3 Phase crossing
		Boom truck		P-20	
68	55 (GPS) Weil Rd Dover Plains, NY 12522				
69					
70	108 (GPS) Weil Rd Dover Plains, NY 12522				
71	73 Lake Ellis Rd Wingdale Rd 12594	H-frame	Additional	P-21	3 phase Dist crossing 71-72 off road no access
72	289 Dog Tail Corners Rd Wingdale NY 12594	H-frame		P-21	Road crossing, single phase delta crossing
		Boom truck	Additional	P-21	
73	265 Dog Tail Corners Rd Wingdale NY 12594				
74		Boom truck	Additional	P-22	Single phase
75					

35

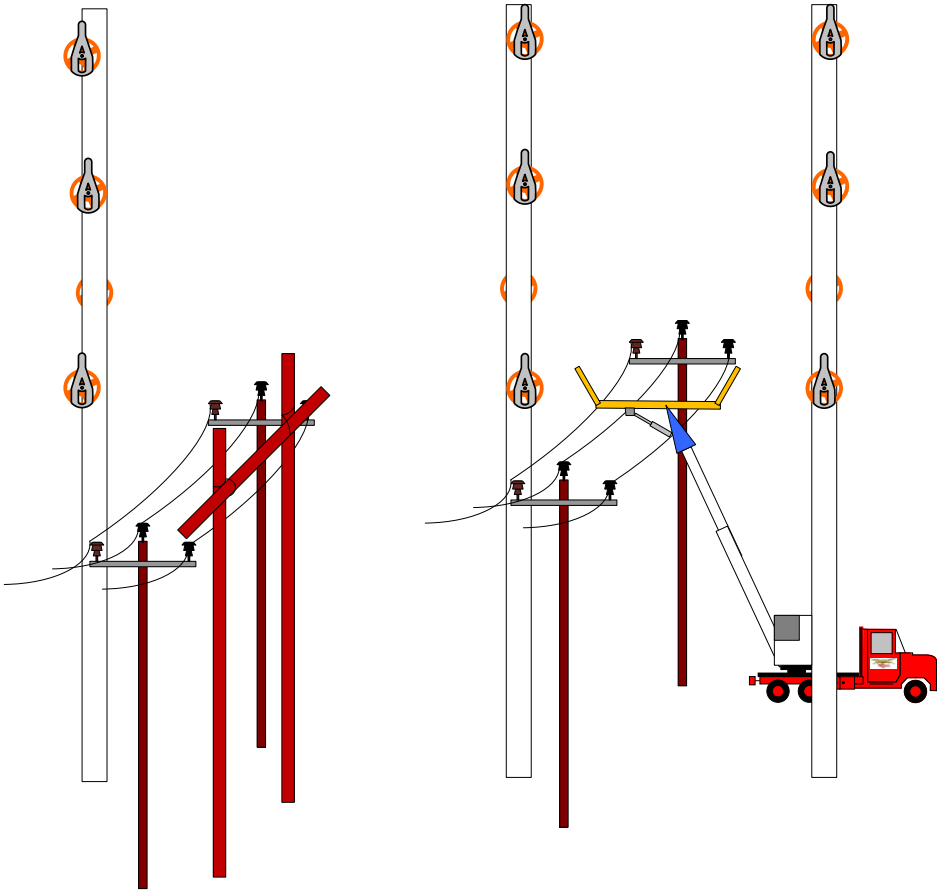
10

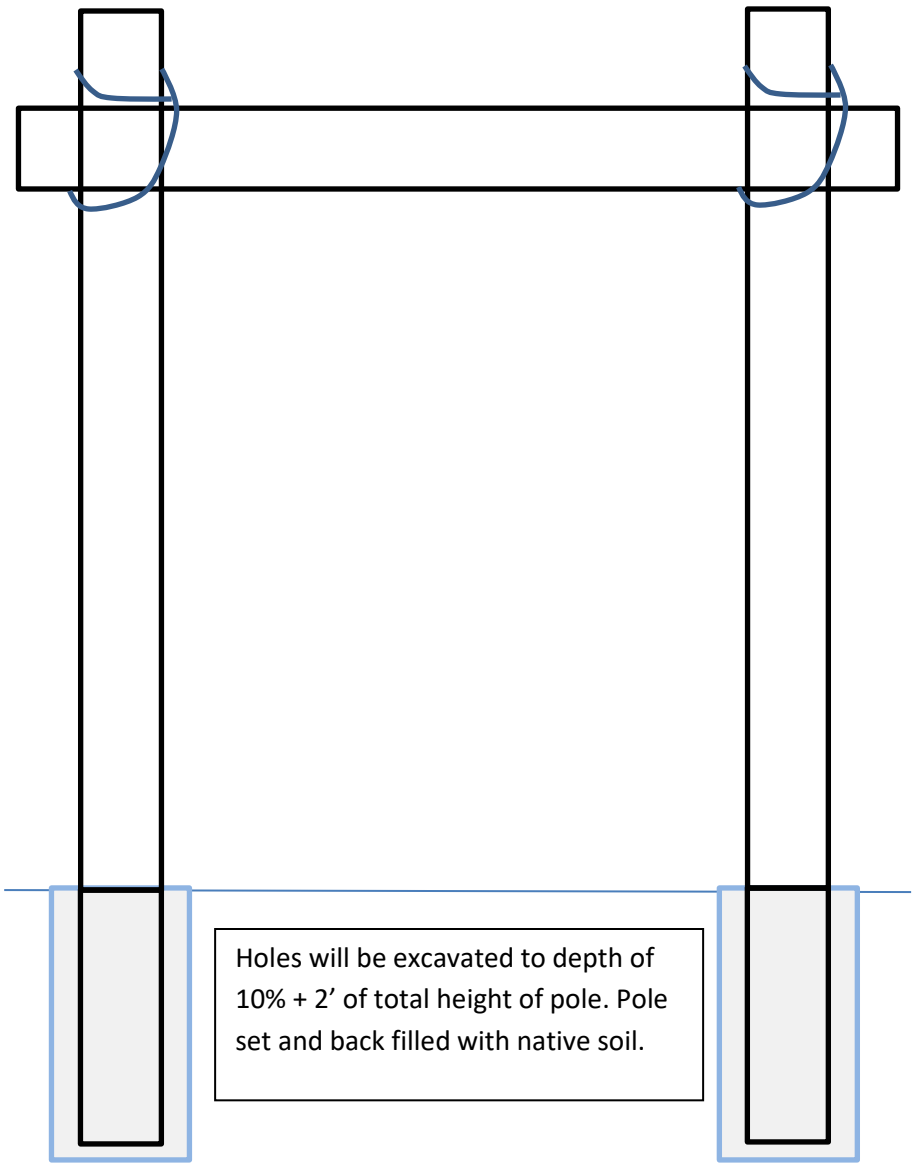
Field Verify acc points numbers for house address

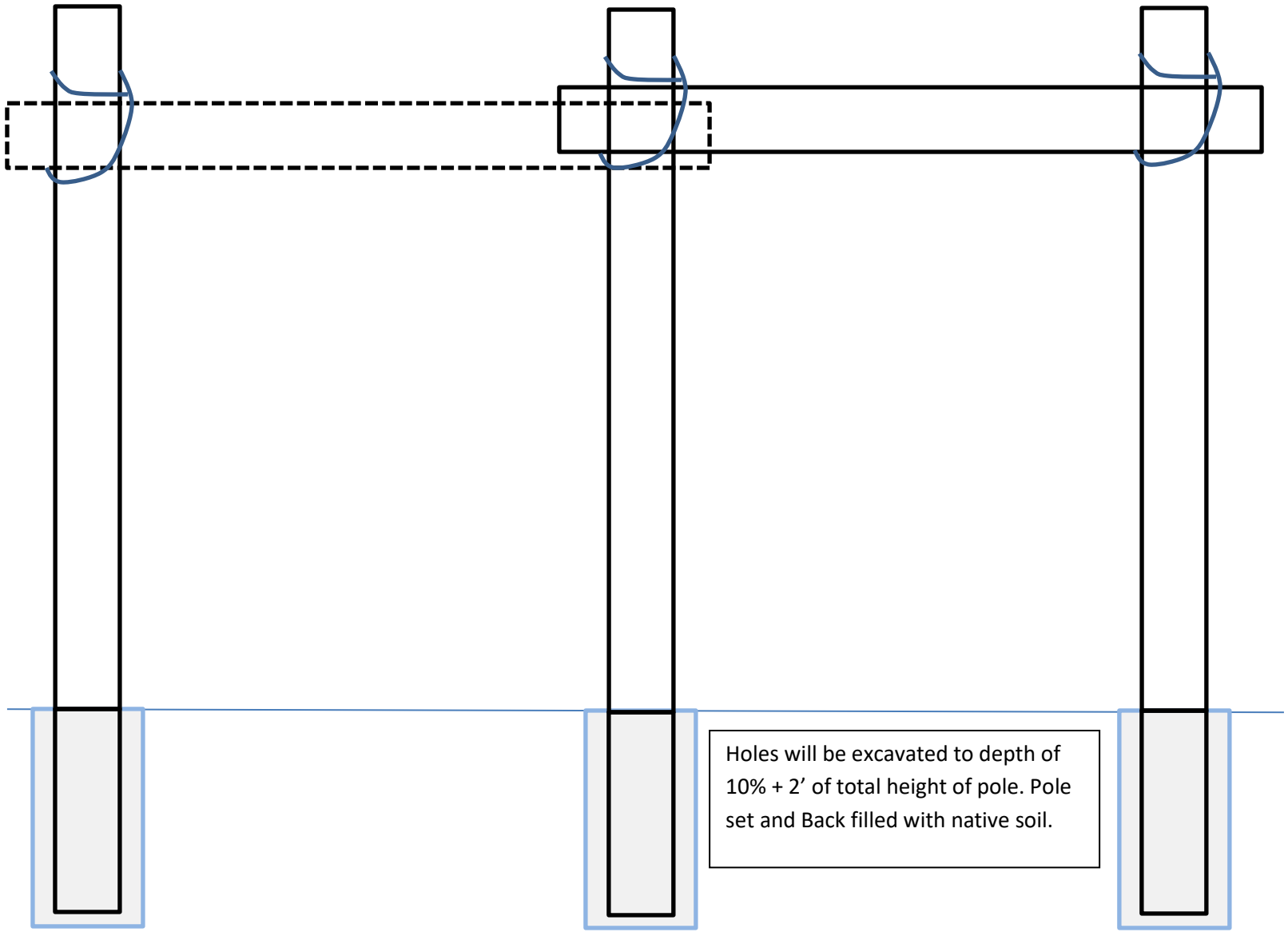
Coordinates are approx.

M. J. ELECTRIC
A QUANTA SERVICES COMPANY

Guard Structures Utilized to protect utility Road crossing




Holes will be excavated to depth of 10% + 2’ of total height of pole. Pole set and back filled with native soil.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